

# Online Academic Staff Booking Appointment Platform

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# Online Academic Staff Booking Appointment Platform

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## Declaration

The project submitted herewith is a result of my own efforts in totality and in every aspect of the project works. All information that has been obtained from other sources had been fully acknowledged. I understand that any plagiarism, cheating or collusion or any sorts constitutes a breach of TAR University College rules and regulations and would be subjected to disciplinary actions.

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## Abstract

A manual time booking procedures is time consuming and ineffective for students or staff to use. An online time booking or a reservation system has been widely used in the world due to its convenience that everyone can access via the Internet and its real-time updated information. This framework main motivation was to propose a web-based online time booking platform for the undergraduates and staff in higher education institution. This platform provides function for undergraduates to book a consultation time of a staff in a simpler way and it also enables the staff to manage their booking time schedule in a more effective way. The proposed system is also able to simplify the process for a user to complete a booking process. Thus, this platform is able to solve the problem faced using the manual booking process and it can provide a better user experience to the academic staff and the students in an academic institution. This project covers five main modules of the platform proposed which are login module, booking module, time schedule module, feedback module and chatbox module. The features of the platform included user authentication, enable users to make consultation appointment requests, view appointment booking schedule, write a feedback after an appointment is ended, leave a message to users to communicate with each other and etc. The software development used to develop the platform is Incremental Model which core product functions are developed first and supplementary functions will be developed in the next version until the desired system is released. The tools used to develop the web-based platform is Visual Studio 2017 software and the database of system will be coded using My Structured Query Language (MySQL).

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# Chapter 1

## **Introduction**

# 1 Introduction

This framework is aimed to propose a web-based online booking appointment platform for undergraduates and staff in higher education institutions. The platform is proposed to provide a more time efficient way for students to book a consultation time and for staff to manage their booking time schedule in a more effective way. Hence, the target users for the platform are the academic staff which includes tutors, lecturers, academic advisors, counsellors and also the undergraduate students. The fundamental modules and functions that should be implemented in the online academic staff time booking platform will be discussed in this chapter.

## 1.1 Objective

The objective of this project is to propose a web-based online academic staff booking appointment platform in a higher education institution in order to minimize the time usage for the students and the staff to book or manage the consultation time booking.

- To identify functions needed in the appointment booking platform for staff and students.
- To simplify the process of consultation time booking.
- To generate reports of user experience and user satisfaction on current manual time booking procedures.
- To identify improvements and suggestions for simplifying the time booking procedures in a new computerized time booking platform.

## 1.2 Traditional Booking Appointment Background

All organizations are required to schedule and manage their appointments and reservations effectively in the daily operations of business. However, the process taken to schedule an appointment may require more effort and time than it seems especially when a traditional way is used in an organization (Appointment-Plus, 2012). The traditional method to book an appointment such as using phone calls and messages normally used up a large amount of

time (Bookly Team, 2019). This is because some people may be too busy to pick up phone calls which cause others need to call many times until the person is free to pick up the call and discuss about the appointment schedule details. Moreover, manual appointment scheduling method which is written down the meeting date, time and person to be met on paper is also a time consuming method (Appointment Care, 2018). This is because using the manual scheduling method can be a tedious process especially when there are numerous people asking for time appointment and this will easily cause overbooking cases happen too. Hence, manual booking appointment system was eliminated and changed into a computerized system by many organizations nowadays (Oluwaseun Bello et al., 2016).

### **1.3 Advantages and Contribution**

This project is aimed to propose a web-based online academic staff booking appointment platform in a higher education institution which enables the computer devices to access it via the Internet. The significance of this framework is that an online academic staff booking appointment platform is proposed and developed for the academic institutions. This platform enables the consultation time booking process to become computerized which is more time saving and its advantage is able to minimize the cases of overbookings occurred. Thus, this proposed platform is able to solve the problem faced using the manual booking process and it can provide a better user experience to the academic staff and the students in an academic institution.

## 1.4 Project Plan

This project is written to propose a web-based online academic staff booking appointment platform in a higher education institution. Hence, the scope of project focused are the functional requirements for the web-based online booking appointment platform which the platform should provide correct and appropriate functions that are needed for both academic staff and students.

### 1.4.1 Introduction of Modules

There are 2 different users targeted for this online booking appointment platform which are academic staff and students. Both users will have the same modules but perform different functions. The essential modules for the web-based online booking appointment platform are:

- Login module
- Booking module
- Time schedule module
- Feedback module
- Chat box module

### 1.4.2 Functional Requirements for Each Module

#### 1. Login Module

- The system shall validate user login details and redirect to user authorized page according to user roles.
- When the username and password does not match, the system should display an error message and prompt the user to input again.
- When the username or password is left empty, the system shall prompt the user that username or password is not allowed to be left blank.
- The system should enable the user to logout.

## 2. Booking Module

- The system shall allow students to make consultation time requests with title and reasons of the request.
- The system should enable the students to choose their preferable time as their consultation time.
- The system should enable staff to accept or reject a consultation time request with reason.
- The system shall allow staff to determine the consultation venue.
- The system shall let the students and staff cancel their time booking slot.
- The system should let the students and staff check a consultation time booking status.

## 3. Time schedule Module

- The system should allow the students to view staff available consultation time.
- The system shall allow staff to choose their available consultation time.
- The system shall allow staff to modify their available consultation time.
- The system should allow students and staff to view their booking time schedule.
- The system should display the title of consultation time request, persons involved and booking status in the correct date of time schedule.

## 4. Feedback Module

- The system should allow the students to read feedback from staff in a consultation.
- The system should enable the staff to choose to write feedback for a consultation.

- The system should send a notification to a student when feedback is written to the student.

#### 5. Chat Box Module

- The system should allow the students to message staff.
- The system should allow staff to message students.
- The system should allow students and staff to reply in the message box.

## **1.5 Project Team and Organization**

This project has a total of 2 members which is Chew Jane and Cheng Qing Xiang. This online academic staff booking appointment platform will be developed in web-based and app-based. The web-based and app-based online booking appointment platform will share the same modules but the functions of each module may be different in the web-based and app-based platform. The organization of the project is Chew Jane will handle and develop for the web-based online booking appointment platform while Cheng Qing Xiang will develop the app-based online booking appointment platform.

### 1.5.1 Task Allocation of Module Functions

| No. | Modules       | Functions                                                                                                | Member Name |                  |
|-----|---------------|----------------------------------------------------------------------------------------------------------|-------------|------------------|
|     |               |                                                                                                          | Chew Jane   | Cheng Qing Xiang |
| 1.  | Login         | Validate user login                                                                                      | /           | /                |
| 2.  | Booking       | Allow students to make consultation time request                                                         | /           | /                |
|     |               | Allow students to choose preferable time as consultation time                                            | /           | /                |
|     |               | Allow staff accept or reject consultation time request with reason                                       | /           | /                |
|     |               | Allow staff to determine consultation venue                                                              | /           | /                |
|     |               | Allow user to cancel their time booking slot                                                             | /           | /                |
|     |               | Allow user to check consultation time booking status                                                     | /           | /                |
|     |               | Send notification to students when consultation time booking request is accepted or is rejected by staff |             | /                |
| 3.  | Time Schedule | Allow students to view staff available consultation time                                                 | /           | /                |
|     |               | Allow staff to choose their available consultation time                                                  | /           |                  |
|     |               | Allow staff to modify their available consultation time                                                  | /           |                  |
|     |               | Allow users to view their booking time schedule in a calendar form                                       | /           |                  |

|    |          |                                                                            |   |   |
|----|----------|----------------------------------------------------------------------------|---|---|
|    |          | Allow users to view their booking time schedule in a list form             |   | / |
| 4. | Feedback | Allow students to read feedback from staff in a consultation               | / | / |
|    |          | Allow students to rate a consultation                                      |   | / |
|    |          | Allow staff to write feedback for a consultation                           | / | / |
|    |          | Send a notification to students when a feedback is written to the student  |   | / |
| 5. | Chatbox  | Allow students to message to a staff                                       | / | / |
|    |          | Allow staff to message to a student                                        | / | / |
|    |          | Allow users to reply to a message                                          | / | / |
|    |          | Send a notification to recipients when there is a message sent to the user |   | / |

## 1.6 Chapter Summary and Evaluation

*In this chapter, a short introduction and objectives are included to discuss the main purpose of proposing the online academic staff booking appointment platform for academic institutions. Background which is related to the traditional booking appointment method is also included in this chapter. Moreover, the project plan is written which is focused on the functional requirements for the proposed web-based online academic staff booking appointment platform. The project team and organization is also written which specifies the team members and modules handled.*

*For the next chapter, we will discuss the background and related work for this project. Moreover, we will analyse the methodology and requirements needed to be used in this project. The remaining chapters will also include the User Interface (UI) design of the online academic staff booking appointment platform, coding implementation and system testing documentation.*

## Chapter 2

# **Background and Related Work**

## 2 Background and Related Work

This framework is aimed to propose a web-based online booking appointment platform for undergraduates and staff in higher education institutions. The platform is proposed to provide a more time efficient way for students to book a consultation time and for staff to manage their booking time schedule in a more effective way. Hence, the target users for the platform are the academic staff which includes tutors, lecturers, academic advisors, counsellors and also the undergraduate students. Background of this project and literature review will be discussed in this chapter.

### 2.1 Company Background

The web-based online booking appointment platform is proposed to the higher education institutions to enable the users which are academic staff and undergraduates could manage and book a consultation hour in a time efficient way. The academic institution targeted is the main campus of Tunku Abdul Rahman University College (TARUC). The main campus of TARUC is located at the town of Setapak within the Gombak district (Tunku Abdul Rahman University College, 2019).

Currently, the procedures to book a consultation time are performed manually in Tunku Abdul Rahman University College (TARUC). Through this manual time booking procedure, the students had faced some difficulties when they wanted to book a consultation time with their tutor or other academic staff such as lecturer, academic advisor and counsellor. The students had complained that it is hard to book a time to have a consultation with the staff. Moreover, academic staff, especially the counsellor had faced a problem that they could not make the time arrangement effectively, especially when there are too many students requested for consultation privately. This problem had caused students' consultation time arrangement to be arranged for a few weeks later, which has to wait for a long time.

Hence, the manual booking appointment process should be eliminated due to human error occurring commonly such as many instances of overbooking and users forget to cancel an appointment which could not be attended and this

had led to the free appointment time slot being wasted (Qaffas & Barker, 2012). Beside that, an online booking appointment system is using a real-time based system which the system information is updated in real time so that the system information can be updated immediately which is more efficient compared to the manual booking appointment process (Oluwaseun Bello et al., 2016). Thus, we had introduced an online academic staff time booking platform in this research study in order to let students save more time by simplifying the appointment booking procedures and let staff can manage their booking time schedule more effectively.

## 2.2 Project Background

Nowadays, the Internet has become a fundamental part of human life. The Internet has brought a revolution, and it has modernized our lives to an easier and simpler way (Sharma, 2016). The use of the Internet has benefited the education aspect enormously. It provides a platform for educators to share their knowledge and experience (Public Policy, 2017). Moreover, educators such as lecturers and tutors can prepare online learning materials and share with the students. Thus, the students can view and learn from online materials everywhere and anytime. To quote an example, Google Classroom is a famous platform used by many universities or colleges in the world. Google Classroom can be used to share the notes and tutorials to the students and enable them to access the learning materials according to their course registered respectively (Lynch, 2018).

Furthermore, the Internet has a significant function which is made reservations or booking via online. Through the use of the Internet, the online reservation or booking system can undergo as a real-time system. A real-time system is the data or information is processed within a short period of time, which usually is processed in milliseconds (Cambridge Dictionary, 2019). This makes the system is able to respond effectively and immediately to the event or process that occurs (Cambridge Dictionary, 2019). An online booking system will update all the changes made in booking schedule and booking information and reflects in real time (Stfalcon.com, 2018). Hence, this online time booking or reservation system has been widely used in the

world due to its convenience that everyone can access via the Internet and its real time updated information (Chang, 2019). Thus, we had proposed an online booking appointment platform which is more efficient and convenient to use in this project.

## **2.3 Literature Review**

In this new era, there is no doubt that many tasks and goals are done with the help of modern technology and the Internet. With the use of the Internet, many tasks can be completed in just a short period of time. The Internet is commonly used in the online booking system in different aspects such as the hotel booking system which enables the customers to access online and the system information is also updated in real-time (Bemile, Achampong, & Danquah, 2014). Online booking system is a paperless web-based application that is designed to provide a flexible and sustainable service for an organization to handle the meetings efficiently (Omar Abdullah M. Al-Maktari, Rozinah Jamaludin, & Al-Samarraie Hosam, 2012). Hence, the online booking appointment system used in different academic institutions is reviewed in order to find out the requirements that should be implemented in the online academic staff booking appointment platform.

### **2.3.1 Related Works of Online Booking Appointment System in Academic Institutions**

An online booking appointment system within a college or a university is a rapid trend. This is due to the system is able to minimize the time usage of handling administrative issues and it can avoid the cases of overbookings in a same time slot for different users. Hence, case studies from different research papers are found to prove that online academic staff booking appointment platforms are applicable in academic institutions.

According to Oluwaseun Bello et al. (2016) have developed a student-teacher online appointment booking web application in order to make the appointment booking process more efficient. The application developed is using a real-time based system which the system information is updated in real time so that the students and academic staff can receive and access the system information immediately. Moreover, the application developed utilized 2-tier techniques of client-server architecture software and used My Structured Query Language (MySQL) server as the database of the online booking appointment system (Oluwaseun Bello et al., 2016). Hence, the client can request for the specific services required through the Internet and the system can store and organize large amounts of information using the MySQL database management system. Thus, the database implementation method used in the application is able to be used in the online academic staff booking appointment platform proposed in this framework since the database implementation method benefits the platform which can help to manage information and data in a more effective way.

According to Oluwaseun Bello et al. (2016), the application was set up on multiple platforms which included laptops to mobile devices to ensure the application is user-friendly to the clients so that they can access the system using different electronic devices. Furthermore, the application developed included all the basic function requirements for an online booking appointment system such as login, request appointment, approve or cancel appointment and schedule appointment (Oluwaseun Bello et al., 2016). The application developed also included a feature which enables the students and academic staff to drop messages to each other which is a useful feature since the student might want to message the respective academic staff when there

is an emergency. In addition, graphical user interfaces of each page that will be implemented on the application proposed are shown and descriptions of the pages are written in detail which enables readers to understand better the functions performed by the particular website page (Oluwaseun Bello et al., 2016). Hence, the user interface and description of web pages can be referenced when developing the online academic staff booking appointment platform to ensure the platform developed is easy to learn and use by end-users.

Apart from that, an online appointment management system is developed and evaluated to facilitate the task of appointment booking from students (Qaffas & Barker, 2012). An online appointment management system is required to store a large amount of data in a database, hence a creation of database tables which is to demonstrate the database relationship of the system is created in order to clearly define the relationship among each entity and attribute (Qaffas & Barker, 2012). The database relationship diagram can also be referenced and applied in the platform proposed in this framework in order to define clearly the relationship among each entity and attributes so that the platform developed had a well-organized database management system.

Besides that, Qaffas and Barker (2012) had shown the web pages implemented on the online appointment management system with description of functions performed by the web pages. The system developed included all the basic function requirements for an online booking appointment system such as user registration, login, request appointment, approve or cancel appointment and schedule appointment (Qaffas & Barker, 2012). The system features also included staff can write results at the end of each appointment and staff can prohibit a student from making appointments if the student cancels an appointment more than five times (Qaffas & Barker, 2012). The system's strength which enables a staff can write results or provide feedback after an appointment can be applied to the online academic staff booking platform proposed in this framework since the feedback written can be a short conclusion of the appointment and this function enables users to keep track their appointment progress especially when the appointment is booked for project discussion purpose.

## **2.4 Feasibility Study**

Feasibility study is important to a project development because it analyses the project relevant factors which include technical, operational, organizational and schedule considerations and this information enables the project manager to determine the pros and cons of undertaking the project before they invest a lot of money and time on it (Kenton, 2019). Hence, the risk of project failure can be reduced since a project's feasibility study had analyzed the project practicality which include the aspects of tools, technology, manpower and time needed for this project to complete.

### **2.4.1 Technical Feasibility**

In this project, an online academic staff booking appointment platform will be created to allow users to access it through a web browser using any electronic devices such as computers. Furthermore, Visual Studio 2017 software with high-level programming language, C# is used to create the online booking appointment platform. The platform development will also include programming languages such as Cascading Style Sheets (CSS), Personal Home Page (PHP) and JavaScript to make the web pages created have an ideal style and appearance.

### **2.4.2 Operational Feasibility**

The online academic staff booking appointment platform proposed in this framework is able to solve the problem of using a manual appointment booking procedures which caused overbooking cases happen commonly and long waiting time needed to complete the procedures manually. Hence, the new online platform proposed benefits the users which include the academic staff and students in academic institutions. This is due to academic staff can manage their booking time schedule more effectively and the students can book an appointment with simpler procedures performed using the online system.

### **2.4.3 Organizational Feasibility**

This project has a total of 2 members involved which is Chew Jane and Cheng Qing Xiang. This online academic staff booking appointment platform will be developed in web-based and app-based. The organization of the project is Chew Jane will handle and develop for the web-based online booking appointment platform while Cheng Qing Xiang will develop the app-based online booking appointment platform. Moreover, they will perform system testing on their platform to ensure the platform developed is error-free and the functions satisfied the users. Hence, the proposed platform has sufficient human resources to develop the system.

### **2.4.4 Schedule Feasibility**

A firm timetable is established for this project which approximately 7 months are given to complete the documentation parts of the project first and approximately 4 months are given to complete the system and perform testing on the system developed. The documentation parts of the project consists of 4 chapters and the deadlines given are reasonable. Hence, the schedule risk on project documentation is low and can be avoided. Apart from that, the system developing and testing time is required to be completed within 4 months. The short time constraint of system developing and testing time causes the risk of project overruns become higher since this project is required to develop a properly functioning real-time updated schedule system. Since the schedule given is desirable rather than mandatory, some changes may be made on system testing schedule in order to have more time to develop the system completely. Thus, the consequences of project overruns can be limited or even no effect at the system at all.

## 2.5 Chapter Summary and Evaluation

*In this chapter, the company background with the current system used which is manual time booking procedures by the company is discussed. The solution to solve the disadvantages brought by manual time booking procedures is a real-time online academic staff booking appointment platform is proposed in this framework. Furthermore, project background which is related to Internet and online time booking services are written and the advantage of an online booking appointment platform is also included in this chapter. Literature review is written based on the comments given on the functionality and characteristics of similar systems found in different journal articles and research papers.*

*For the next chapter, we will analyse the methodology and requirements needed to be used in this project. The remaining chapters will also include the User Interface (UI) design of the online academic staff booking appointment platform, coding implementation and system testing documentation.*

## Chapter 3

# **Methodology and Requirements Analysis**

## 3 Methodology and Requirements Analysis

In this chapter, methodology of research and software development model were identified to ensure the platform proposed is needed by the users and it can be developed within the schedule. Moreover, the functions required to be implemented in the platform is also analyzed through the requirements gathering techniques. Functional requirements section in this chapter also included the use case diagrams and use case descriptions for each module proposed to be implemented in the platform.

### 3.1 Methodology

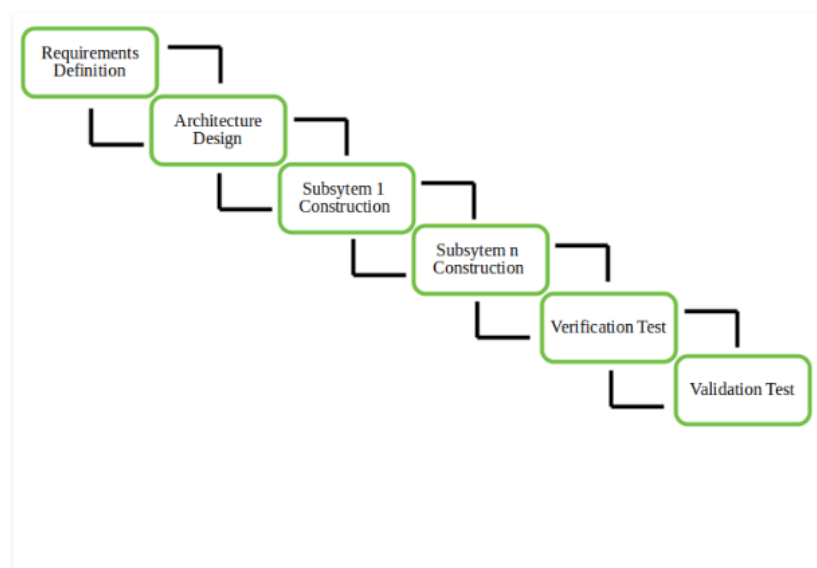
Methodology is defined as the theoretical analysis of the methods applied to a particular area of study to ensure the research is valid and reliable (Cambridge Dictionary, 2020). This framework was based on qualitative study and quantitative study. The qualitative study taken in this research is literature review method and the quantitative study taken in this research is online questionnaire survey.

#### 3.1.1 Software Development Model - Incremental Model

Furthermore, software development models are described as the series of processes or methodologies that are being used in development of a project depending on the project's goals (Tryqa.com , 2020). There are different types of software development model or methodologies which include waterfall model, incremental model, rapid application development (RAD) model, agile model, prototyping model and spiral model (Tryqa.com , 2020).

In this framework, the software development model that is chosen for this project is an incremental model. Incremental model is a software development model that delivers the core product functions at the first version and delivers supplementary function in the next version until the desired system is released.

Generally, there are two types of incremental model in software development life cycle which are staged delivery model and parallel development model (GeeksforGeeks, 2020). Staged delivery model is a software process model where only one part of a project is constructed at a time and this construction process is repeated until all the requirements are implemented. Parallel development model is different from staged delivery model where parallel development model involves different subsystems being developed at a time by different developer teams. Thus, the parallel development model requires more human resources compared to the staged delivery model. Due to the constraint of human resources in this project, we decided to use a staged delivery model to incrementally develop the online academic staff booking appointment platform.



*Diagram 1: Staged Delivery Model*

Diagram 1 shows the concept of a staged delivery model in which the construction of a system is divided into phases (GeeksforGeeks, 2020). This model is chosen as the development model of the platform proposed is due to this online platform is suitable to be implemented by phases.

The online booking appointment platform proposed in this platform contains two core functions which are booking module and time schedule module. Hence, these two modules can be developed at the first phase together with the login module. Then, the feedback module and chat box module which are supplementary functions can be developed in the second phase. Staged delivery model enables the system to develop phase by phase where each phase partial systems are built until all designed functions have been implemented and satisfied the users. This enables the platform to be developed in a fast way during the software life cycle since the subsystems developed in earlier phases are able to release to the users first.

Furthermore, the staged delivery model enables high user involvement in the software development process. By applying this model in the project, an initial implementation of the new system can be developed and is exposed to the project stakeholders which are TARUC students and staff to get feedback from them. The feedback gained from stakeholders are analysed and the system can be modified until all the user requirements are completely satisfied in the particular construction phase.

### **3.2 Requirements Gathering Techniques**

There are many requirements gathering techniques that can be used in a project. The requirements gathering are one of the most important pieces to ensure the system developed is needed by the users and the functions performed are useful to help users to complete a specific task (Eid, 2015). The examples of requirements gathering techniques included one-on-one interview, group interview, questionnaire or survey, literature review or document review, and observation (Eid, 2015).

Two types of requirements gathering techniques are used to analyse the user acceptance and the functional requirements of the online academic staff booking appointment platform. The techniques used in this framework are literature review and questionnaire.



### 3.2.1 Literature Review Technique

The literature review technique is a method of data collection involving the review of articles. Literature review is applied in this research by analysing other scholarly papers which are relevant to the topic of research which is the online appointment booking system in academic institutions. Literature review is used because it is able to provide a conceptual framework for a research and this helps to build up new research topics since the concepts on prior research in the particular field is studied. In chapter 2, literature review is applied by reviewing two articles papers with a similar system towards the proposed platform in this framework. Both articles found had discussed the implementation of online appointment management systems in academic institutions. The implementation part of the articles paper reviewed included functional requirements, non-functional requirements, database management, user interface (UI) design and detailed description of implemented web pages. After reviewing both articles, an identical issue is found which both articles do not consider notification function in their system developed. However, notification function is quite important in an online appointment booking system because this enables the users to be aware of their time schedule or news in a push-up notification. Thus, the notification functionality is considered as a function requirement in the app-based online academic staff booking appointment platform which is to display push-up notification when necessary such as an appointment booking request is accepted by respective staff.

### 3.2.2 Questionnaire Technique

Furthermore, the technique used to gather requirements of the platform proposed included questionnaire surveys. The questionnaire is designed as a mix of closed questions and open questions in order to know the facts on the current manually booking appointment procedures and get users' opinions on it. Questionnaire is chosen as the technique of requirement gathering in this project is due to the questionnaire is able to gain the opinions from a large group of users on a particular issue or

problem. For instance, questionnaires are distributed to some users of the platform which include both academic staff and students to investigate the problem and difficulty faced by the current booking appointment procedures using 1 - 5 points of Likert scale where 1 is strongly disagree and 5 is strongly agree. Thus, a questionnaire is able to provide quantifiable answers for the research topic that are relatively easier to analyse.

### **3.3 Requirements Analysis**

The target population for this study is the undergraduate students and academic staff from the academic institutions. This questionnaire survey is conducted at Tunku Abdul Rahman University College (TARUC) KL main campus. This framework had used a sampling method which is simple random sampling to select the sample from the target population (Ahmed, 2009). Every possible sample of the same size has an equal possibility to be chosen.

#### **3.3.1 Fact Recording**

The questionnaire survey is created using Google Form. Google Form is used because it enables us to collect responses online which have saved our time in fact-gathering. In addition, Google Form is able to create an Excel spreadsheet based on the respondents' answer which is convenient for collecting respondents data.

We had used an online questionnaire survey to collect users acceptance level and users requirements towards the platform proposed in this framework. A sample survey of questionnaires for students is attached in Appendix I while a sample survey of questionnaires for staff is attached in Appendix II. The questionnaire survey is made available for a month to collect data from the students and staff.

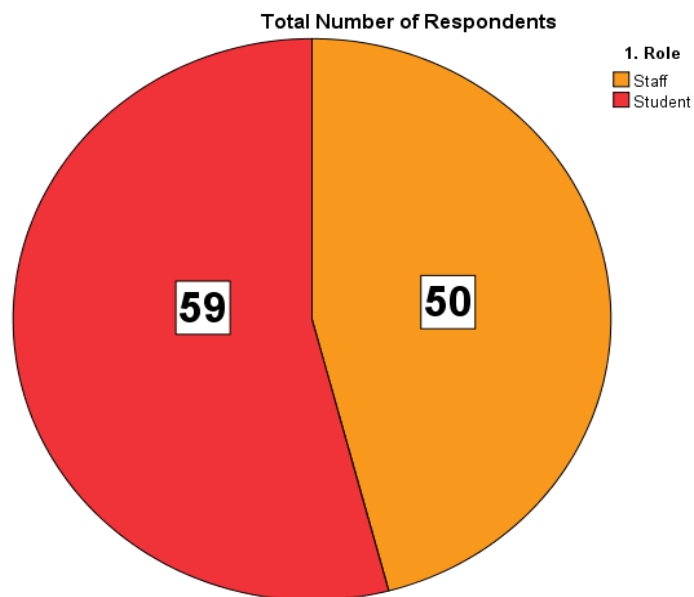
The respondents are required to answer three different sections regarding (1) their frequency of booking or receiving a consultation

appointment. Moreover, the section (2) of questionnaire is related to the current consultation appointment booking and the section (3) is related to the functionality and functional requirements of the new consultation booking appointment platform.

The data collected are analyzed using IBM Statistical Package for the Social Sciences(SPSS) Statistics Software. SPSS provides functions such as hypothesis testing, predictive analysis and ad hoc analysis to solve research and business problems (IBM, 2020). These functions enable users to understand their data and write an accurate conclusion based on their data analysis (IBM, 2020).

### 3.3.2 Fact Analysis

In this section, we will analyze the data collected based on the questionnaire survey that had been distributed to the staff and students.



*Diagram 1: Pie Chart of Total number of respondents*

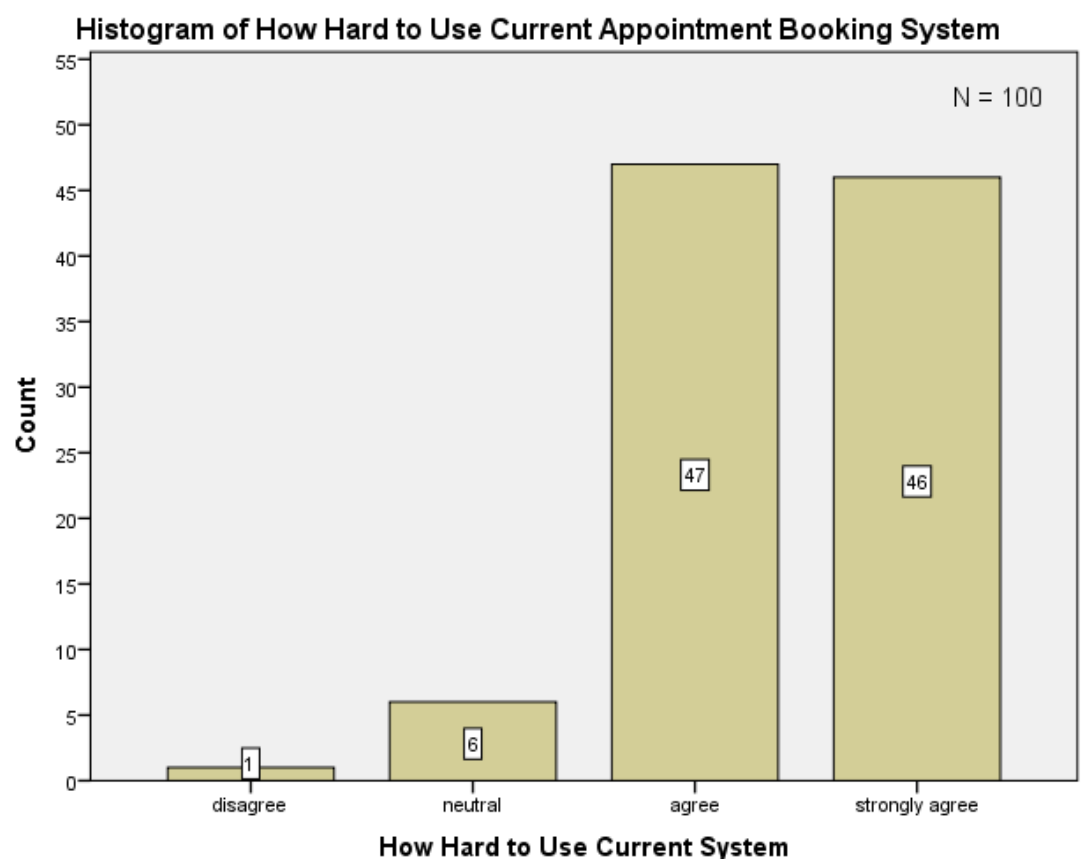
Diagram 1 above shows the pie chart of total respondents received from the questionnaires. The pie chart had used different colors to differentiate the roles of respondents. Orange color is represented as the staff which we had collected 50 sets of valid data from the TARUC staff. Red color is represented as the students who had collected 59 sets of data but only 50 sets of data is counted as valid due to there being nine students said that they never book an appointment with staff before. Hence, these nine students' data collected are counted as invalid and would not be used during data analysis since they do not have any experience on booking an appointment with staff. Thus, the online questionnaire had collected a total of 100 valid respondents data which are 50 undergraduate students and 50 academic staff of TARUC.

In this project, we had used histograms and independent t-test to analyze the data collected. The histogram is used to investigate the

overall opinion from both of the users which are students and staff. Moreover, the independent t-test is conducted to compare two independent groups which are academic staff and students whether there is statistical evidence that the two samples taken from the same population have significantly different opinions or results. The t-test result is analyzed based on the t-statistic value, the degrees of freedom (df) and the significance value of the test (p-value) (Laerd Statistics, 2020). Apart from that, a margin of error is set to 5% with a 95% confidence level and the significance value,  $p$  corresponding to the test statistic is set as 0.05. Then, a conclusion is made whether the hypothesis written is accepted or rejected.

1.  $H_0$  : There is no significant difference in how hard to use the current manual appointment booking system between students and staff.

Diagram 2.1 shows the histogram of users' counts on how hard to use the current manual appointment booking system. Diagram 2.2 represents the independent t-test results based on the two tested groups and the users' opinion on how hard to use the current manual appointment booking system.



*Diagram 2.1: Histogram of How Hard to Use the Current Appointment Booking System*

Diagram 2.1 shows the histogram of counts on how hard to use the current appointment booking system. The histogram is drawn based on both students and staff opinion in order to investigate the overall opinion from samples taken on how hard to use the current manual

appointment booking system. From the histogram, we can clearly see that there is 1 user disagree that the system is hard to use and 6 users had a neutral opinion. Moreover, 47 users agree and 46 users strongly agree that the current appointment booking system is hard to use. The evidence showed that 93 users agreed that the current appointment booking system is hard to use out of the 100 users. Thus, this result can be concluded as most of the users felt the current manual appointment booking system is hard to use.

| Independent Samples Test                |                                                 | Equal variances assumed | Equal variances not assumed |
|-----------------------------------------|-------------------------------------------------|-------------------------|-----------------------------|
| Levene's Test for Equality of Variances | F                                               | 1.985                   |                             |
|                                         | Sig.                                            | .162                    |                             |
| t-test for Equality of Means            | t                                               | -.925                   | -.925                       |
|                                         | df                                              | 98                      | 94.905                      |
|                                         | Sig. (2-tailed)                                 | .357                    | .357                        |
|                                         | Mean Difference                                 | -.120                   | -.120                       |
|                                         | Std. Error Difference                           | .130                    | .130                        |
|                                         | 95% Confidence Interval of the Lower Difference | -.377                   | -.377                       |
|                                         | 95% Confidence Interval of the Upper Difference | .137                    | .137                        |

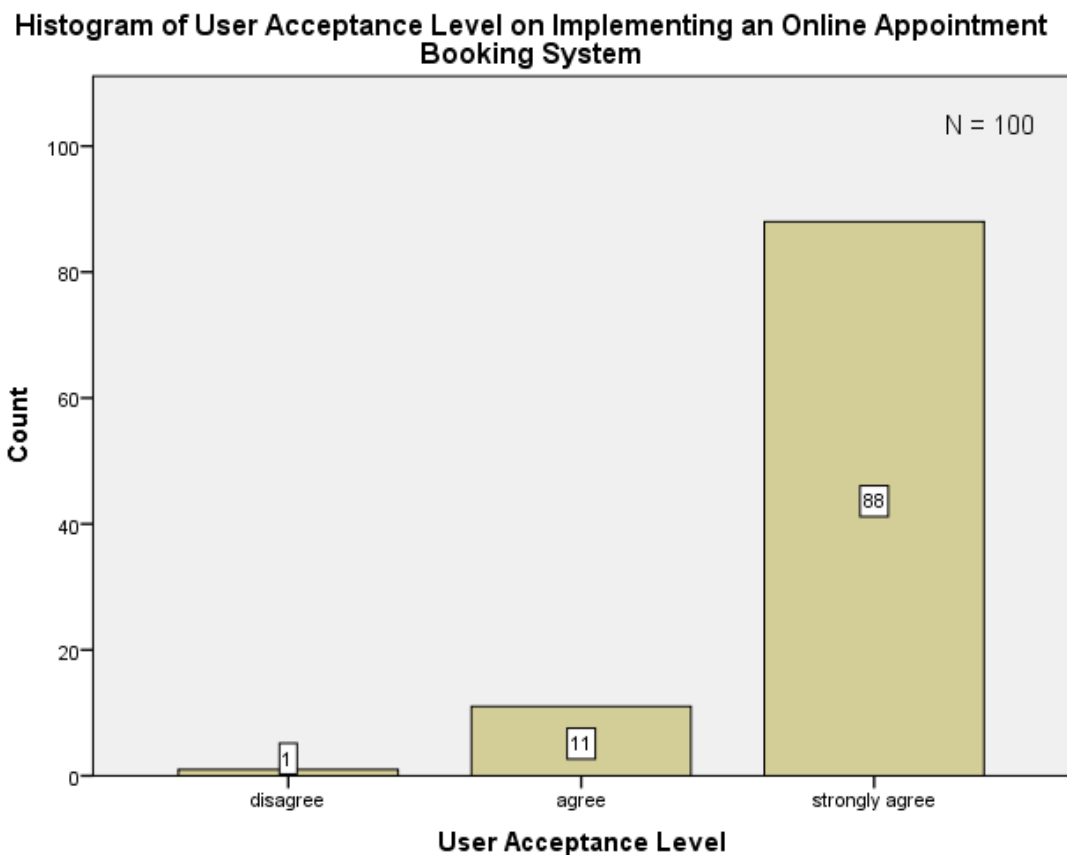
*Diagram 2.2: Independent T-test Result of How Hard to Use the Current Appointment Booking System*

Diagram 2.2 represents the independent t-test results based on the two tested groups and the users' opinion on how hard to use the current manual appointment booking system. From the result of Levene's test for equality of variances, we can clearly see that the significance value is 0.162 which is greater than 0.05, so the two tested groups are assumed to have equal variances. According to the equal variances assumed in the column in diagram 2.2, the significance value,  $p$  is 0.357 which is greater than 0.05. Thus, a

conclusion can be made that the hypothesis is accepted with a 95% confidence level (CI) that there is no significant difference among students and staff opinions which most of them agreed that the current appointment booking system is hard to use.

**2.  $H_0$  : Students and staff's acceptance level on implementing an online appointment booking system do not have significant differences.**

Diagram 3.1 shows the histogram of users' counts on user acceptance level to implement an online appointment booking system. Diagram 3.2 represents the independent t-test results based on the two tested groups and the users' acceptance level on implementing an online appointment booking system.



*Diagram 3.1: Histogram of User Acceptance Level on Implementing an Online Appointment Booking System*

Diagram 3.1 above displays the histogram of counts on user acceptance level on implementing an online appointment booking system in TARUC. The histogram shows that there are 11 users agreed and 88 users strongly agreed to implement an online appointment booking system. Thus, this result can be concluded as most of the users are agreed to implement an online academic staff booking appointment platform in TARUC.

| Independent Samples Test                |                                                 | Equal variances assumed | Equal variances not assumed |
|-----------------------------------------|-------------------------------------------------|-------------------------|-----------------------------|
| Levene's Test for Equality of Variances | F                                               | 0.026                   |                             |
|                                         | Sig.                                            | .873                    |                             |
| t-test for Equality of Means            | t                                               | .000                    | .000                        |
|                                         | df                                              | 98                      | 88.239                      |
|                                         | Sig. (2-tailed)                                 | 1.000                   | 1.000                       |
|                                         | Mean Difference                                 | .000                    | .000                        |
|                                         | Std. Error Difference                           | .086                    | .086                        |
|                                         | 95% Confidence Interval of the Lower Difference | -.170                   | -.171                       |
|                                         | 95% Confidence Interval of the Upper Difference | .170                    | .171                        |

*Diagram 3.2: Independent T-test Result of User Acceptance Level on Implementing an Online Appointment Booking System*

The diagram 3.2 above shows the independent t-test results from two tested groups of users who are students and academic staff on the user acceptance level of implementing an online appointment booking system at TARUC. The Levene's test for equality of variances had shown that the significance value is 0.873 which is greater than 0.05. This indicates the two tested groups are assumed to have equal variances. According to the equal variances assumed in the column in diagram 3.2, the significance value,  $p$  is 1.000 which is

greater than 0.05. Thus, a conclusion can be made that the hypothesis is accepted with a 95% confidence level (CI) that user acceptance level on implementing an online appointment booking system between students and staff has no significant difference.

## **3.4 Functional Requirements**

In this section, the functional and non-functional requirements needed for the online academic staff booking appointment platform are gathered and analyzed.

### **3.4.1 Introduction of Modules**

The online booking appointment platform is proposed to be implemented in academic institutions and hence, 2 different users targeted for the platform are academic staff and students. Both users will have the same modules but perform different functions. The essential modules for the web-based online booking appointment platform are:

- Authentication module
- Appointment Booking module
- Time schedule module
- Feedback module
- Chatbox module

### 3.4.2 Overview Use Case Diagram

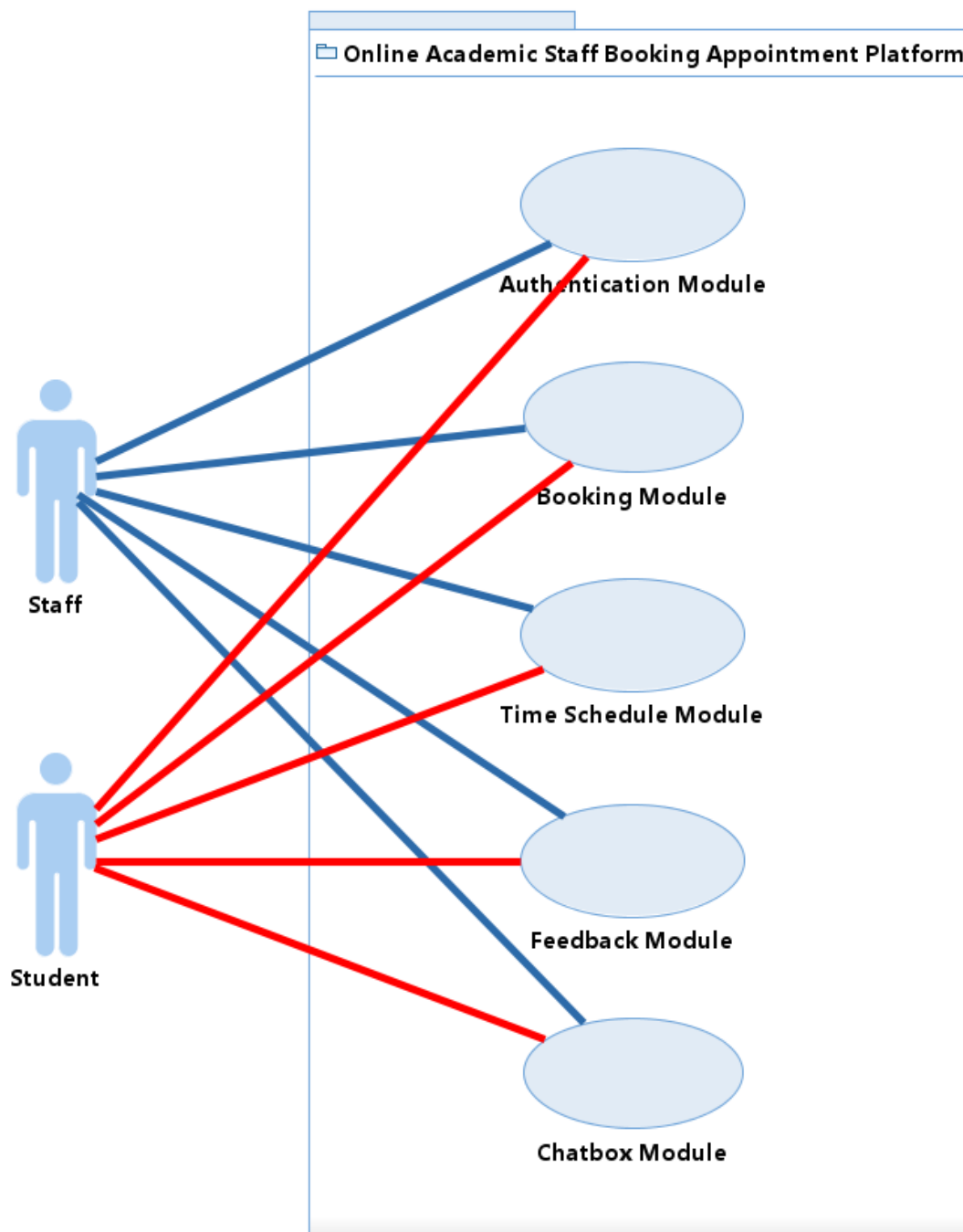


Diagram 4: Overview Use Case Diagram

### 3.4.3 Functional Requirements for Each Module

#### **1. Authentication Module**

This module is used to verify a user by validating the user login details and redirect to the user authorized page according to the user roles.

##### **I. Login**

The login module is designed to allow only authorized users to access the platform. When the username and password do not match, the system will display an error message and prompt the user to input again.

##### **II. Logout**

The logout module is designed to enable users to log out their session when they have completed their tasks using the platform. This helps to secure the current user's details and also prevent unauthorized actions on the current login session.

#### **2. Appointment Booking Module**

The booking module is designed to allow students to make appointment requests with the academic staff and this module also allows the staff to accept or reject a consultation time request. The booking processes performed in this module are shown below.

##### **I. Make consultation appointment requests**

The system shall allow students to make consultation time requests with title and reasons of the request.

##### **II. Choose appointment time**

The system should enable the students to choose their preferable time as their consultation time.

##### **III. Search a staff to make appointment request**

The system shall allow students to search for a staff to make appointments.

IV. Accept or reject appointment request

The system should enable staff to accept or reject a consultation time request.

V. Determine appointment venue

The system shall allow staff to determine the consultation venue for each appointment.

Moreover, the booking module also allows students and staff to cancel their appointment with reason provided.

### **3. Time Schedule Module**

This time schedule module is designed to allow users to manage their appointment booking schedule. This module enables users to view their appointment booking schedule. Furthermore, staff can choose and update their available consultation time in this module too.

I. View appointment booking schedule

The system should allow students and staff to view their appointment booking schedule. The appointment booking schedule should display the details information of appointment booked such as title of consultation appointment request, persons involved, expected duration of the appointment, appointment booking status, date and time of the appointment.

II. View staff available consultation time

The system should allow students to view staff available consultation time.

III. Choose available consultation time

The system shall allow staff to choose their available consultation time so that the students can make appointments based on the available time shown. The system shall allow the staff to update their available consultation time too.

IV. Check appointment booking status

The system shall allow students to check an appointment booking status whether the appointment request is pending, accepted or rejected.

#### **4. Feedback Module**

Feedback module introduced in an online booking appointment platform is to allow users to write and read feedback based on an appointment. The feedback written can be used as a short conclusion of the appointment for future use.

I. Write feedback

The system should enable the staff to choose to write feedback for an appointment.

II. Read feedback

The system should allow students and staff to read feedback for an appointment.

#### **5. Chat Box Module**

This chat box module is designed to enable users to communicate with each other by leaving a message.

I. Write message

The system should enable the users to message with other users they desire.

II. Reply message

The system should enable the users to reply to messages.

### 3.5 Details Use Case Diagrams

Individual use case diagrams are used to gather system requirements and actors who will use the system specification (Tutorialspoint.com, 2020). In this project, individual use case diagrams are drawn for each module required for the web-based online academic staff booking appointment platform. The use case diagrams are drawn using software application, IBM Rational Software Architect for WebSphere Software 8.5.

#### 3.5.1 Authentication Module

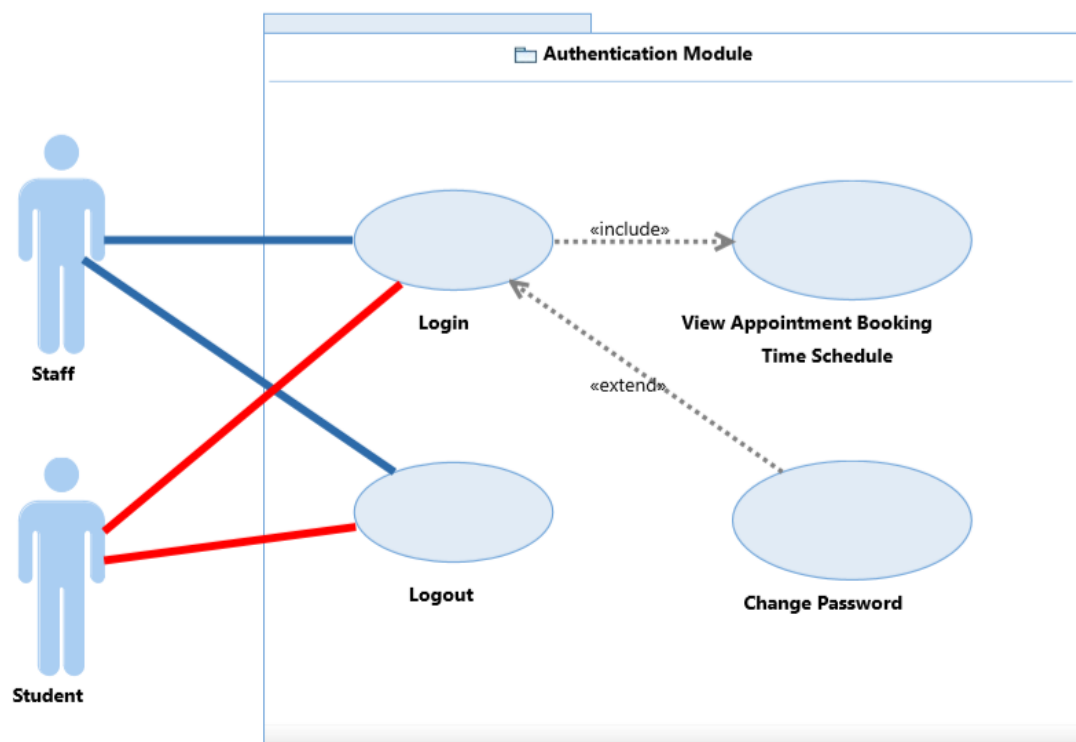


Diagram 5.1: Authentication Module Individual Use Case Diagram

### 3.5.2 Appointment Booking Module

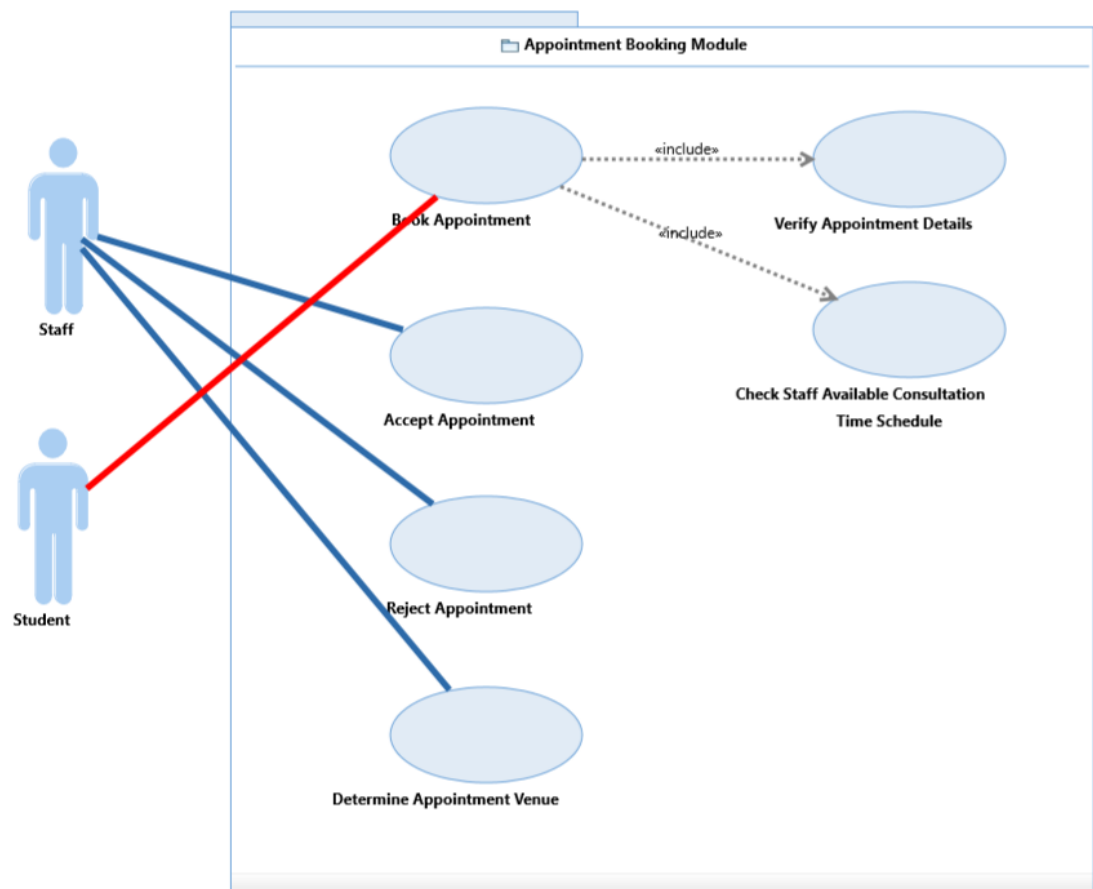


Diagram 5.2: Appointment Booking Module Individual Use Case Diagram

### 3.5.3 Time Schedule Module

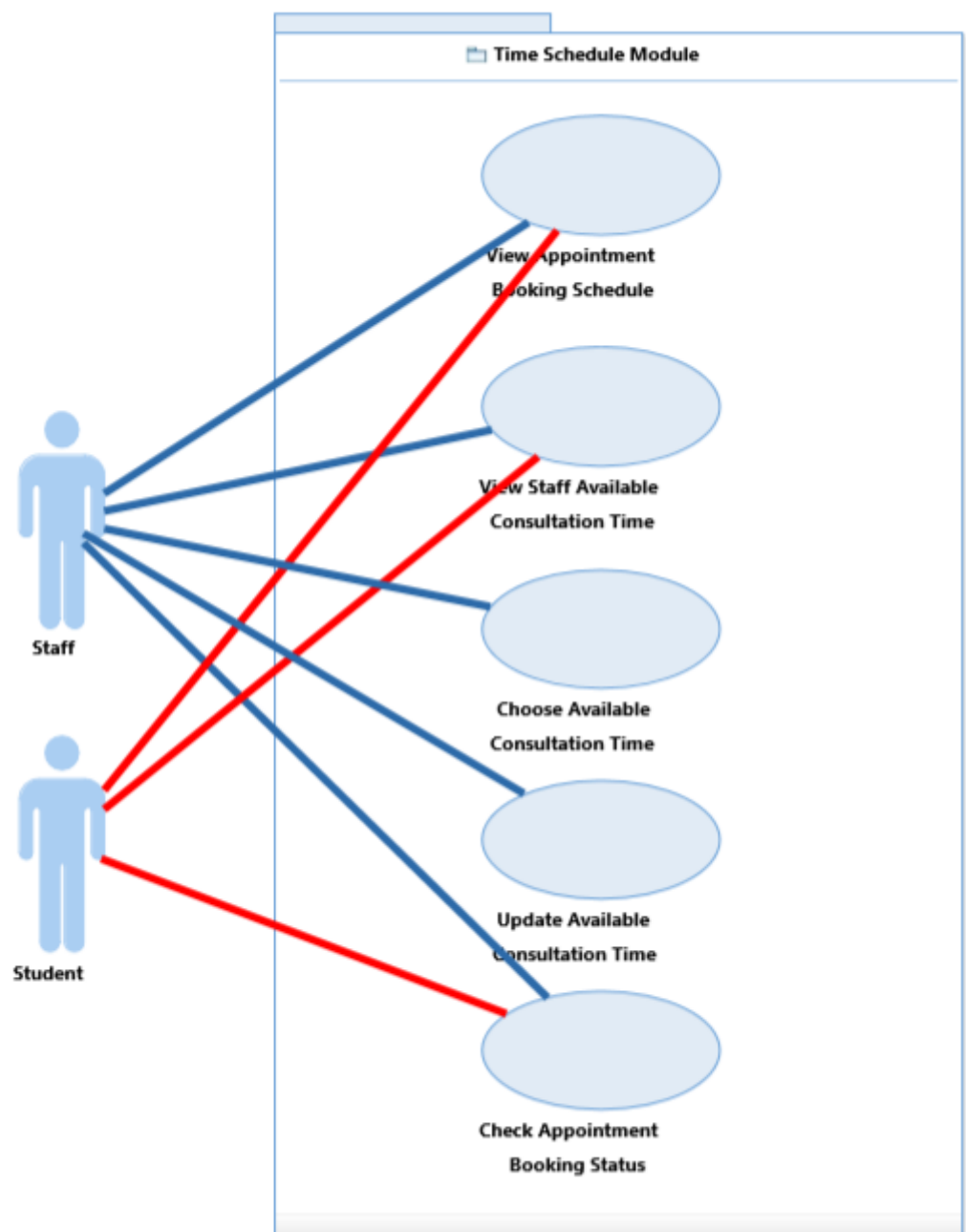


Diagram 5.3: Time Schedule Module Individual Use Case Diagram

### 3.5.4 Feedback Module

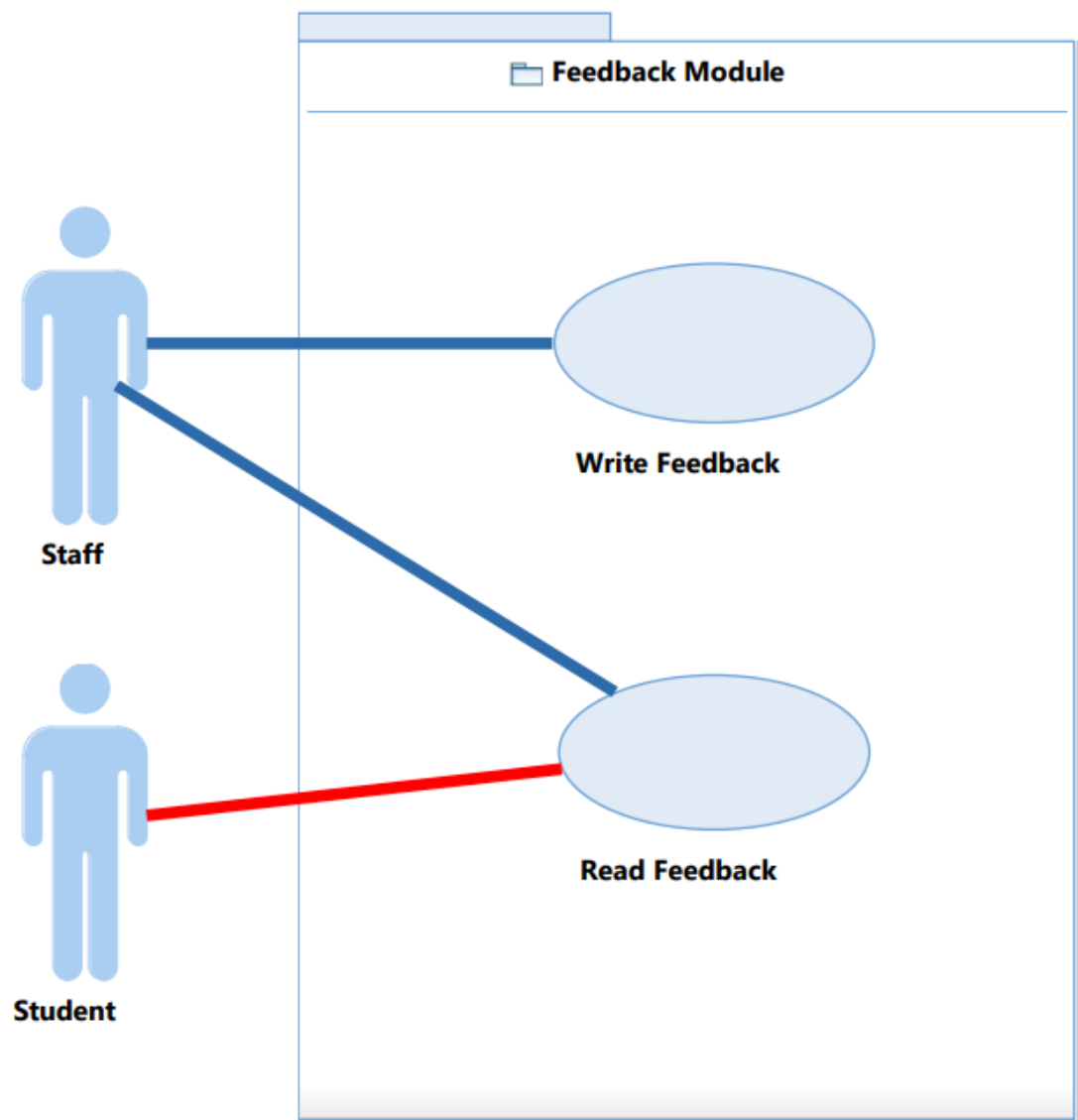
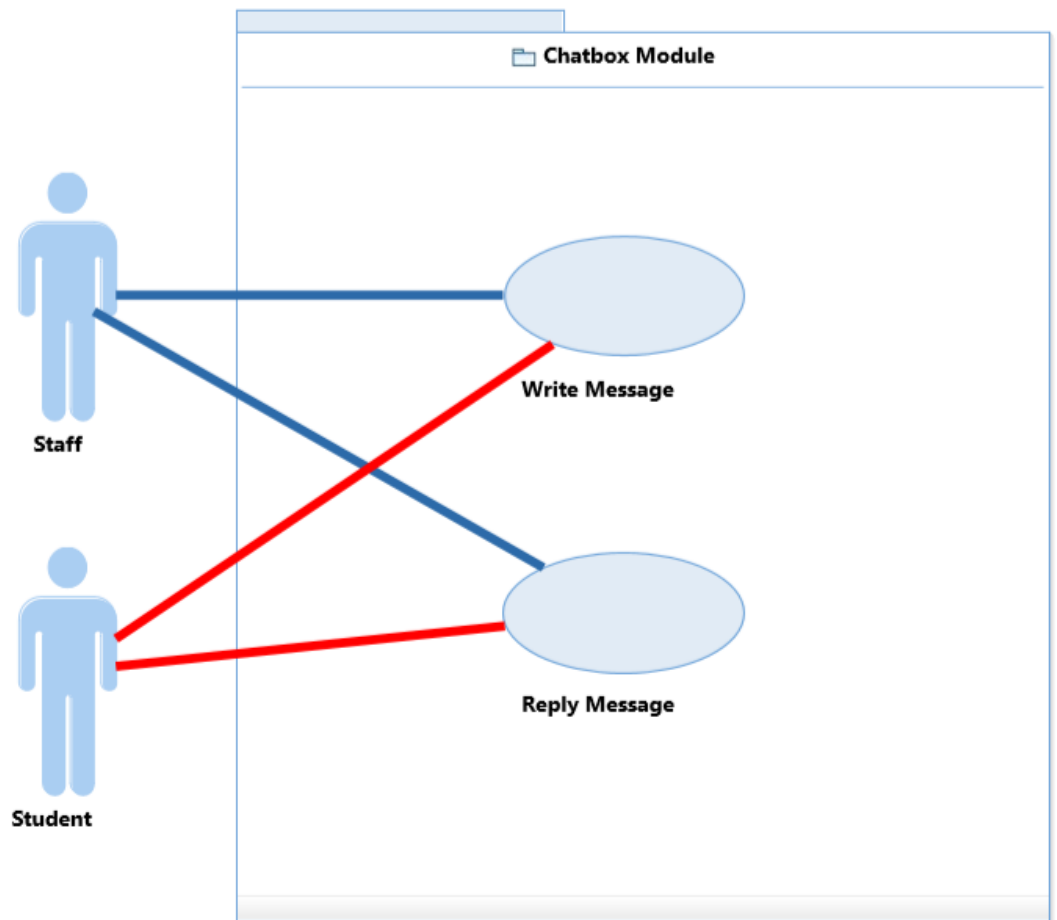


Diagram 5.4: Time Schedule Module Individual Use Case Diagram

### 3.5.5 Chat Box Module



*Diagram 5.5: Chat Box Module Individual Use Case Diagram*

### 3.6 Use Case Descriptions and Activity Diagram

Use case description is represented as a series of steps which start from a user's goal and end when the goal is achieved (usability.gov, 2020). Use case description is included in this chapter because it helps to define how the system should perform in the process which enables to know how the website should respond to an action (usability.gov, 2020). Moreover, the activity diagram is also included in this chapter. Activity diagram is aimed to express the flow from one activity to another activity just like a flow chart.

#### 1. Authentication Module

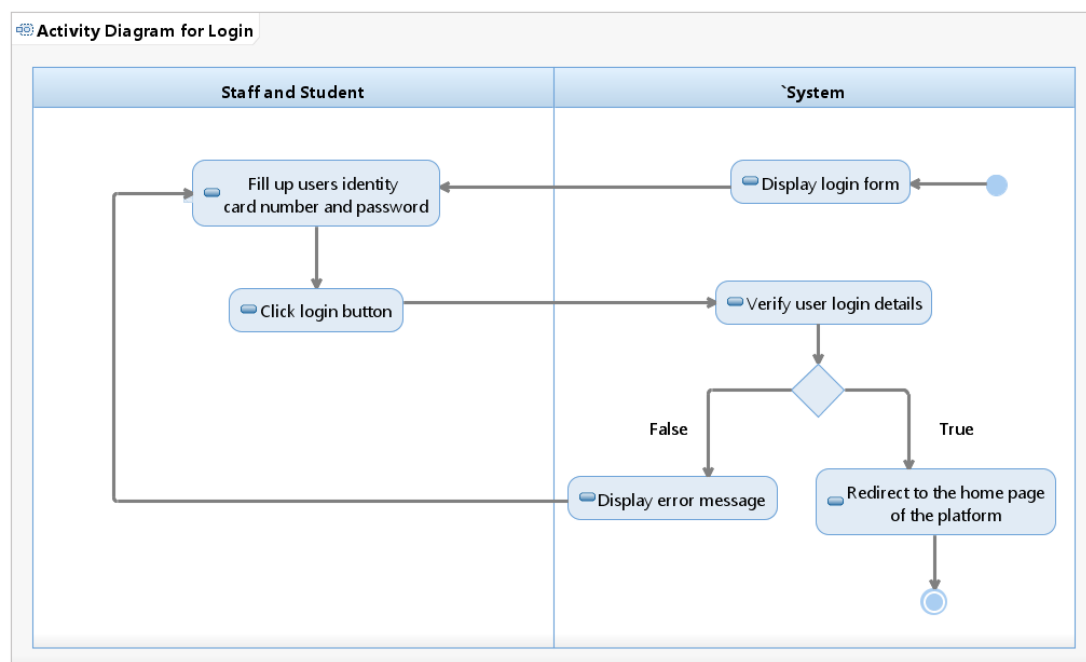


Diagram 6.1: Activity Diagram for Login

Diagram 6.1 shows the activity diagram for login. Refer table 1.1 below for details of login use case descriptions.

A. Login

|                                                     |                                                                                                                                                                                               |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case Name :</b>                              | Login                                                                                                                                                                                         |
| <b>Actor :</b>                                      | Student and staff                                                                                                                                                                             |
| <b>Brief Descriptions :</b>                         | Allow student and staff login to their user account which password is required to authenticate the users.                                                                                     |
| <b>Preconditions :</b>                              | Users had a TARUC identity card.                                                                                                                                                              |
| <b>Main Event Flows :</b>                           |                                                                                                                                                                                               |
| <b>Actor Action</b>                                 | <b>System Response</b>                                                                                                                                                                        |
|                                                     | 1. Display login form.                                                                                                                                                                        |
| 2. Fill up users identity card number and password. |                                                                                                                                                                                               |
| 3. Click login button                               | 4. Verify user login details.                                                                                                                                                                 |
|                                                     | 5. Redirect to the home page of the platform.                                                                                                                                                 |
| <b>Alternative Event Flows :</b>                    |                                                                                                                                                                                               |
| <b>A1. Step 4</b>                                   | If user login details are failed to verify, the system will display an error message that indicates the user's ID or password is incorrect and users are prompt to enter login details again. |

Table 1.1: Use Case Description for Login

## 2. Appointment Booking Module

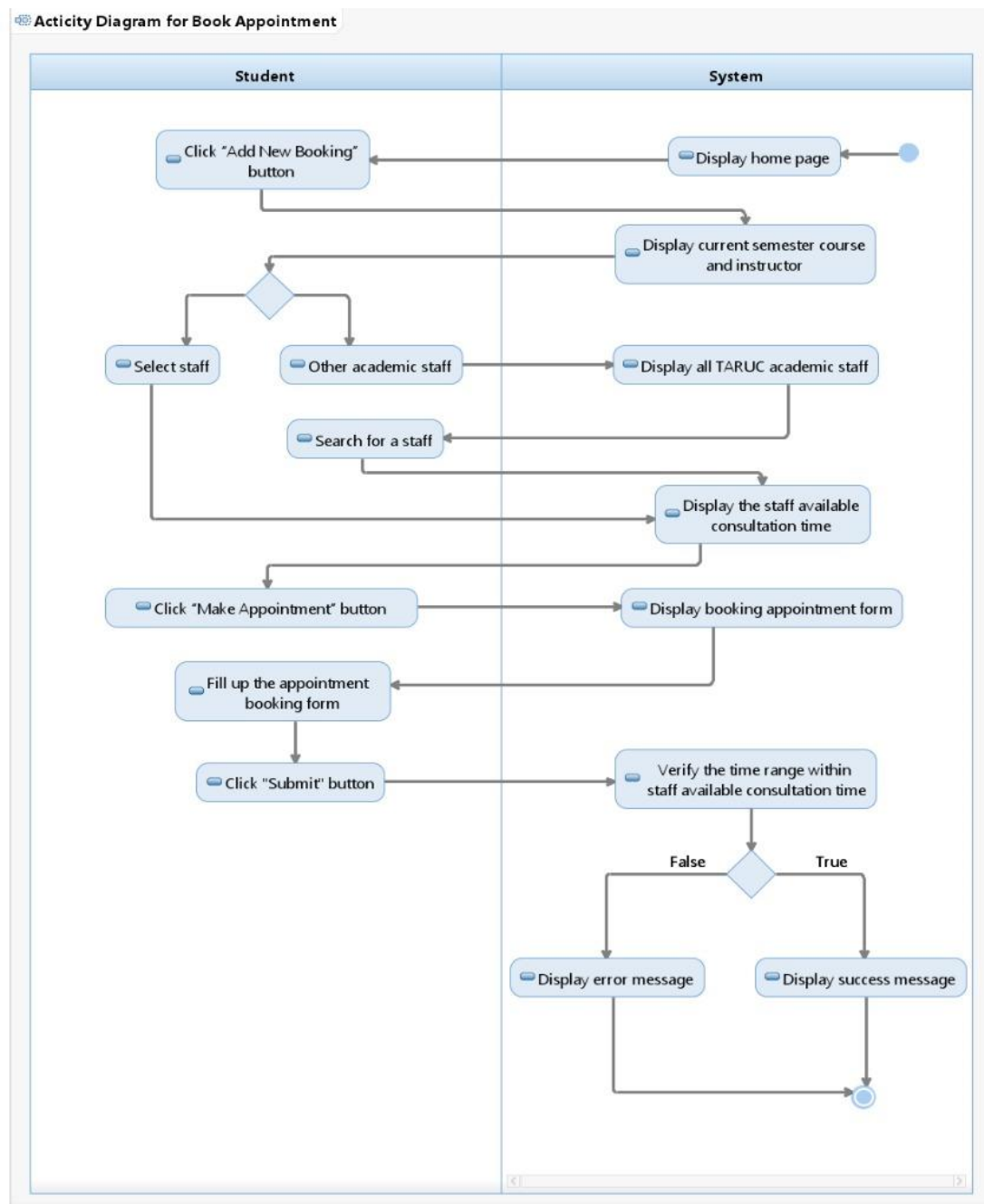


Diagram 6.2: Activity Diagram for Book Appointment

Diagram 6.2 shows the activity diagram for book appointment. Refer table 1.2 below for details of book appointment.

A. Book Appointment

|                                                            |                                                                                                                                                                                                                                                                |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case Name :</b>                                     | Book Appointment                                                                                                                                                                                                                                               |
| <b>Actor :</b>                                             | Student                                                                                                                                                                                                                                                        |
| <b>Brief Descriptions :</b>                                | Allow students to book an appointment with academic staff for consultation purposes.                                                                                                                                                                           |
| <b>Preconditions :</b>                                     | Students had login to their account.                                                                                                                                                                                                                           |
| <b>Main Event Flows :</b>                                  |                                                                                                                                                                                                                                                                |
| <b>Actor Action</b>                                        | <b>System Response</b>                                                                                                                                                                                                                                         |
|                                                            | 1. Display home page.                                                                                                                                                                                                                                          |
| 2. Click the “Add New Booking” button.                     | 3. Display current semester course and instructor.                                                                                                                                                                                                             |
| 4. Select a staff to book an appointment.                  | 5. Display the staff available consultation time.                                                                                                                                                                                                              |
| 6. Click the “Make Appointment” button.                    | 7. Display booking appointment form for students to input appointment request purpose and reasons.                                                                                                                                                             |
| 8. Fill up the booking form and click the “Submit” button. | 9. Verify whether the time range entered is within the staff available consultation time.                                                                                                                                                                      |
|                                                            | 10. Display a message to notify the students that the appointment request is sent to the staff successfully.                                                                                                                                                   |
| <b>Alternative Event Flows :</b>                           |                                                                                                                                                                                                                                                                |
| <b>A1. Step 4</b>                                          | If students click the “other academic staff” button, the system will display all TARUC academic staff for students by categorized from departments and faculties. Then, students can search for a staff member of TARUC to book an appointment with the staff. |
| <b>A1. Step 8</b>                                          | If students click the “Cancel” button, the system will return to the previous process which displays the respective searched staff available consultation time.                                                                                                |
| <b>A1. Step 9</b>                                          | If the time range entered is invalid, the system will display an error message and enable the students to enter the time range again.                                                                                                                          |

Table 1.2: Use Case Description for Book Appointment

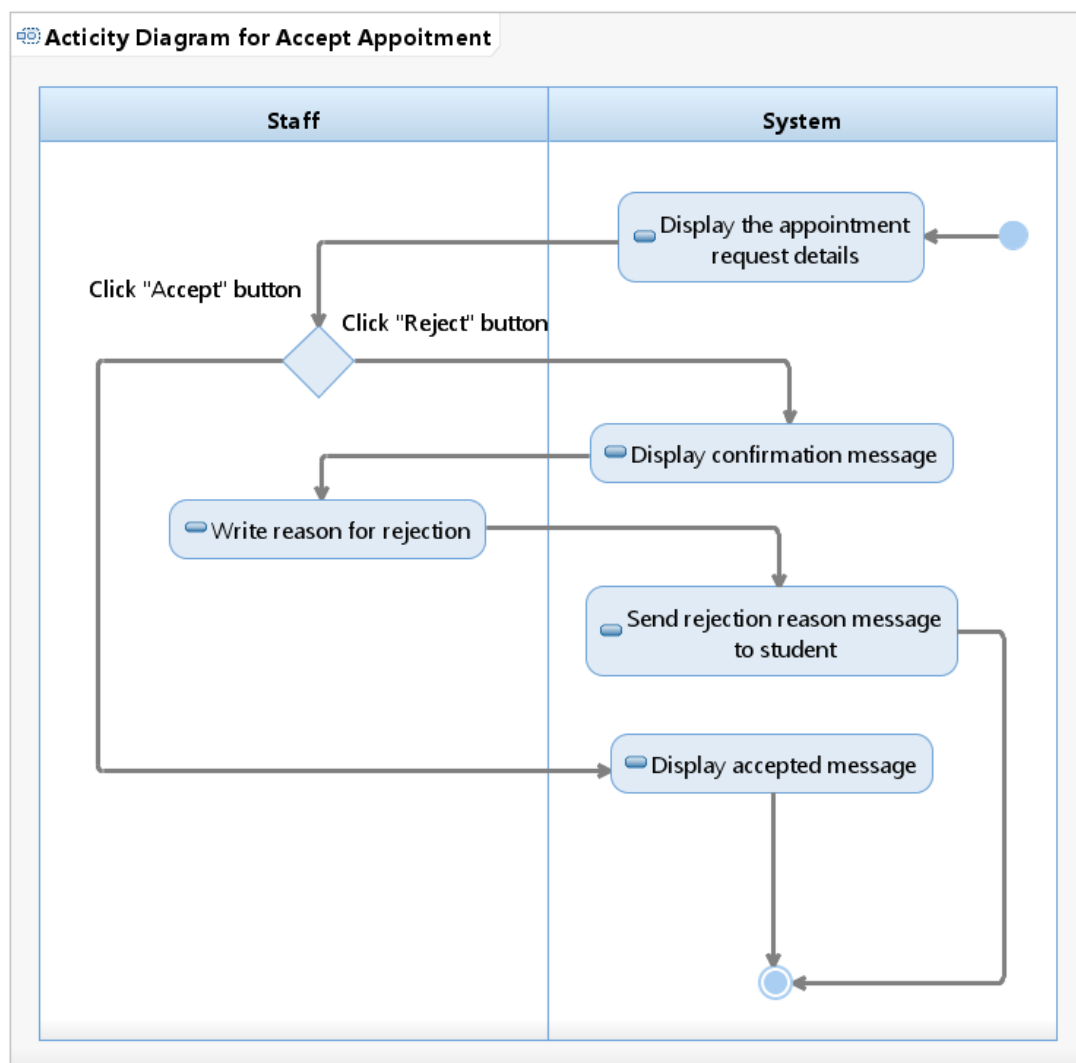


Diagram 6.3: Activity Diagram for Accept Appointment

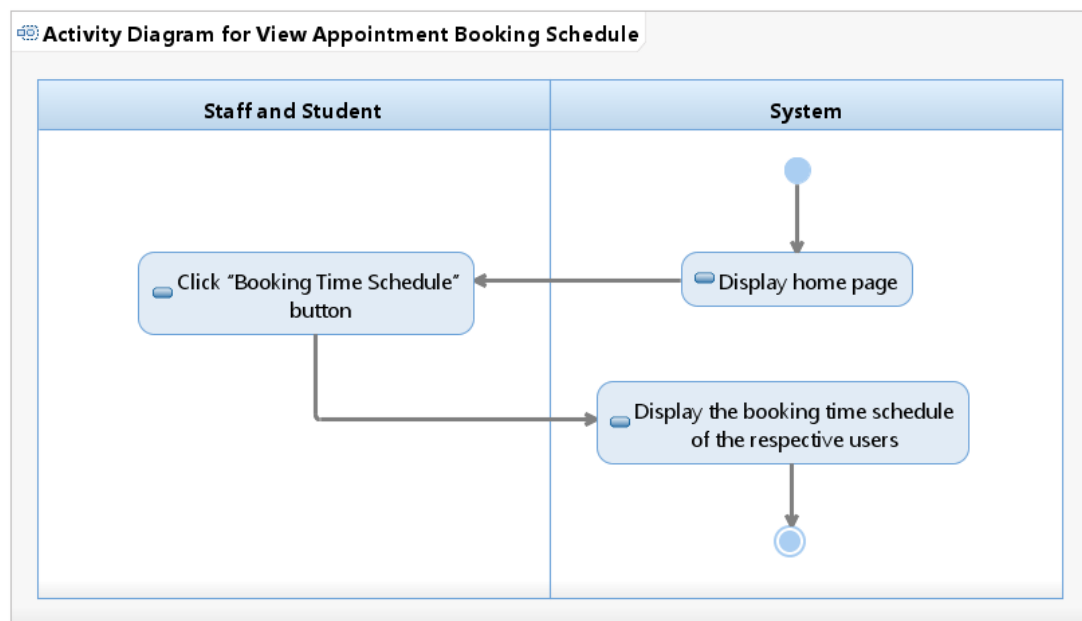
Diagram 6.3 shows the activity diagram for accepting an appointment. Refer table 1.3 below for details of accepting an appointment.

B. Accept Appointment

|                                                         |                                                                                                                                                     |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case Name :</b>                                  | Accept Appointment                                                                                                                                  |
| <b>Actor :</b>                                          | Staff                                                                                                                                               |
| <b>Brief Descriptions :</b>                             | Allow staff to accept an appointment request sent by a student.                                                                                     |
| <b>Preconditions :</b>                                  | Staff had received an appointment booking request and opened it through push-up notification.                                                       |
| <b>Main Event Flows :</b>                               |                                                                                                                                                     |
| <b>Actor Action</b>                                     | <b>System Response</b>                                                                                                                              |
|                                                         | 1. Display the appointment request details.                                                                                                         |
| 2. Click the “Accept” button to accept the appointment. | 3. Display a message the appointment request by student name with request title is accepted.                                                        |
| <b>Alternative Event Flows :</b>                        |                                                                                                                                                     |
| <b>B1. Step 2</b>                                       | If staff click the reject button, the system will prompt a confirmation message and the reason for rejection can be written to inform the students. |

Table 1.3: Use Case Description for Accept Appointment

### **3. Time Schedule Module**



*Diagram 6.4: Activity Diagram for View Appointment Booking Schedule*

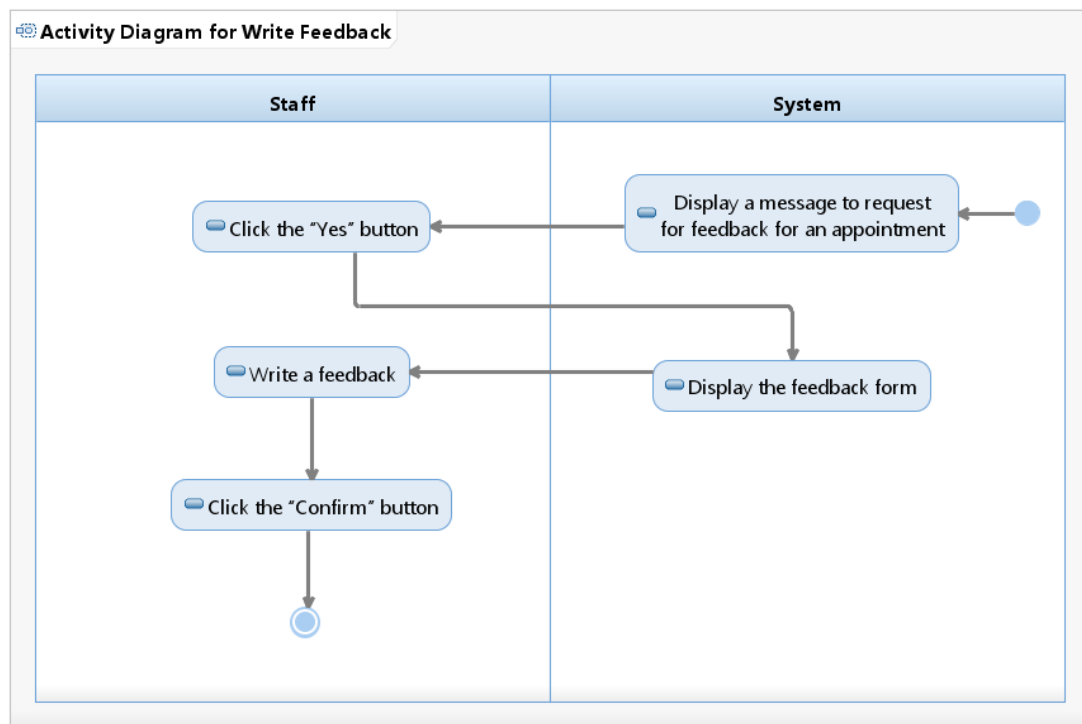
Diagram 6.4 shows the activity diagram for viewing appointment booking schedule. Refer table 1.4 below for details of viewing appointment booking schedule.

A. View Appointment Booking Schedule

|                                              |                                                                      |
|----------------------------------------------|----------------------------------------------------------------------|
| <b>Use Case Name :</b>                       | View appointment booking schedule                                    |
| <b>Actor :</b>                               | Student and staff                                                    |
| <b>Brief Descriptions :</b>                  | Allow students and staff to view their appointment booking schedule. |
| <b>Preconditions :</b>                       | Students and staff had login to their account.                       |
| <b>Main Event Flows :</b>                    |                                                                      |
| <b>Actor Action</b>                          | <b>System Response</b>                                               |
|                                              | 1. Display home page.                                                |
| 2. Click the “Booking Time Schedule” button. | 3. Display the booking time schedule of the respective users.        |

Table 1.4: Use Case Description for View Appointment Booking Schedule

## **4. Feedback Module**



*Diagram 6.5: Activity Diagram for Write Feedback*

Diagram 6.5 shows the activity diagram for writing feedback. Refer table 1.5 below for details of writing feedback.

A. Write Feedback

|                                                                         |                                                                  |
|-------------------------------------------------------------------------|------------------------------------------------------------------|
| <b>Use Case Name :</b>                                                  | Write feedback                                                   |
| <b>Actor :</b>                                                          | Staff                                                            |
| <b>Brief Descriptions :</b>                                             | Allow staff to write feedback after a consultation.              |
| <b>Preconditions :</b>                                                  | The appointment consultation duration is end.                    |
| <b>Main Event Flows :</b>                                               |                                                                  |
| <b>Actor Action</b>                                                     | <b>System Response</b>                                           |
|                                                                         | 1. Display a message to request for feedback for an appointment. |
| 2. Click the “Yes” button to write feedback for a consultation.         | 3. Display the feedback form.                                    |
| 4. Write feedback.                                                      |                                                                  |
| 5. Click the “Confirm” button to save and send feedback to the student. |                                                                  |

*Table 1.5: Use Case Description for Write Feedback*

## 5. Chat Box Module

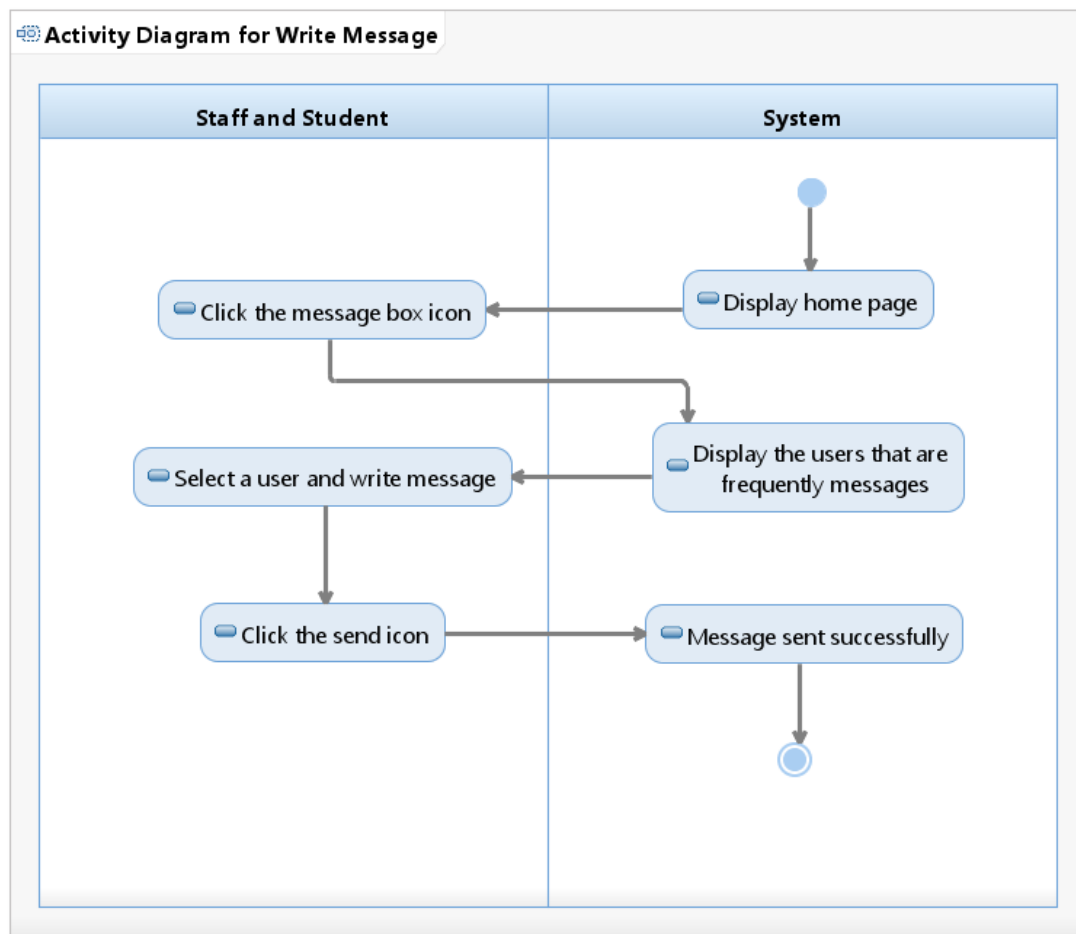


Diagram 6.6: Activity Diagram for Write Message

Diagram 6.6 shows the activity diagram for writing messages. Refer table 1.6 below for details of writing messages.

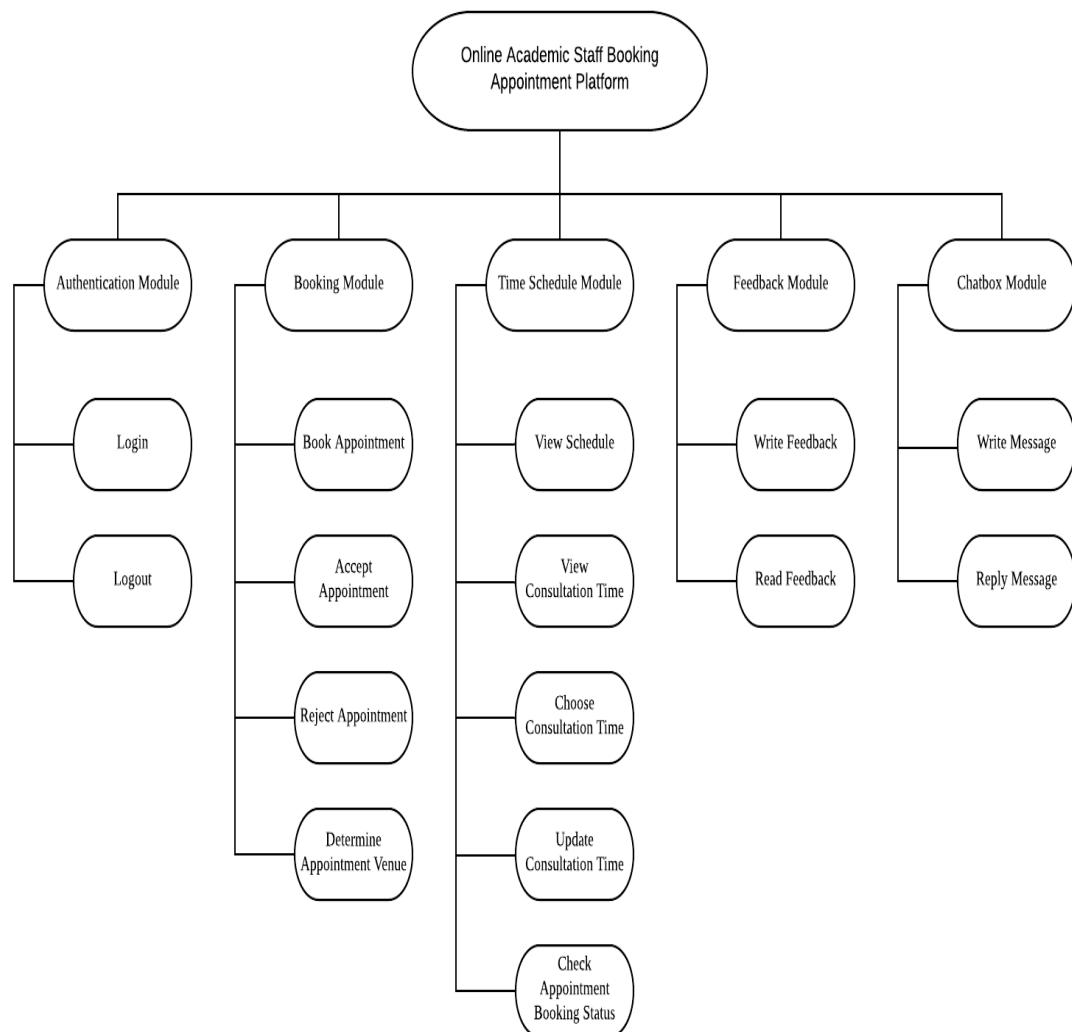
A. Write Message

|                                      |                                                                            |
|--------------------------------------|----------------------------------------------------------------------------|
| <b>Use Case Name :</b>               | Write message                                                              |
| <b>Actor :</b>                       | Students and staff                                                         |
| <b>Brief Descriptions :</b>          | Allow users to write messages to communicate with each other.              |
| <b>Preconditions :</b>               | -                                                                          |
| <b>Main Event Flows :</b>            |                                                                            |
| <b>Actor Action</b>                  | <b>System Response</b>                                                     |
|                                      | 1. Display home page.                                                      |
| 2. Click the message box icon.       | 3. Display the users that are frequently messaged.                         |
| 4. Select a user and write messages. |                                                                            |
| 5. Click the send icon.              | 6. The message is sent successfully with a label “sent” under the message. |

*Table 1.6: Use Case Description for Write Message*

## 3.7 Project Scope and Analysis Class Diagram

### 3.7.1 Project Scope



*Diagram 7.1: Online Academic Staff Booking Appointment Platform Project Scope and Modules*

### 3.7.2 Analysis Class Diagram

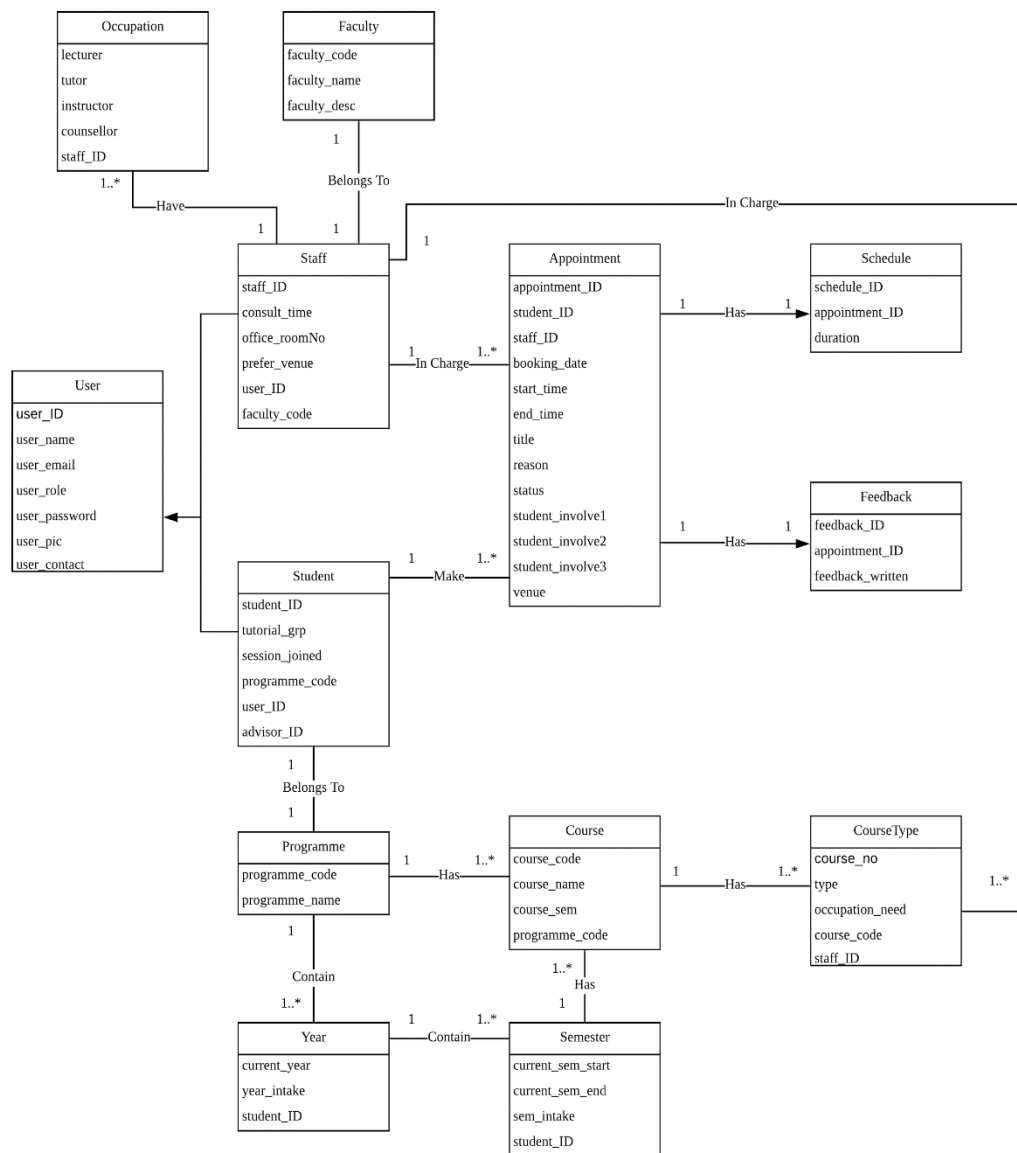


Diagram 7.2: Analysis Class Diagram

## 3.8 Non-functional Requirements

As an online booking appointment platform, there are some non-functional requirements that should be focused in order to have a high-quality platform which is satisfied by users. The non-functional requirements are analyzed and listed below.

### **1. Availability Requirement**

Availability is one of the most important non-functional requirements for an online system. This is because an online booking appointment platform should be available for 24 hours the whole day so that the user can access the system at any time.

### **2. Reliability Requirement**

An online system should have high reliability which the reliability requirements refer to how often a system fails. Hence, an online booking appointment platform should have a low system down time which should not be more than 0.05%.

### **3. Security Requirement**

Since all users information and booking details are stored in the database, thus it is important to secure all data stored from accessed by third parties. An online booking appointment platform should be secure so that confidential data will not be stolen.

### **4. Functional Requirement**

A system should perform the functions that it is required to help users to complete their tasks. Hence, an online booking appointment platform should perform the functions that it is

required

to.

### **5. Efficiency Requirement**

As an online system, the system should be efficient on its response time since all information is required to update in real-time. Thus, an online booking appointment platform should be efficient which the website response time should take less than 5 seconds.

### 3.9 Chapter Summary and Evaluation

*This chapter has discussed the methodology used in this research which are literature review technique and questionnaire. We also covered the software development model in the methodology section which staged delivery model is chosen where the system functions are developed incrementally. The data collected in the questionnaire had been analysed by interpreting the histogram and independent t-test table results generated using software RSA. The functional requirements section also included use case diagrams and activity diagrams to represent the requirements in a clearer way.*

*In the next chapter, we will design the User Interface (UI) and the data design for the proposed platform. The remaining chapters will also include the coding implementation and system testing documentation.*

# Chapter 4

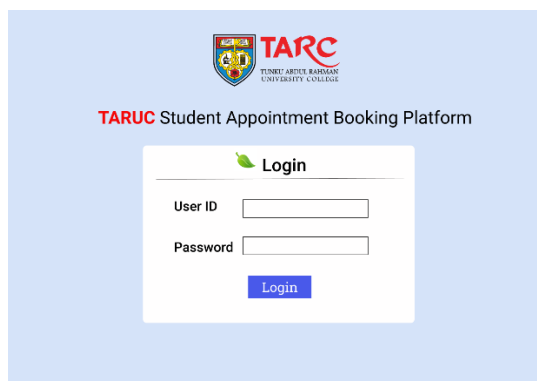
## **Design**

## 4 Design

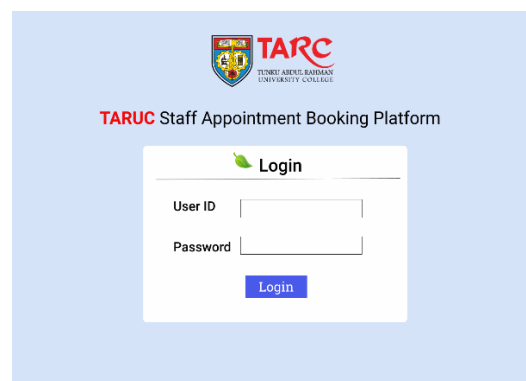
Chapter 4 had covered the design aspects of the online academic staff booking appointment platform proposed in this framework. The design aspects included the user interface design of the web-based online booking appointment platform, data design such as entity relationship diagram (ERD) and report design which shows different reports generated by the platform. Apart from that, this chapter also included process design which is state chart diagram and software architecture design.

### 4.1 System and User Interface(UI) Design

Figure 1.1 and 1.2 shows the starting screen UI of the online academic staff booking appointment platform. Both UI requires the user to enter their user ID and their password correctly.

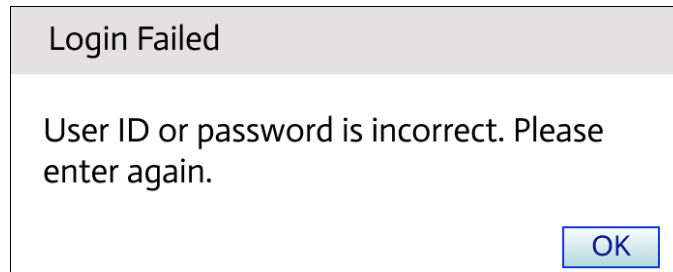


*Figure 1.1 Student Login Screen*



*Figure 1.2 Staff Login Screen*

If the users enter their user ID or password wrongly, an error message as shown in Figure 1.3 will be displayed and the users are required to enter the information again.



*Figure 1.3 Login Failed Error Message*

After a user successfully login to their account, the homepage will be displayed as shown in Figure 2.1 and Figure 2.2. The difference of the homepage between staff and students is that students are allowed to make new appointments whereas the staff account does not include this function.

In the homepage, there is an active and a history button which the active button shows the appointments that are coming in the future, but the history button shows the appointments that are already past. The appointments are shown in a table form which the table had included the appointment details such as date, time, duration, persons involved in the appointment, appointment title, appointment status, and the appointment venue. Besides that, the users can sort the appointment bookings by the appointment status which are all, accepted, pending, and rejected.

| No | Date     | Time              | Duration | Persons                                        | Title        | Status   | Venue |
|----|----------|-------------------|----------|------------------------------------------------|--------------|----------|-------|
| 1  | 02/04/20 | 03.00PM - 04.00PM | 1 hour   | Ms. CHIN CHAI LIM, CHEW JANE, CHENG QING XIANG | FYP C3 Check | ACCEPTED | B103  |

Figure 2.1 Student Homepage

The screenshot shows the Staff Homepage of the Online Academic Staff Booking Appointment Platform. The interface features a dark blue header with a hamburger menu, notification and chat icons, the platform name 'Book Appointment' with a crest, and a user profile section for 'Welcome, Chew Jane'. A light blue bar below the header displays the date and time 'Fri, Mar 30, 12:00PM'. A dark blue sidebar on the left contains navigation links for 'HOME', 'SCHEDULE', and 'CONSULTATION RECORD'. The main content area, titled 'Home', shows 'My Appointment Bookings' with tabs for 'Active' and 'History'. The 'Active' tab is selected, displaying a table of bookings. The table has columns for No, Date, Time, Duration, Persons, Title, Status, and Venue. One booking is listed with status 'ACCEPTED'. Navigation buttons 'Previous' and 'Next' are at the bottom right of the table.

Book Appointment

Welcome, Chew Jane

Fri, Mar 30, 12:00PM

[Home](#)

My Appointment Bookings

Active History

Show 10 entries

| No | Date     | Time              | Duration | Persons                                        | Title        | Status   | Venue |
|----|----------|-------------------|----------|------------------------------------------------|--------------|----------|-------|
| 1  | 02/04/20 | 03.00PM - 04.00PM | 1 hour   | Ms. CHIN CHAI LIM, CHEW JANE, CHENG QING XIANG | FYP C3 Check | ACCEPTED | B103  |

Previous Next

Figure 2.2 Staff Homepage

Users can also select the respective row in the table to view the appointment details. In this appointment details web page, users can choose to cancel the appointment if they wish too. The appointment details web page design is same for both academic staff and students. The UI design of appointment details web page is shown in Figure 2.3 as below.

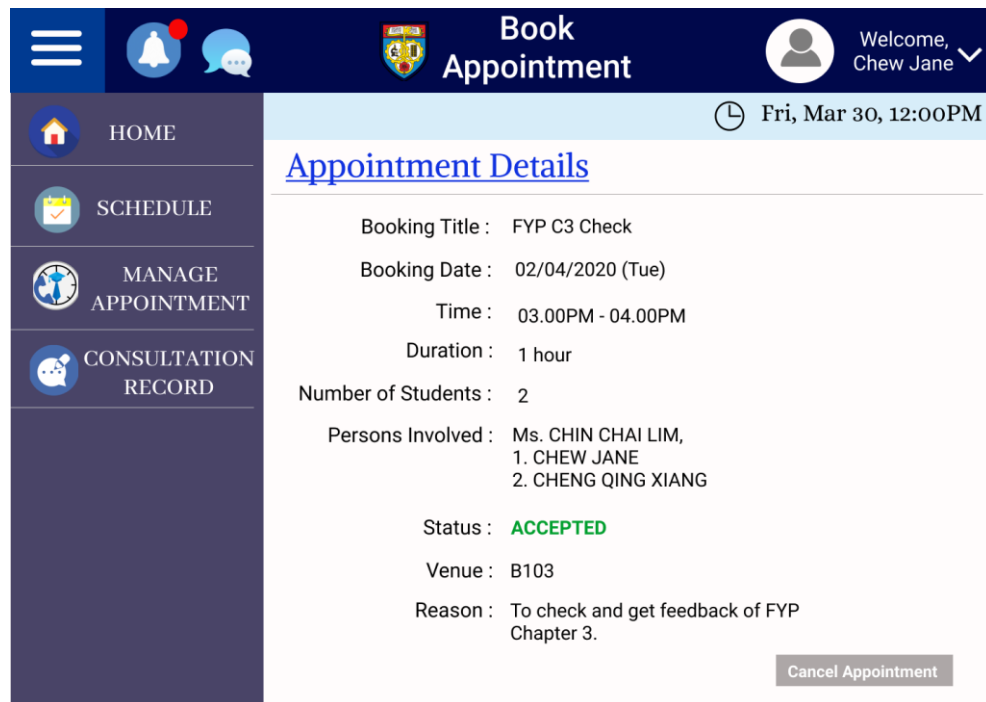
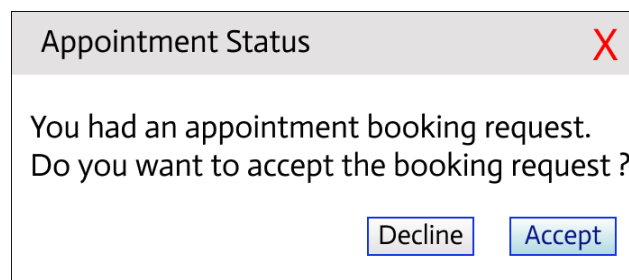


Figure 2.3 Appointment Details

In the appointment details page, academic staff are allowed to accept or reject the appointment request if the appointment status is pending. By clicking on the “pending” status in the web page, a window will be popped up to allow staff to change the appointment status as shown in Figure 2.4.



*Figure 2.4 Change Appointment Status*

If staff choose to accept the appointment request, the small window will be closed and the appointment details web page will be updated by changing the appointment status to “accepted”. However, if the staff choose to decline the appointment request, the staff is then required to enter reason of rejecting the appointment request.

**Book Appointment**

Welcome, Chew Jane ✓

Fri, Mar 30, 12:00PM

### Book Appointment

**202003 Semester**  
(Monday, 2020-03-02 ~ Sunday, 2020-04-19)

**Semester Course & Instructor :**

| No | Course                                           | Class                                  | Lecturer / Instructor                                                                |
|----|--------------------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------|
| 1  | BACS 2103 Software Quality Assurance And Testing | Lecture :<br>Tutorial :<br>Practical : | Ms Mazlinda Binti Nezam Mudeen<br>Dr Lim Siew Mooi<br>Ms Mazlinda Binti Nezam Mudeen |
| 2  | BAIT 3153 Software Project Management            | Lecture :<br>Tutorial :<br>Practical : | Ms Kathleen Tan Swee Neo<br>Ms Kathleen Tan Swee Neo<br>Ms Kathleen Tan Swee Neo     |
| 3  | BACS 3403 Project I                              | Course Without Class                   | Ms Chin Chai Lim                                                                     |

Academic Advisor : [PUAN FONG WAI SHAM](#)

... Other Academic Staff

Figure 3.1 Book Appointment Step 1 - Select a Staff

Book an appointment function can only be performed by students. At first, the web page will show the student's current semester, courses, respective instructor and student's academic advisor as shown in Figure 3.1 as above. Then, students can make an appointment easily by clicking the academic staff name. If the students want to make an appointment with other academic staff, the student can choose to click the blue button with the label "other academic staff" which enables the students to search for a staff and make appointments with the staff.

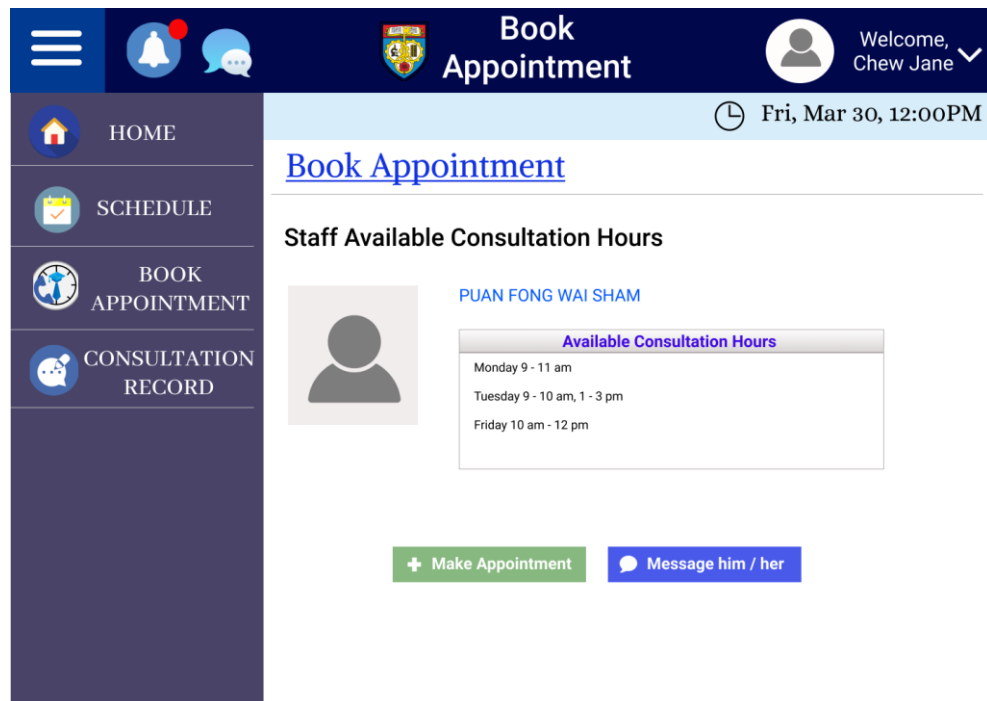


Figure 3.2 Book Appointment Step 2 - View Staff Available Consultation Hours

After a student has selected a staff, the staff available consultation hours will be displayed as shown in Figure 3.2 above. Then, the student can click on the green button to make an appointment with the staff.

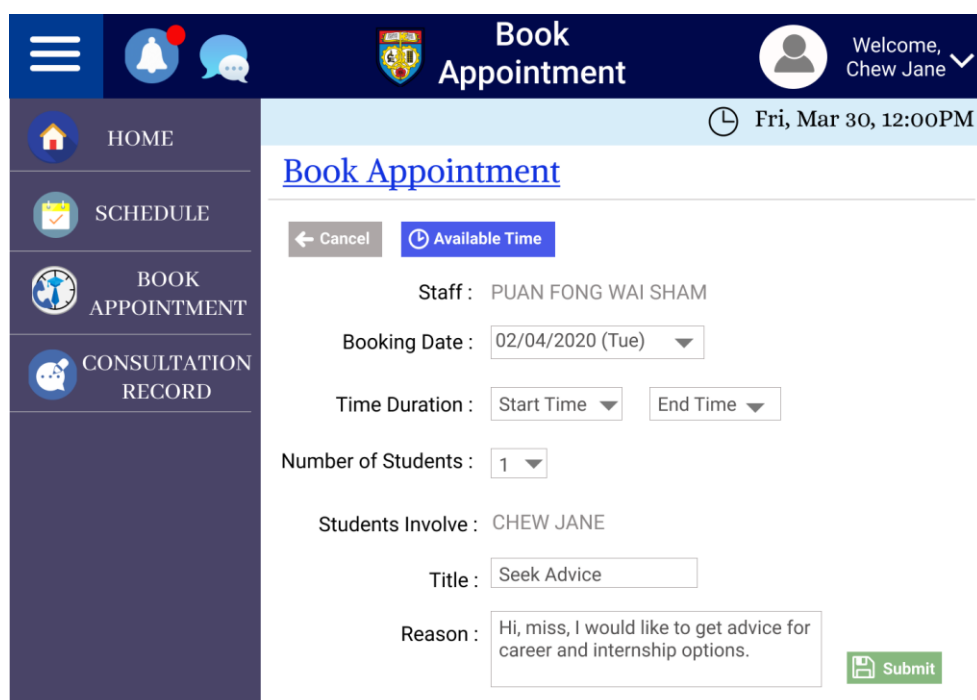


Figure 3.3 Book Appointment Step 3 - Submit Appointment Request Form

Figure 3.3 shows the appointment request form which will be displayed after the student clicks to make an appointment button with the respective staff. In the appointment request form, the staff name is displayed automatically and could not be changed by students. Then, the students need to fill up the form and click on the submit button.

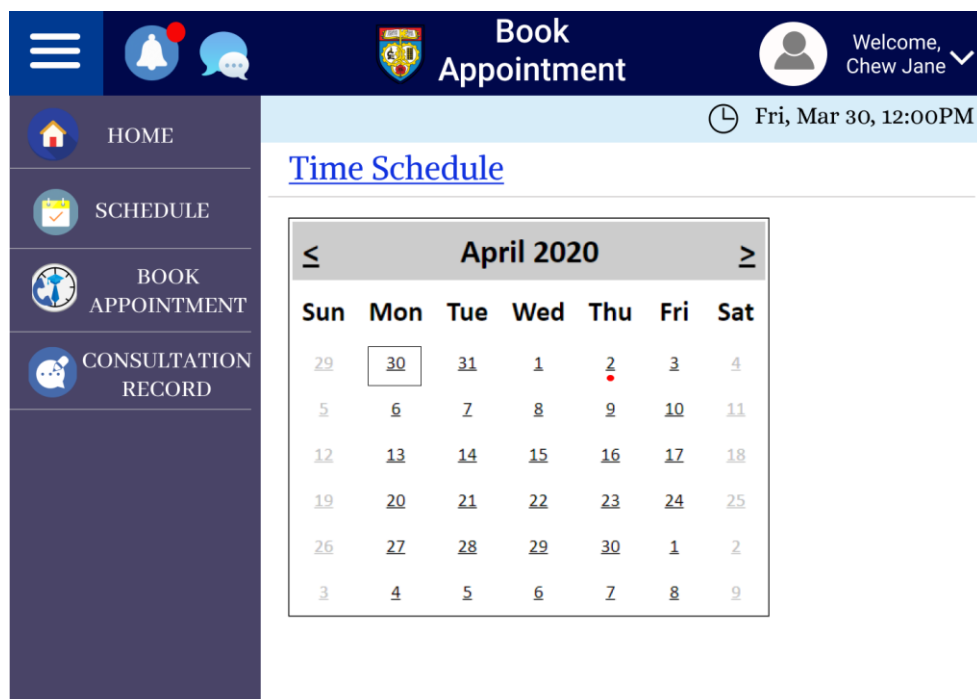


Figure 4 Time Schedule

Figure 4 shows the time schedule UI design which allows students and staff to view which date they have appointment(s). In the calendar, the date with appointment(s) will contain a red dot. The users can click on the date to view the appointments booked on the respective date.

**Book Appointment**

Welcome, Chew Jane ✓

Fri, Mar 30, 12:00PM

### Consultation Record

Show 10 entries

| No | Date     | Time              | Persons                                              | Title        | Feedback / Conclusion                         |
|----|----------|-------------------|------------------------------------------------------|--------------|-----------------------------------------------|
| 1  | 02/04/20 | 03.00PM - 04.00PM | Ms. CHIN CHAI LIM,<br>CHEW JANE,<br>CHENG QING XIANG | FYP C3 Check | Changes required on C3. C4 briefing is given. |

Previous Next

*Figure 5 Consultation Record*

Furthermore, the online academic staff booking appointment platform allows users to view their previous consultation record. Every staff is required to write their feedback or conclusion to conclude the ended appointment. Feedback or conclusion written will be sent to respective students and saved on the consultation record. This enables the students and staff to keep track of their appointment progress especially when the appointment is booked for project discussion purposes.

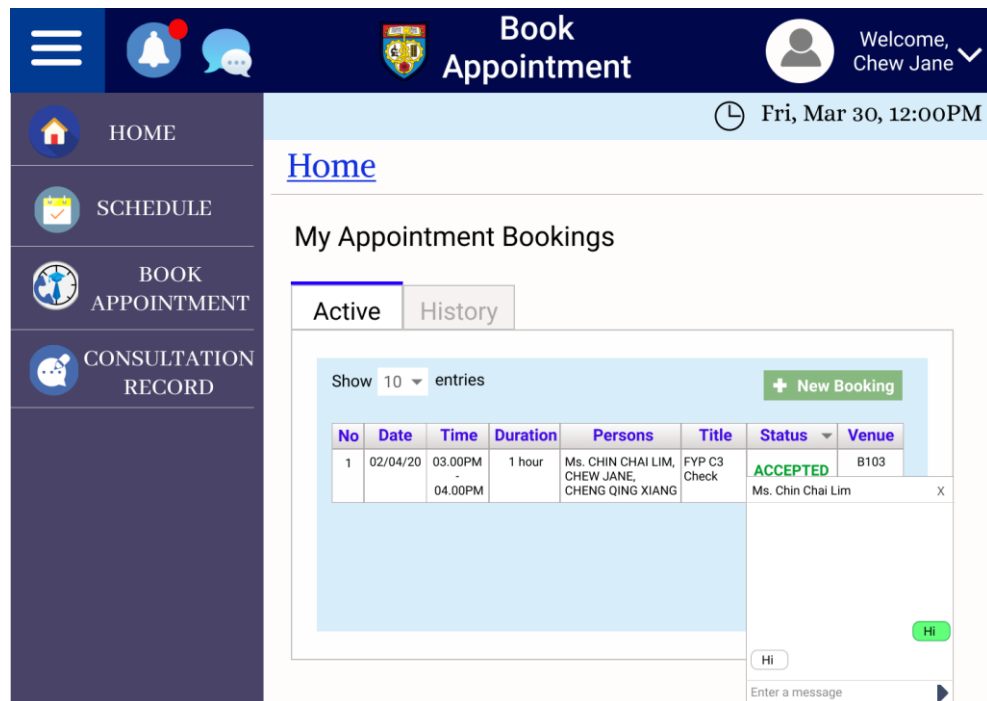


Figure 6 Chat Box Function

Moreover, the chat box function is designed for the platform proposed in this framework. This function enables the students and staff to message with each other especially if there is an emergency. A student can search for a staff and message them as shown in Figure 6 above.

## 4.2 Data Design

In this section, database design language and data dictionary will be included. Moreover, this section will also include the entity relationship diagram (ERD) and final class diagram designed for the online academic staff booking appointment platform.

### 4.2.1 Database Design Language

*Note: Underlined - Primary Key, End with \* - Foreign Key*

**User** (user\_ID, user\_name, user\_email, user\_role, user\_password, user\_pic, user\_contact)

**Student** (student\_ID, tutorial\_grp, session\_joined, programme\_code\*, user\_ID\*, advisor\_ID\*)

**Staff** (staff\_ID, consult\_time, slot\_durationStart, slot\_durationEnd, office\_roomNo, prefer\_venue, user\_ID\*, faculty\_code\*)

**Occupation** (staff\_ID\*, lecturer, tutor, instructor, counsellor)

**Faculty** (faculty\_code, faculty\_name, faculty\_desc)

**Programme** (programme\_code, programme\_name)

**Year** (current\_year, year\_intake, student\_ID\*)

**Semester** (current\_sem\_start, current\_sem\_end, sem\_intake, student\_ID\*)

**Course** (course\_code, course\_name, course\_sem, programme\_code\*)

**CourseType** (course\_no, type, occupation\_need, course\_code\*, staff\_ID\*)

**Appointment** (appointment\_ID, student\_ID\*, staff\_ID\*, booking\_date, start\_time, end\_time, TimeBook, title, reason, status, rejected\_reason, student\_involve1, student\_involve2, student\_involve3, venue)

**Feedback** (feedback\_ID, appointment\_ID\*, feedback\_written)

### 4.2.2 Data Dictionary

Data dictionary shows the attributes stored in the tables and each table contains the description of attributes and data type of the attributes.

#### 1. Table Name: User

**Description:** To store the individual user information that is shared by both students and staff.

| Field Name    | Description                                        | PK/FK | Data Type   | Length       | Nulls | Rules/Remark           |
|---------------|----------------------------------------------------|-------|-------------|--------------|-------|------------------------|
| user_ID       | User unique ID number that used to identify a user | PK    | Integer     | -            | No    | Auto increment         |
| user_name     | Name of user                                       | -     | VARCHAR(80) | Max 80 chars | No    | -                      |
| user_email    | E-mail address of user                             | -     | VARCHAR(80) | Max 80 chars | No    | -                      |
| user_role     | Role of user in the system                         | -     | VARCHAR(10) | Max 10 chars | No    | Default value: student |
| user_password | Password for user to access the system             | -     | VARCHAR(20) | Max 20 chars | No    | Min 6 chars            |
| user_pic      | Profile picture of user                            | -     | LOB         | -            | No    | -                      |
| user_contact  | Mobile phone contact number of user                | -     | VARCHAR(20) | Max 20 chars | No    | -                      |

**2. Table Name:** Student**Description:** To store the student information.

| Field Name     | Description                                          | PK/FK | Data Type   | Length         | Nulls | Rules/Remark                       |
|----------------|------------------------------------------------------|-------|-------------|----------------|-------|------------------------------------|
| student_ID     | Student ID number that uniquely identify a student   | PK    | Integer     | Auto Increment | No    | Auto increment from value: 2000000 |
| tutorial_grp   | Student tutorial group of their programme            | -     | Integer(20) | Max 20 counts  | No    | -                                  |
| session_joined | Student session joined date of their programme       | -     | VARCHAR(10) | Max 10 chars   | No    | -                                  |
| programme_code | Key that identify student belongs to which programme | FK    | VARCHAR(10) | Max 10 chars   | No    | Reference Table: Programme         |
| user_ID        | Key that identify student user details               | FK    | Integer     | -              | No    | Reference Table: User              |
| advisor_ID     | Key that identify student's academic advisor         | FK    | Integer     | -              | No    | Reference Table: Staff             |

**3. Table Name:** Staff**Description:** To store the staff information.

| Field Name         | Description                                                    | PK/FK | Data Type    | Length         | Nulls | Rules/Remark                    |
|--------------------|----------------------------------------------------------------|-------|--------------|----------------|-------|---------------------------------|
| staff_ID           | Staff ID number that uniquely identify a staff                 | PK    | Integer      | Auto Increment | No    | Auto increment from value: 1000 |
| consult_time       | Staff available consultation time                              | -     | VARCHAR(300) | Max 300 chars  | No    | -                               |
| slot_durationStart | Staff consultation time set is available start from which date | -     | Date         | -              | No    | -                               |
| slot_durationEnd   | Staff consultation time set is available until which date      | -     | Date         | -              | No    | -                               |
| office_roomNo      | Staff office room number in campus                             | -     | VARCHAR(20)  | Max 20 chars   | Yes   | -                               |
| prefer_venue       | Staff prefer consultation venue                                | -     | VARCHAR(80)  | Max 80 chars   | No    | -                               |
| user_ID            | Key that identify staff user details                           | FK    | Integer      | -              | No    | Reference Table: User           |
| faculty_code       | Key that identify staff belongs to which faculty               | FK    | VARCHAR(10)  | Max 10 chars   | No    | Reference Table: Faculty        |

**4. Table Name:** Occupation**Description:** To store staff occupation

| Field Name | Description                                                  | PK/FK  | Data Type  | Length      | Nulls | Rules/Remark           |
|------------|--------------------------------------------------------------|--------|------------|-------------|-------|------------------------|
| lecturer   | Staff that gives lecture                                     | -      | VARCHAR(5) | Max 5 chars | Yes   | Default value:<br>Yes  |
| tutor      | Staff that teaches a small group of students                 | -      | VARCHAR(5) | Max 5 chars | Yes   | Default value:<br>Yes  |
| instructor | Staff that instructs student to complete a task              | -      | VARCHAR(5) | Max 5 chars | Yes   | Default value:<br>Yes  |
| counsellor | Staff that give advice on personal or psychological problems | -      | VARCHAR(5) | Max 5 chars | Yes   | Default value:<br>Yes  |
| staff_ID   | Key that identify staff have what occupations                | PK, FK | Integer    | -           | No    | Reference Table: Staff |

**5. Table Name:** Faculty**Description:** To store faculty information.

| Field Name   | Description                                          | PK/FK | Data Type   | Length       | Nulls | Rules/Remark |
|--------------|------------------------------------------------------|-------|-------------|--------------|-------|--------------|
| faculty_code | Faculty code number that uniquely identify a faculty | PK    | VARCHAR(10) | Max 10 chars | No    | -            |
| faculty_name | Name of faculty                                      | -     | VARCHAR(80) | Max 80 chars | No    | -            |
| faculty_desc | Description of faculty                               | -     | LONGTEXT    | -            | No    | -            |

**6. Table Name:** Programme**Description:** To store programme information provided by college.

| Field Name     | Description                                       | PK/FK | Data Type        | Length       | Nulls | Rules/Remark |
|----------------|---------------------------------------------------|-------|------------------|--------------|-------|--------------|
| programme_code | Programme code that uniquely identify a programme | PK    | VARCHAR(10)<br>) | Max 10 chars | No    | -            |
| programme_name | Name of programme                                 | -     | VARCHAR(80)<br>) | Max 80 chars | No    | -            |

**7. Table Name:** Year

**Description:** To identify which year a student studied for their programme.

| Field Name   | Description                                | PK/FK  | Data Type  | Length      | Nulls | Rules/Remark             |
|--------------|--------------------------------------------|--------|------------|-------------|-------|--------------------------|
| current_year | Current year according to the calendar     | -      | YEAR(4)    | Max 4 chars | No    | Default Value: 2020      |
| year_intake  | Student joined the programme on which year | -      | VARCHAR(5) | Max 5 chars | No    | -                        |
| student_ID   | Key that identify student year intake      | PK, FK | Integer    | -           | No    | Reference Table: Student |

**8. Table Name:** Semester

**Description:** To identify which semester a student studied for their programme.

| Field Name        | Description                                      | PK/FK  | Data Type  | Length      | Nulls | Rules/Remark             |
|-------------------|--------------------------------------------------|--------|------------|-------------|-------|--------------------------|
| current_sem_start | Student current semester start date              | -      | VARCHAR(5) | Max 5 chars | No    | -                        |
| current_sem_end   | Student current semester end date                | -      | VARCHAR(5) | Max 5 chars | No    | -                        |
| sem_intake        | Student joined the first semester on which month | -      | VARCHAR(5) | Max 5 chars | No    | -                        |
| student_ID        | Key that identify student year intake            | PK, FK | Integer    | -           | No    | Reference Table: Student |

**9. Table Name:** Course**Description:** To store the course information.

| Field Name     | Description                                             | PK/FK | Data Type    | Length        | Nulls | Rules/Remark               |
|----------------|---------------------------------------------------------|-------|--------------|---------------|-------|----------------------------|
| course_code    | Course code that uniquely identify a course             | PK    | VARCHAR(10)  | Max 10 chars  | No    | -                          |
| course_name    | Name of course                                          | -     | VARCHAR(200) | Max 200 chars | No    | -                          |
| course_sem     | Year and semester of the course taken by students       | -     | VARCHAR(20)  | Max 20 chars  | No    | -                          |
| programme_code | Key that identify the course belongs to which programme | FK    | VARCHAR(10)  | Max 10 chars  | No    | Reference Table: Programme |

**10. Table Name:** CourseType**Description:** To identify class type of a course.

| Field Name      | Description                                                     | PK/FK | Data Type   | Length       | Nulls | Rules/Remark            |
|-----------------|-----------------------------------------------------------------|-------|-------------|--------------|-------|-------------------------|
| course_no       | Course number that used to uniquely identify a row of data      | PK    | Integer     | -            | No    | Auto Increment          |
| course_code     | Course code that uniquely identify a course                     | FK    | VARCHAR(10) | Max 10 chars | No    | Reference Table: Course |
| type            | Type of course which included lecture, tutorial and practical.  | -     | VARCHAR(20) | Max 20 chars | No    | -                       |
| occupation_need | The staff's occupation needed for the particular type of course | -     | VARCHAR(20) | Max 20 chars | No    | -                       |
| staff_ID        | Key that identify                                               | FK    | Integer     | -            | No    | Reference               |

|  |                               |  |  |  |  |              |
|--|-------------------------------|--|--|--|--|--------------|
|  | staff in charge of the course |  |  |  |  | Table: Staff |
|--|-------------------------------|--|--|--|--|--------------|

### 11. Table Name: Appointment

**Description:** To store the appointment information made by students.

| Field Name     | Description                                                        | PK/FK | Data Type          | Length         | Nulls | Rules/Remark                       |
|----------------|--------------------------------------------------------------------|-------|--------------------|----------------|-------|------------------------------------|
| appointment_ID | Appointment ID number that uniquely identify a booking information | PK    | Integer            | -              | No    | Auto Increment                     |
| student_ID     | Key that identify students who made the booking                    | FK    | Integer            | -              | No    | Reference Table: Student           |
| staff_ID       | Key that identify staff in charge of the booking                   | FK    | Integer            | -              | No    | Reference Table: Staff             |
| booking_date   | Date of the booking made                                           | -     | Date               | -              | No    | -                                  |
| start_time     | Start time of the booking made                                     | -     | Time               | -              | No    | -                                  |
| end_time       | End time of the booking made                                       | -     | Time               | -              | No    | -                                  |
| TimeBook       | The date and time of booking request made.                         | -     | Timestamp          | -              | No    | Default value: current timestamp() |
| title          | Title of appointment requested                                     | -     | VARCHAR(100)       | Max 100 chars  | No    | -                                  |
| reason         | Reason of requesting the appointment                               | -     | VARCHAR(5000)<br>) | Max 5000 chars | No    | -                                  |

|                  |                                                |   |               |                |     |                        |
|------------------|------------------------------------------------|---|---------------|----------------|-----|------------------------|
| status           | Status of the booking made                     | - | VARCHAR(20)   | Max 20 chars   | No  | Default value: PENDING |
| rejected_reason  | Reason of the appointment is rejected by staff | - | VARCHAR(1000) | Max 1000 chars | Yes | -                      |
| student_involve1 | Other student that involve in the appointment  | - | Integer       | -              | Yes | -                      |
| student_involve2 | Other student that involve in the appointment  | - | Integer       | -              | Yes | -                      |
| student_involve3 | Other student that involve in the appointment  | - | Integer       | -              | Yes | -                      |
| venue            | Venue of the appointment to be held            | - | VARCHAR(30)   | Max 30 chars   | Yes | -                      |

**12. Table Name:** Feedback

**Description:** To store the feedback information of an appointment booking.

| Field Name       | Description                                                           | PK/FK | Data Type      | Length         | Nulls | Rules/Remark                 |
|------------------|-----------------------------------------------------------------------|-------|----------------|----------------|-------|------------------------------|
| feedback_ID      | Feedback ID number that uniquely identify feedback for an appointment | PK    | Integer        | -              | No    | Auto Increment               |
| appointment_ID   | Key that identify the appointment booking information                 | FK    | Integer        | -              | No    | Reference Table: Appointment |
| feedback_written | Feedback written to student after an appointment                      | -     | VARCHAR(5000 ) | Max 5000 chars | No    | -                            |

### 4.2.3 Entity Relationship Diagram (ERD)

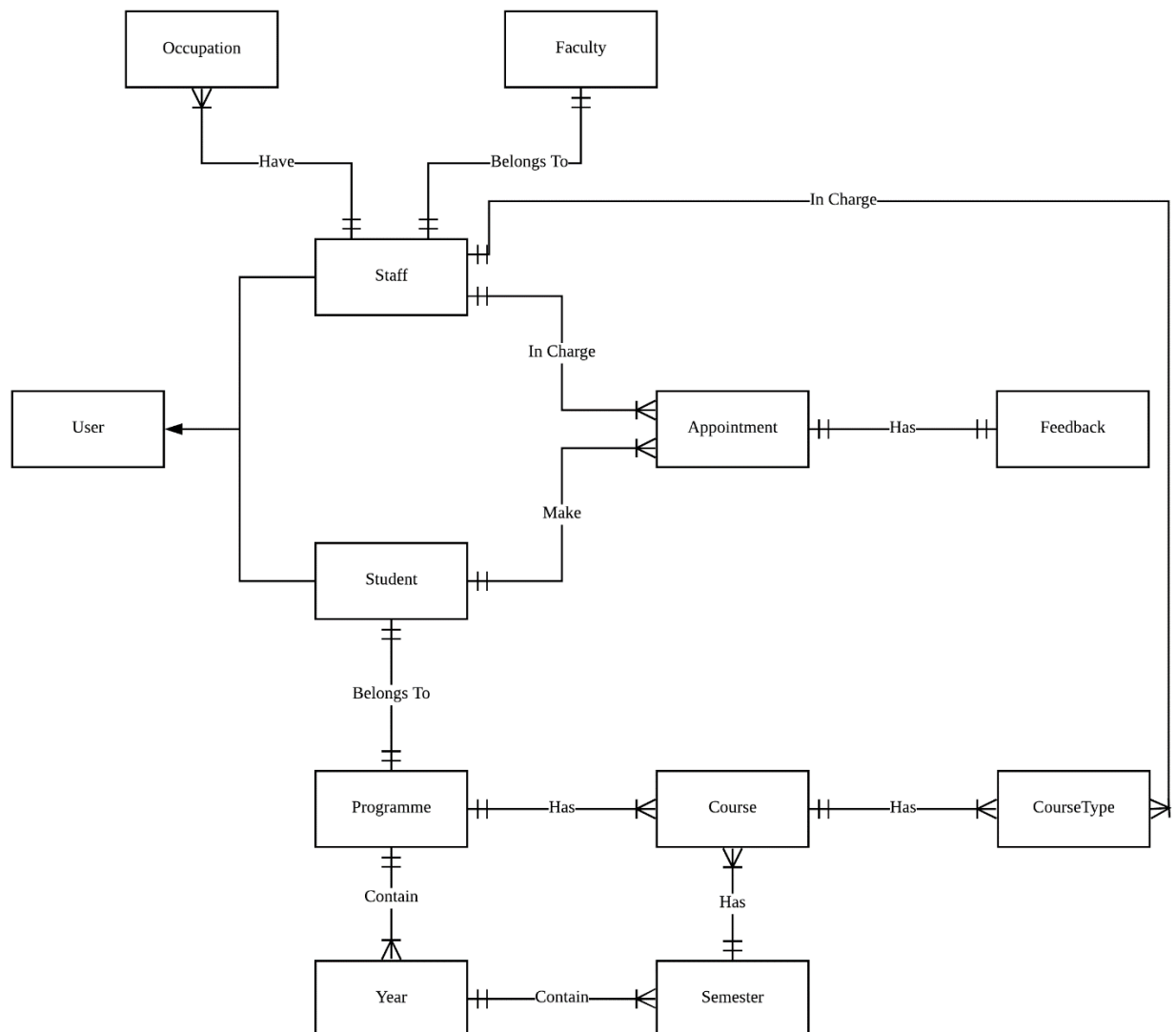


Diagram 1: Entity Relationship Diagram (ERD)

## 4.2.4 Final Class Diagram

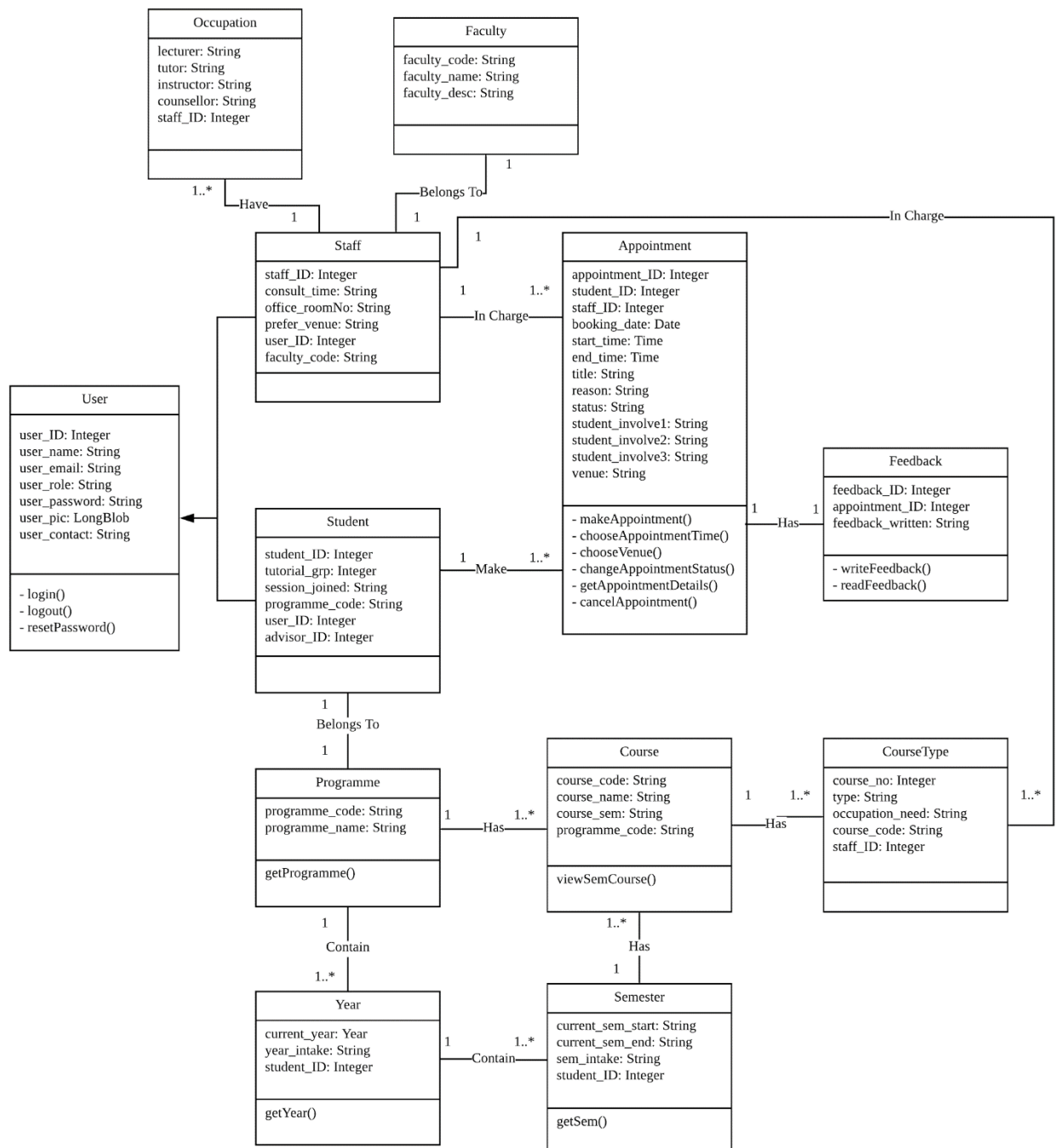


Diagram 2: Final Class Diagram

## 4.3 Report Design

The reports generated in the online academic staff booking appointment platform will be included in this section. The reports are active appointment bookings report, history appointment bookings report, and cancellation appointment bookings report.

### 4.3.1 Active Appointment Bookings Report

In the homepage, there is an active button which shows the appointments that are coming in the future which the appointment's status is either pending or accepted. This active appointment bookings report is aimed to let users view their appointment booking that is coming in the future in an arrangeable and easier way.

The screenshot shows the 'Book Appointment' web application. The top header is dark blue with a hamburger menu, notification bell, chat bubble, university crest, 'Book Appointment' title, and user profile 'Welcome, Chew Jane'. Below the header is a light blue bar showing the date and time 'Fri, Mar 30, 12:00PM'. The left sidebar is dark blue with icons and labels for 'HOME', 'SCHEDULE', 'BOOK APPOINTMENT', and 'CONSULTATION RECORD'. The main content area is white and titled 'Home'. Under 'My Appointment Bookings', there are two tabs: 'Active' (selected) and 'History'. Below the tabs is a light blue box containing a table of active bookings. The table has columns: No, Date, Time, Duration, Persons, Title, Status, and Venue. A single booking is listed with status 'ACCEPTED'. There is a '+ New Booking' button and 'Previous/Next' navigation buttons at the bottom of the table area.

| No | Date     | Time              | Duration | Persons                                        | Title        | Status   | Venue |
|----|----------|-------------------|----------|------------------------------------------------|--------------|----------|-------|
| 1  | 02/04/20 | 03.00PM - 04.00PM | 1 hour   | Ms. CHIN CHAI LIM, CHEW JANE, CHENG QING XIANG | FYP C3 Check | ACCEPTED | B103  |

Figure 7.1 Active Appointment Bookings Report

### 4.3.2 History Appointment Bookings Report

Apart from that, there is a history button in the homepage which shows the appointments that are past. This history appointment bookings report will display the past appointments and enables users to click on it to view the respective consultation feedback.

The screenshot displays the 'Book Appointment' web application interface. The top navigation bar includes a menu icon, notification and chat icons, the application logo, the title 'Book Appointment', and a user profile section with the text 'Welcome, Chew Jane' and a dropdown arrow. A clock icon indicates the current time as 'Fri, Mar 30, 12:00PM'. The left sidebar contains four main menu items: 'HOME', 'SCHEDULE', 'BOOK APPOINTMENT', and 'CONSULTATION RECORD'. The main content area is titled 'My Appointment Bookings' and features two tabs: 'Active' and 'History'. The 'History' tab is selected, showing a table of past appointments. Above the table, there is a 'Show 10 entries' dropdown and a '+ New Booking' button. The table has columns for 'No', 'Date', 'Time', 'Duration', 'Persons', 'Title', 'Status', and 'Venue'. A single entry is visible with the status 'ACCEPTED'. At the bottom of the table, there are 'Previous' and 'Next' pagination buttons.

| No | Date     | Time              | Duration | Persons                                        | Title        | Status   | Venue |
|----|----------|-------------------|----------|------------------------------------------------|--------------|----------|-------|
| 1  | 22/03/20 | 03.00PM - 04.00PM | 1 hour   | Ms. CHIN CHAI LIM, CHEW JANE, CHENG QING XIANG | FYP C2 Check | ACCEPTED | B103  |

Figure 7.2 History Appointment Bookings Report

### 4.3.3 Cancellation Appointment Bookings Report

In addition, the cancellation appointments will also be recorded under the history appointment bookings. The appointment status will be marked as “rejected” in red color. When users pointed their mouse to the “rejected” status, the reason for the appointment being rejected will be displayed as shown in Figure 7.3. Thus, this cancellation appointment bookings report enables the users to view the rejected appointments and the reason written by the staff for the particular appointment is failed to organize.

The screenshot displays the 'Book Appointment' web application interface. The top navigation bar includes a menu icon, notification and chat icons, the application title 'Book Appointment', and a user profile section for 'Welcome, Chew Jane'. The left sidebar contains navigation links: HOME, SCHEDULE, BOOK APPOINTMENT, and CONSULTATION RECORD. The main content area shows the 'Home' page with a clock indicating 'Fri, Mar 30, 12:00PM'. Below this is the 'My Appointment Bookings' section, which has tabs for 'Active' and 'History'. The 'History' tab is selected, showing a table of booking entries. The first entry is marked as 'REJECTED' in red. A tooltip is displayed over the 'REJECTED' status, stating: 'I had a meeting on that time. Pls book the appointment again in another time. TQ'. The table has columns for No, Date, Time, Duration, Persons, Title, Status, and Venue.

| No | Date     | Time              | Duration | Persons                                        | Title        | Status   | Venue |
|----|----------|-------------------|----------|------------------------------------------------|--------------|----------|-------|
| 1  | 22/03/20 | 03.00PM - 04.00PM | 1 hour   | Ms. CHIN CHAI LIM, CHEW JANE, CHENG QING XIANG | FYP C2 Check | REJECTED | B103  |

Figure 7.3 Cancellation Appointment Bookings Report

## 4.4 Process Design

Process design section shows the state chart diagram of status changes of a booked appointment.

### 4.4.1 State Chart Diagram

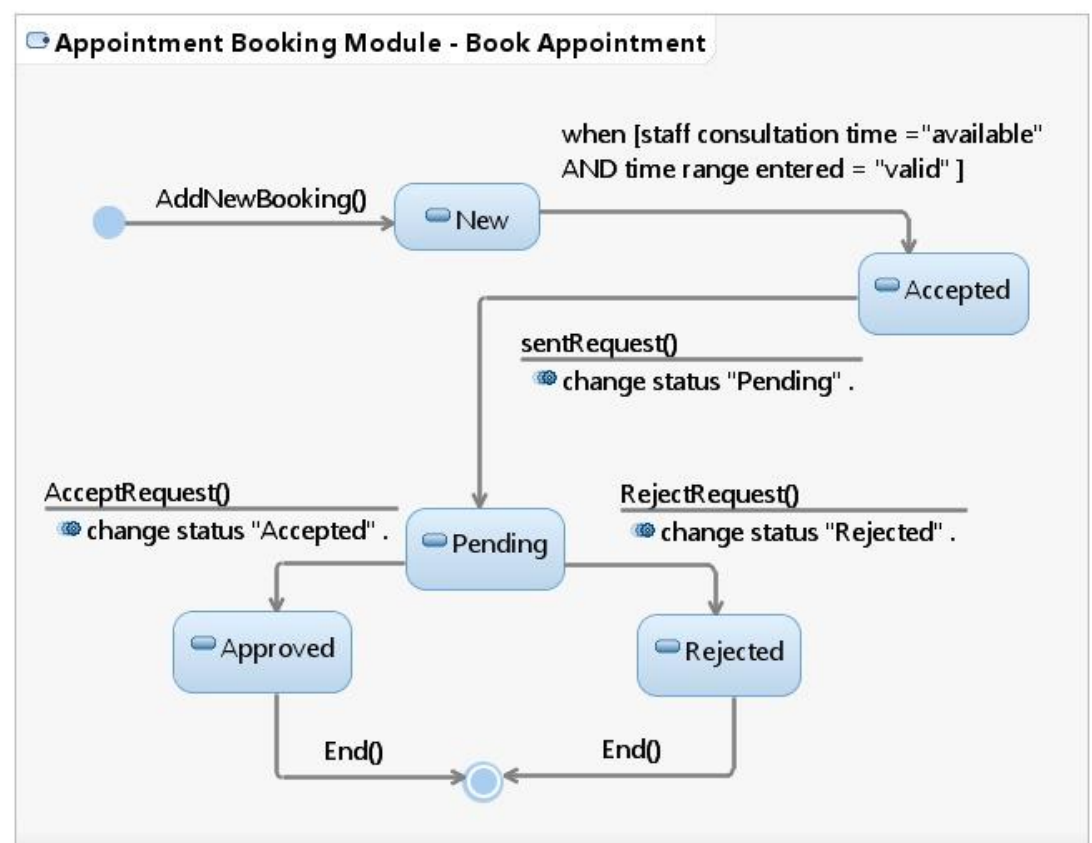


Diagram 3: State Chart Diagram of Book Appointment

## 4.5 Software Architecture Design

The component diagram and deployment diagram are included in this section as shown below.

### 4.5.1 Component Diagram

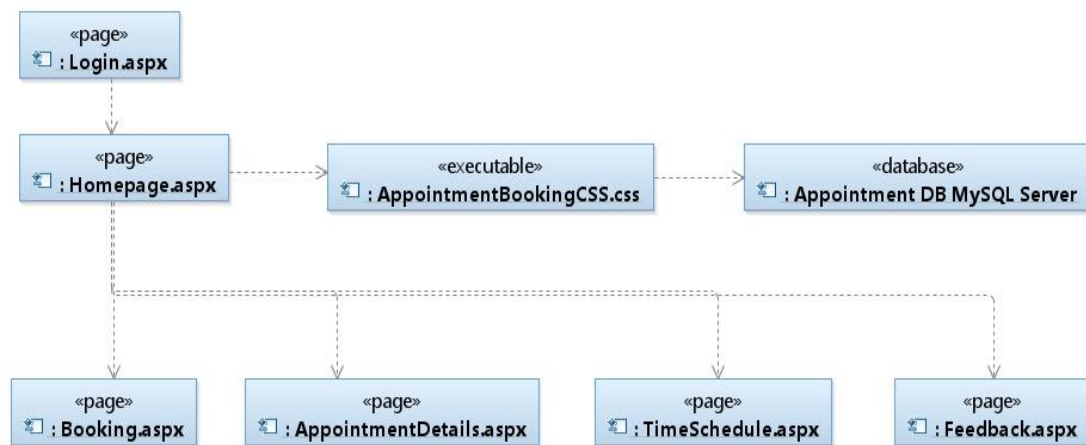


Diagram 4: Component Diagram

### 4.5.2 Deployment Diagram

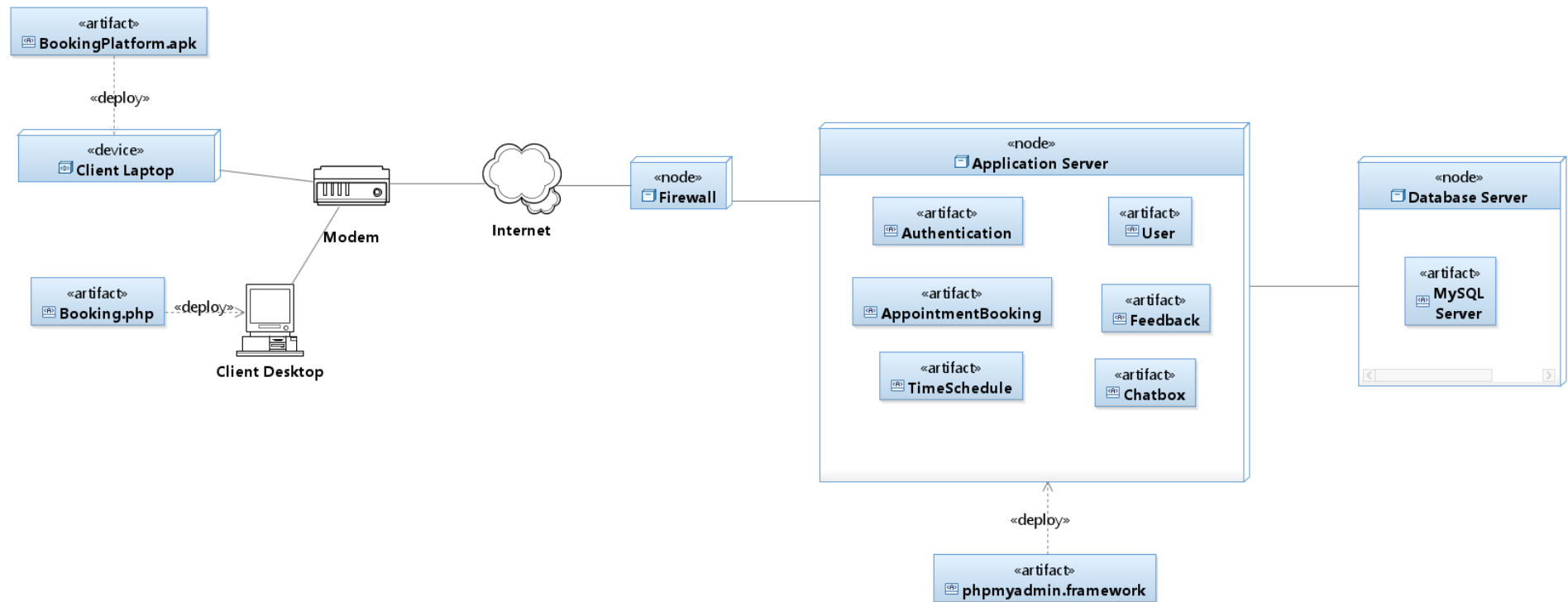


Diagram 5: Deployment Diagram

## 4.6 Chapter Summary and Evaluation

*Chapter 4 had discussed the design aspects of the online academic staff booking appointment platform proposed in this framework. User interface(UI) design with explanation is written in the first section, while the second section describes the data design of the system needed. Furthermore, report design, process design and the software architecture design had been included in this chapter.*

*The next chapter will discuss the implementation and testing of systems where test cases are created to test the coding written. This is to ensure the system is working correctly as expected and performs all the functions as required.*

## Chapter 5

# Implementation and Testing

## 5 Implementation and Testing

Chapter 5 had included the implementation method of the system and the testing aspects of the online academic staff booking appointment platform. The testing aspects covered the testing strategy that is applied on the system. Moreover, test plan and test cases are created to test whether the system is able to function correctly as expected.

### 5.1 Implementation

Implementation of a system is the process of installing the software and hardware to the intended operating environment. The software required is Microsoft Visual Studio 2017 and XAMPP local server. Microsoft Visual Studio 2017 is chosen instead of Microsoft Visual Studio 2019 due to the version 2017 being more stable than the newly released version 2019. XAMPP local server is used as the MySQL database of the online academic staff booking appointment platform. Furthermore, the web-based platform is required to run on a web browser which the web browser selected to use is Google Chrome. Moreover, the web-based platform required internet connection during its operation hour to operate the XAMPP local server in a real-time manner.

### 5.2 Coding

In this framework, the online academic staff booking appointment platform is coded using C#. Besides that, Hyper Text Markup Language (HTML) and Cascading Style Sheet (CSS) are used to create and improve the appearance design of web pages. MySQL database language is also used to create the system database as mentioned in chapter 4.



## 5.3 Testing Strategies

Software testing is important to be conducted on a system because it is able to reduce the mistakes in software production before releasing the system to the end users (Institute of Electrical and Electronics Engineers (IEEE) Societies, 2013). The testing strategy that will be applied on the platform proposed is the five stages in the testing process which included unit testing, module testing, sub-system testing, system testing and acceptance testing. The five stages in the testing process will be described in detail in this section.

### 5.3.1 Five Stages in Testing Process

The five stages in the testing process will be carried out in the system in order to let the platform proposed meet the requirements and to discover faults of the system where the system is behaved in an incorrect way.

#### 1. Unit Testing

Unit testing is defined as a software testing level in which individual components or units are tested to validate each unit of software operates correctly as designed (Software Testing Fundamentals, 2020). A unit is the smallest testable part of a system. For example, the user login component and staff accept an appointment request component. In order to perform unit testing on the platform proposed, test cases are designed and unit testing is organized together with white box testing.

White box testing is a software testing technique which verifies the inputs and outputs by analyzing the software code structure (Guru99, 2020). Moreover, test cases designed included the test data, test condition, expected result and actual result. Thus, unit testing applied together with white box testing is able to derive test data by analyzing the code structure. This can guarantee a given level of test coverage on the test cases created and it is able to ensure the actual

testing result obtained from the platform proposed is the same with the expected result.

## **2. Module Testing**

Module testing is one of the software testing methods which checks for individual subprograms, an object class, or some looser collection of procedures in a program (Guru99, 2020). Hence, module testing is able to test the collection of dependent components instead of testing the whole software program at once.

Moreover, module testing will be tested by applying the testing technique, bottom up testing. Bottom up testing is an integration testing technique where testing is carried out with the lower level modules and works upwards until all modules are tested (Guru99, 2020). Module testing carried out with bottom-up testing technique enables the fault localization becomes easier and it is time saving since it does not need to wait for all modules to be completely developed which enables us to test for the significant modules in the first software development version.

## **3. Sub-system Testing**

Sub-system testing is the third stage of the testing process. This phase involves testing on a collection of modules which have been integrated into sub-systems. It is aimed to test the specific function within a sub-system and check the correctness of data transmission between sub-system (Integrated Publishing, 2020). Thus, sub-system testing will also be tested by applying the software testing technique, interface testing.

Interface testing is a software testing technique which is to verify the communication between sub-system whether the sub-system is able to pass data correctly (Guru99, 2020). During the testing process of interface testing, we can verify whether the procedures encapsulated

in a sub-system can be successfully called by other sub-system. We can also check for message passing among sub-system by testing the correctness of data transmission when a sub-system request for services from other sub-system. Thus, interface testing will be applied together with sub-system testing to test the platform proposed in this framework to ensure the data transmission among sub-system are correct and accurate.

#### **4. System Testing**

System testing is a software testing level which tests the entire system as a full integration of each sub-system (Guru99, 2020). System testing is able to test the discrepancies between the platform proposed and the system specification before releasing the system. System testing will be carried out together with white box testing and black box testing.

Black box testing is a software testing technique in which the system functionality is tested by studying its inputs and related outputs without looking at the internal code structure (Guru99, 2020). However, white box testing verifies the system inputs and outputs by analyzing the code structure of the system (Guru99, 2020). System testing is applied with the black box testing technique and white box testing technique in order to ensure the system has a correct output and appropriate internal code structure.

#### **5. Acceptance Testing**

Acceptance testing is the final stage in the testing process where a system is tested for its acceptability by end users (Software Testing Fundamentals, 2020). Acceptance testing is also known as user acceptance testing where testing is performed by the end users of the software (Software Testing Fundamentals, 2020).

## 5.4 Test Plan

Test plan is defined as a document that contains a detailed description of software testing scope, setting out standards for the testing process and organizing the testing activities for a set of test items (Software Testing Fundamentals, 2020).

There contents identified in this test plan are:

1. Testing Process
2. Requirement Traceability
3. Tested Items
4. Testing Schedule
5. Hardware and Software Requirements
6. Constraints

### 5.4.1 Testing Process

The testing process that will be carried out to test the system contain five stages which are :

1. Unit Testing
2. Module Testing
3. Sub-system Testing
4. System Testing
5. Acceptance Testing

The five stages in the testing process had been defined in the last section, 5.3 Testing Strategies. These five stages in the testing process enable us to find the defects, correct the defects and also improve the system quality that is proposed in this framework.

### 5.4.2 Requirement Traceability

Requirement traceability is defined as the ability to track the requirements and check whether the current project requirements are met (Guru99, 2020). In a project, a plenty of test cases will be created to test the functionality and specification of the system. Hence, requirement traceability is important which is used to trace user requirements with test cases to ensure all system functionality is checked during software testing phase (Guru99, 2020). All test cases had been assigned a specific test case ID and each test case is categorized according to their requirement specification. This is to help the system testers to understand better on different levels of testing activities.

In this framework, the requirement traceability included several parameters as listed below.

1. Requirement Category (Req\_Category)
2. Requirement ID (Req\_ID)
3. Requirement Description (Req\_Desc)
4. Test Case ID (Test\_ID)

5. Test Case Title (Test\_Title)
6. Test Case Status (Test\_Status)

The requirement traceability of testing activities in this project is shown in table 1 below.

| Req_Category                             | Req_ID  | Req_Desc                                                          | Test_ID                      | Test_Title                                                              | Test_Status |
|------------------------------------------|---------|-------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------|-------------|
| Authentication module test category      | Auth_TC | Test login and logout of the platform.                            | Valid_Login_01               | Login using valid student login details                                 | Pass        |
|                                          |         |                                                                   | Valid_Login_02               | Login using valid staff login details                                   | Pass        |
|                                          |         |                                                                   | Invalid_Login_01             | Login using valid student ID but invalid password                       | Pass        |
|                                          |         |                                                                   | Invalid_Login_02             | Login using valid staff ID but invalid password                         | Pass        |
|                                          |         |                                                                   | Logout_01                    | Log out of current user account                                         | Pass        |
| Appointment booking module test category | Book_TC | Test make appointment requests and update the appointment record. | Valid_BookAppointment_01     | Make a valid appointment request                                        | Pass        |
|                                          |         |                                                                   | Invalid_BookAppointment_01   | Make an invalid appointment request                                     | Pass        |
|                                          |         |                                                                   | Valid_CancelAppointment_01   | Cancel an appointment                                                   | Pass        |
|                                          |         |                                                                   | Valid_AcceptAppointment_01   | Accept an appointment                                                   | Pass        |
|                                          |         |                                                                   | Invalid_AcceptAppointment_01 | Accept an appointment which the appointment status is invalid to accept | Pass        |
|                                          |         |                                                                   | Valid_RejectAppointment_01   | Reject an appointment                                                   | Pass        |

|                                    |             |                                                                                           |                                      |                                                                             |      |
|------------------------------------|-------------|-------------------------------------------------------------------------------------------|--------------------------------------|-----------------------------------------------------------------------------|------|
|                                    |             |                                                                                           | Valid_UpdateAppointmentVenue_01      | Update an appointment venue                                                 | Pass |
| Time schedule module test category | Schedule_TC | Test view staff available consultation hours and update the available consultation hours. | ViewStaffConsultationTime_01         | View a staff available consultation time                                    | Pass |
|                                    |             |                                                                                           | Valid_UpdateStaffConsultationTime_01 | Update staff available consultation time                                    | Pass |
|                                    |             |                                                                                           | Valid_UpdateStaffConsultationTime_01 | Update staff available consultation time                                    | Pass |
|                                    |             |                                                                                           | Valid_UpdateStaffConsultationTime_02 | Update staff available consultation time with more than 1 time slot         | Pass |
| Feedback module test category      | Feedback_TC | Test to write feedback.                                                                   | Valid_WriteFeedback_01               | Write and save the feedback for an appointment                              | Pass |
|                                    |             |                                                                                           | Invalid_WriteFeedback_01             | Write a feedback which the appointment status is invalid to accept feedback | Pass |
| Chat Box module test category      | Chat_TC     | Test to write a message.                                                                  | MessageStaff_01                      | Write a message to a staff                                                  | Pass |
|                                    |             |                                                                                           | ReplyStudent_01                      | Reply a message to a student                                                | Pass |

Table 1: Requirement Traceability of Testing Activities

### 5.4.3 Tested Items

In this section, the test items included program modules, user interface involved, and procedures written in the system. The tested items involved in this project are shown in table 2 below.

| Program Modules            | User Interfaces         | Procedure Written                                                                                      |
|----------------------------|-------------------------|--------------------------------------------------------------------------------------------------------|
| Authentication Module      | loginPage.aspx          | - Login()<br>- closeForm()                                                                             |
| Appointment Booking Module | BookAppointment1.aspx   | - getSem()<br>- viewSemCourse()                                                                        |
|                            | AppointmentDetails.aspx | - getAppointmentDetails()                                                                              |
|                            | Book2_StaffTime.aspx    | - viewStaffTime()<br>- viewAdvisorTime()                                                               |
|                            | Book3_Form.aspx         | - getStaffName()<br>- getStudName()<br>- validateNull()<br>- submitForm()                              |
|                            | CancelReason.aspx       | - cancelAppointment()                                                                                  |
| Time Schedule Module       | Homepage.aspx           | - viewActiveAppointment()<br>- activeNowBorder()<br>- viewHistoryAppointment()<br>- historyNowBorder() |
|                            | OtherStaff.aspx         | - getStaff()<br>- DataListPaging(DataTable dt)                                                         |
| Feedback Module            | Feedback.aspx           | - viewConsultRec()                                                                                     |
| Chat Box Module            | Book2_StaffTime.aspx    | - getChatID()<br>- createChatID()                                                                      |
|                            | ChatRoom.aspx           | - showChatHistory()                                                                                    |
|                            | ChatRoom2.aspx          | - showChatRoom()<br>- showChatHis()<br>- readMesDate()<br>- updateMes()                                |

Table 2: Tested Items in Student Appointment Booking Platform

| Program Modules            | User Interfaces         | Procedure Written                                                                                                                                                    |
|----------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authentication Module      | loginPage.aspx          | - Login()<br>- closeForm()                                                                                                                                           |
| Appointment Booking Module | AppointmentDetails.aspx | - getAppointmentDetails()                                                                                                                                            |
|                            | RejectedReason.aspx     | - rejectAppointment()                                                                                                                                                |
|                            | CancelReason.aspx       | - cancelAppointment()                                                                                                                                                |
| Time Schedule Module       | Homepage.aspx           | - viewActiveAppointment()<br>- activeNowBorder()<br>- viewHistoryAppointment()<br>- historyNowBorder()                                                               |
|                            | StaffAvailable.aspx     | - viewStaffTime()<br>- viewPreferVenue()                                                                                                                             |
|                            | UpdateVenue.aspx        | - getPreferVenue()<br>- venueNowBorder()<br>- getText()<br>- updatePreVenue()<br>- getAvaConsultTime()<br>- timeNowBorder()<br>- validateNull()<br>- updateConsult() |
| Feedback Module            | Feedback.aspx           | - viewConsultRec()                                                                                                                                                   |
|                            | feedbackForm.aspx       | - writeFeedback()                                                                                                                                                    |
| Chat Box Module            | ChatRoom.aspx           | - showChatHistory()                                                                                                                                                  |
|                            | ChatRoom2.aspx          | - showChatRoom()<br>- showChatHis()<br>- readMesDate()<br>- updateMes()                                                                                              |

Table 3: Tested Items in Staff Appointment Booking Platform

#### 5.4.4 Testing Schedule

Testing schedule for a project is important because it indicates the duration of a specific testing activity which is to ensure the project can be completed within a given time constraint. Hence, an overall test schedule is created for this project as shown in table below.

| Test Activities               | Duration (Week) |   |   |   |   |   |   |   |   |    |    |    |    |
|-------------------------------|-----------------|---|---|---|---|---|---|---|---|----|----|----|----|
|                               | 1               | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 |
| Analyze Software Requirements |                 |   |   |   |   |   |   |   |   |    |    |    |    |
| Test Plan Creation            |                 |   |   |   |   |   |   |   |   |    |    |    |    |
| Requirement Traceability      |                 |   |   |   |   |   |   |   |   |    |    |    |    |
| Test Specification Creation   |                 |   |   |   |   |   |   |   |   |    |    |    |    |
| Environment Setup             |                 |   |   |   |   |   |   |   |   |    |    |    |    |
| Test Cases Design             |                 |   |   |   |   |   |   |   |   |    |    |    |    |
| Test Execution (All 2 cycles) |                 |   |   |   |   |   |   |   |   |    |    |    |    |
| Completion of Report          |                 |   |   |   |   |   |   |   |   |    |    |    |    |

Table 4: Testing Schedule

#### 5.4.5 Hardware and Software Requirements

There are some hardware and software required in order to execute the online academic staff booking appointment platform proposed in this framework. Hence, the hardware and software required to test the platform proposed are listed as below.

##### Hardware Requirement (Laptop Requirements) :

- 2.7 GHz i5 or better processor
- 4 GB or bigger RAM
- 4 GB of available hard disk space (SSD will be better)
- Internet Connection

**Software Requirement :**

- Microsoft Visual Studio 2017
- XAMPP for Windows
- Windows 10
- Google Chrome (Web browser)

**5.4.6 Constraints**

There are few constraints faced in this project. The first constraint faced is the project has version incompatibility which the system may encounter issues when it is executed in a higher or lower version of Microsoft Visual Studio. The second constraint faced is that the XAMPP local server requires the devices that execute the platform must have the same Internet Protocol (IP) address in order to connect to the database of the system. Moreover, time constraint is faced which the project is required to complete the testing phase in a duration of 13 weeks as specified in the testing schedule.

## 5.5 Test Data

Before performing tests on a system, the input data types and requirements are needed to identify in order to apply them in the test case. In this section, the input data requirements to test the platform proposed is listed. Generally, there are two types of input data requirements we used in this project which are valid data and invalid data. Valid data is the legitimate data which should enable the software to work as expected and return a valid result (BBC Bitesize, 2020). However, invalid data is the wrong input data that is not accepted by the system and it will break the system behaviour by displaying a meaningful error message (BBC Bitesize, 2020).

Table below shows the input data requirement of the platform proposed in this framework.

| Module                 | Functions                 | Input Box Name | Input Requirement                                                                                                                                   |
|------------------------|---------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Authentication      | Login                     | txtStudentID   | <ul style="list-style-type: none"> <li>• Must be digit</li> <li>• No special characters</li> <li>• No letters</li> <li>• Input size is 7</li> </ul> |
|                        |                           | txtPW          | <ul style="list-style-type: none"> <li>• Maximum 20 values</li> </ul>                                                                               |
| 2. Appointment Booking | Make appointment requests | txtTitle       | <ul style="list-style-type: none"> <li>• None</li> </ul>                                                                                            |
|                        |                           | txtReason      | <ul style="list-style-type: none"> <li>• None</li> </ul>                                                                                            |
| 3. Chat Box            | Send message              | txtWriteMes    | <ul style="list-style-type: none"> <li>• None</li> </ul>                                                                                            |

Table 5: Test Data in Student Appointment Booking Platform

| Module                 | Functions                            | Input Box Name | Input Requirement                                                                                                                                   |
|------------------------|--------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Authentication      | Login                                | txtStudentID   | <ul style="list-style-type: none"> <li>• Must be digit</li> <li>• No special characters</li> <li>• No letters</li> <li>• Input size is 4</li> </ul> |
|                        |                                      | txtPW          | <ul style="list-style-type: none"> <li>• Maximum 20 values</li> </ul>                                                                               |
| 2. Appointment Booking | Save appointment cancellation reason | txtReason      | <ul style="list-style-type: none"> <li>• None</li> </ul>                                                                                            |
|                        | Save appointment rejected reason     | txtReason      | <ul style="list-style-type: none"> <li>• None</li> </ul>                                                                                            |
| 3. Time Schedule       | Update Prefer Venue                  | txtVenue1      | <ul style="list-style-type: none"> <li>• None</li> </ul>                                                                                            |
|                        |                                      | txtVenue2      | <ul style="list-style-type: none"> <li>• None</li> </ul>                                                                                            |
|                        |                                      | txtVenue3      | <ul style="list-style-type: none"> <li>• None</li> </ul>                                                                                            |
|                        | Update Consultation Hours            | txt1Start1     | <ul style="list-style-type: none"> <li>• Must input in format “hh:mm AM/PM”</li> </ul>                                                              |
|                        |                                      | txt1End1       | <ul style="list-style-type: none"> <li>• Must input in format “hh:mm AM/PM”</li> </ul>                                                              |
|                        |                                      | txt1Start2     | <ul style="list-style-type: none"> <li>• Must input in format “hh:mm AM/PM”</li> </ul>                                                              |
|                        |                                      | txt1End2       | <ul style="list-style-type: none"> <li>• Must input in format “hh:mm AM/PM”</li> </ul>                                                              |
|                        |                                      | txt2Start1     | <ul style="list-style-type: none"> <li>• Must input in format “hh:mm AM/PM”</li> </ul>                                                              |
|                        |                                      | txt2End1       | <ul style="list-style-type: none"> <li>• Must input in format “hh:mm AM/PM”</li> </ul>                                                              |
|                        |                                      | txt2Start2     | <ul style="list-style-type: none"> <li>• Must input in format “hh:mm AM/PM”</li> </ul>                                                              |
|                        |                                      | txt2End2       | <ul style="list-style-type: none"> <li>• Must input in format “hh:mm AM/PM”</li> </ul>                                                              |
|                        |                                      | txt3Start1     | <ul style="list-style-type: none"> <li>• Must input in format “hh:mm AM/PM”</li> </ul>                                                              |
|                        |                                      | txt3End1       | <ul style="list-style-type: none"> <li>• Must input in format “hh:mm AM/PM”</li> </ul>                                                              |

|             |                |             |                                                                                      |
|-------------|----------------|-------------|--------------------------------------------------------------------------------------|
|             |                | txt3Start2  | <ul style="list-style-type: none"> <li>Must input in format “hh:mm AM/PM”</li> </ul> |
|             |                | txt3End2    | <ul style="list-style-type: none"> <li>Must input in format “hh:mm AM/PM”</li> </ul> |
| 4. Feedback | Write feedback | txtFeedback | <ul style="list-style-type: none"> <li>None</li> </ul>                               |
| 5. Chat Box | Send message   | txtWriteMes | <ul style="list-style-type: none"> <li>None</li> </ul>                               |

Table 5: Test Data in Student Appointment Booking Platform

## 5.6 Test Cases

Test cases are created based on a set of variables or conditions which help testers to determine whether a system functions correctly and satisfies the requirements (Software Testing Fundamentals, 2020). The elements and their description of the test case template used in this project is shown in table 5 below (Software Testing Fundamentals, 2020).

| Elements             | Description                                                                |
|----------------------|----------------------------------------------------------------------------|
| Test Case ID         | A unique key that specifically identifies a test case.                     |
| Related Requirement  | A unique requirement ID that a specific test case traces to.               |
| Precondition         | Conditions that must be fulfilled before executing the test case.          |
| Test Procedures      | A step-by-step procedure to execute a test case.                           |
| Test Data            | Sample or real data generated to conduct the test case.                    |
| Expected Result      | The result that probably gets from the test.                               |
| Actual Result        | The real result from the system is recorded after executing the test case. |
| Status (Pass / Fail) | The test case status whether it is pass or fail.                           |
| Notes                | Remarks or comments on the test case or the test execution.                |
| Test Designed By     | The author name who created the test case.                                 |
| Test Designed Date   | The date of the test case is created.                                      |

|                     |                                                               |
|---------------------|---------------------------------------------------------------|
| Test Executed By    | The tester name who executes the test case.                   |
| Test Execution Date | The data of the test case is executed.                        |
| Postcondition       | Conditions or final result get after executing the test case. |

Table 5: Test Cases

In this section, test cases will be created based on the modules of the online academic staff booking appointment platform. There are five essential modules for the web-based online booking appointment platform which included:

- Authentication module
- Appointment Booking module
- Time schedule module
- Feedback module
- Chatbox module

### 5.6.1 Test Cases for Authentication Module

Test cases created for the authentication module is to verify a user login details and redirect the users to the homepage of the system when they successfully login to their user account.

#### Test Case 1 :

|                                                                                                                 |                                           |
|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Test Case ID :</b> Valid_Login_01                                                                            | <b>Test Designed By :</b> Chew Jane       |
| <b>Test Priority (Low / Medium / High) :</b> High                                                               | <b>Test Designed Date :</b> 22 May 2020   |
| <b>Module Name :</b> Authentication Module                                                                      | <b>Test Executed By :</b> Chew Jane       |
| <b>Test Title :</b> Login using valid student login details                                                     | <b>Test Execution Date :</b> 25 June 2020 |
| <b>Description :</b> Test the platform able to let students login to student accounts with valid login details. |                                           |

**Pre-conditions :** User has a valid student account.

| Step | Test Steps                                 | Test Data  | Expected Results                                                                                                               | Actual Result                                                                                                               | Status (Pass/Fail) | Notes |
|------|--------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|--------------------|-------|
| 1.   | Click on the student ID text box provided. |            |                                                                                                                                |                                                                                                                             |                    |       |
| 2.   | Enter a valid student ID number.           | "2000000"  |                                                                                                                                |                                                                                                                             |                    |       |
| 3.   | Click on the password text box provided.   |            |                                                                                                                                |                                                                                                                             |                    |       |
| 4.   | Enter a valid password.                    | "JC123456" |                                                                                                                                |                                                                                                                             |                    |       |
| 5.   | Click the login button.                    |            | System will verify the student login details. The user is verified and successfully logged in and is redirected to the student | System verifies the student login details. The student is logged in successfully and is redirected to the student homepage. | Pass               |       |

---

|  |  |  |           |  |  |  |
|--|--|--|-----------|--|--|--|
|  |  |  | homepage. |  |  |  |
|--|--|--|-----------|--|--|--|

|                                                                                     |
|-------------------------------------------------------------------------------------|
| <b>Post - conditions</b> : Student is successfully logged into the student account. |
|-------------------------------------------------------------------------------------|

## Test Case 2 :

|                                                                                                            |                                           |
|------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Test Case ID :</b> Valid_Login_02                                                                       | <b>Test Designed By :</b> Chew Jane       |
| <b>Test Priority (Low / Medium / High) :</b> High                                                          | <b>Test Designed Date :</b> 22 May 2020   |
| <b>Module Name :</b> Authentication Module                                                                 | <b>Test Executed By :</b> Chew Jane       |
| <b>Test Title :</b> Login using valid staff login details                                                  | <b>Test Execution Date :</b> 25 June 2020 |
| <b>Description :</b> Test the platform able to let staff login to staff accounts with valid login details. |                                           |

**Pre-conditions :** User has a valid staff account.

| Step | Test Steps                               | Test Data  | Expected Results                                                                                                                     | Actual Result                                                                                                       | Status (Pass/Fail) | Notes |
|------|------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------|-------|
| 1.   | Click on the staff ID text box provided. |            |                                                                                                                                      |                                                                                                                     |                    |       |
| 2.   | Enter a valid staff ID number.           | "1000"     |                                                                                                                                      |                                                                                                                     |                    |       |
| 3.   | Click on the password text box provided. |            |                                                                                                                                      |                                                                                                                     |                    |       |
| 4.   | Enter a valid password.                  | "JC123456" |                                                                                                                                      |                                                                                                                     |                    |       |
| 5.   | Click the login button.                  |            | System will verify the staff login details. The user is verified and successfully logged in and is redirected to the staff homepage. | System verifies the staff login details. The staff is logged in successfully and is directed to the staff homepage. | Pass               |       |

**Post - conditions :** Staff is successfully logged in to the staff account.

## Test Case 3 :

|                                                                                                                                                                                |                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Test Case ID :</b> Invalid_Login_01                                                                                                                                         | <b>Test Designed By :</b> Chew Jane       |
| <b>Test Priority (Low / Medium / High) :</b> High                                                                                                                              | <b>Test Designed Date :</b> 22 May 2020   |
| <b>Module Name :</b> Authentication Module                                                                                                                                     | <b>Test Executed By :</b> Chew Jane       |
| <b>Test Title :</b> Login using valid student ID but invalid password                                                                                                          | <b>Test Execution Date :</b> 26 June 2020 |
| <b>Description :</b> Test the platform able to verify the invalid password and restrict the student from login to their student account even though the student ID is correct. |                                           |

**Pre-conditions :** User has a valid student account.

| Step | Test Steps                                 | Test Data | Expected Results                                                                                                                                            | Actual Result                                                                                                                                                                  | Status (Pass/Fail) | Notes |
|------|--------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------|
| 1.   | Click on the student ID text box provided. |           |                                                                                                                                                             |                                                                                                                                                                                |                    |       |
| 2.   | Enter a valid student ID number.           | "2000000" |                                                                                                                                                             |                                                                                                                                                                                |                    |       |
| 3.   | Click on the password text box provided.   |           |                                                                                                                                                             |                                                                                                                                                                                |                    |       |
| 4.   | Enter an invalid password.                 | "ABC123"  |                                                                                                                                                             |                                                                                                                                                                                |                    |       |
| 5.   | Click the login button.                    |           | System will verify the student login details. The student failed to login due to an unmatch of student ID and password. An error message will be displayed. | System verifies the student login details. The student failed to login to his student account. An error message is displayed that student ID or password entered is incorrect. | Pass               |       |

**Post - conditions :** Student failed to log in to the student account. The student is required to re-enter the student ID and password.

## Test Case 4 :

|                                                                                                                                                                          |                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Test Case ID :</b> Invalid_Login_02                                                                                                                                   | <b>Test Designed By :</b> Chew Jane       |
| <b>Test Priority (Low / Medium / High) :</b> High                                                                                                                        | <b>Test Designed Date :</b> 22 May 2020   |
| <b>Module Name :</b> Authentication Module                                                                                                                               | <b>Test Executed By :</b> Chew Jane       |
| <b>Test Title :</b> Login using valid staff ID but invalid password                                                                                                      | <b>Test Execution Date :</b> 26 June 2020 |
| <b>Description :</b> Test the platform able to verify the invalid password and restrict the staff from login to their staff account even though the staff ID is correct. |                                           |

**Pre-conditions :** User has a valid staff account.

| Step | Test Steps                               | Test Data | Expected Results                                                                                                                                      | Actual Result                                                                                                                                                          | Status (Pass/Fail) | Notes |
|------|------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------|
| 1.   | Click on the staff ID text box provided. |           |                                                                                                                                                       |                                                                                                                                                                        |                    |       |
| 2.   | Enter a valid staff ID number.           | "1000"    |                                                                                                                                                       |                                                                                                                                                                        |                    |       |
| 3.   | Click on the password text box provided. |           |                                                                                                                                                       |                                                                                                                                                                        |                    |       |
| 4.   | Enter an invalid password.               | "ABC123"  |                                                                                                                                                       |                                                                                                                                                                        |                    |       |
| 5.   | Click the login button.                  |           | System will verify the staff login details. The staff failed to login due to an unmatch of staff ID and password. An error message will be displayed. | System verifies the staff login details. The staff failed to login to his staff account. An error message is displayed that staff ID or password entered is incorrect. | Pass               |       |

**Post - conditions :** Staff failed to log in to the staff account. The staff is required to re-enter the staff ID and password.

Test Case 5 :

|                                                         |                                           |
|---------------------------------------------------------|-------------------------------------------|
| <b>Test Case ID :</b> Logout_01                         | <b>Test Designed By :</b> Chew Jane       |
| <b>Test Priority (Low / Medium / High) :</b> Low        | <b>Test Designed Date :</b> 22 May 2020   |
| <b>Module Name :</b> Authentication Module              | <b>Test Executed By :</b> Chew Jane       |
| <b>Test Title :</b> Log out of current user account     | <b>Test Execution Date :</b> 26 June 2020 |
| <b>Description :</b> Let users log out of their account |                                           |

**Pre-conditions :** User has a login into their student or staff account successfully.

| Step | Test Steps                             | Test Data | Expected Results                                           | Actual Result                                                     | Status (Pass/Fail) | Notes |
|------|----------------------------------------|-----------|------------------------------------------------------------|-------------------------------------------------------------------|--------------------|-------|
| 1.   | Point on the down arrow on the header. |           | A drop down menu with option "Log Out" should be displayed | A drop down menu with option "Log Out" is displayed successfully. | Pass               |       |
| 2.   | Click "Log Out".                       |           | Users should be redirected to the login page.              | Users are redirected to the login page.                           | Pass               |       |

**Post - conditions :** User is successfully logged out from their user account.

### 5.6.2 Test Cases for Appointment Booking Module

Test cases created for the appointment booking module is to ensure the platform developed is able to let students make appointment requests with the academic staff and the system should allow a staff to accept or reject an appointment request.

#### Test Case 1 :

|                                                                                                                                                                    |                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Test Case ID :</b> Valid_BookAppointment_01                                                                                                                     | <b>Test Designed By :</b> Chew Jane       |
| <b>Test Priority (Low / Medium / High) :</b> High                                                                                                                  | <b>Test Designed Date :</b> 23 May 2020   |
| <b>Module Name :</b> Appointment Booking Module                                                                                                                    | <b>Test Executed By :</b> Chew Jane       |
| <b>Test Title :</b> Make a valid appointment request                                                                                                               | <b>Test Execution Date :</b> 27 June 2020 |
| <b>Description :</b> Test the platform able to let students make an appointment request with a tutor of current studying semester using all valid booking details. |                                           |

**Pre-conditions :** User has a valid student account and had login to their student account successfully.

| Step | Test Steps                                                   | Test Data         | Expected Results                                                                              | Actual Result                                                                            | Status (Pass/Fail) | Notes |
|------|--------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------|-------|
| 1.   | Click on the “Book Appointment” button in the side bar menu. |                   | System should display the student’s current studying semester, courses and the related staff. | System displayed the student's current studying semester, courses and the related staff. | Pass               |       |
| 2.   | Select a tutor name.                                         | “Mr. Ocean Cheng” | System should display the selected staff available consultation hours.                        | The available consultation hours of selected staff is displayed.                         | Pass               |       |
| 3.   | Click on the “Make Appointment” button.                      |                   | System should display the appointment booking form with the selected staff                    | The appointment booking form is displayed with the selected staff name                   | Pass               |       |

|     |                                           |              |                                                                                                            |                                                          |      |  |
|-----|-------------------------------------------|--------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------|--|
|     |                                           |              | name.                                                                                                      | correctly.                                               |      |  |
| 4.  | Click on the booking date button.         |              | System should display a calendar form for students to choose the appointment date.                         | The calendar is displayed.                               | Pass |  |
| 5.  | Select a desired appointment date.        | "21/07/2020" | System should display the selected date in the textbox in format dd/mm/yyyy together with the day of week. | The selected date is displayed in the textbox correctly. | Pass |  |
| 6.  | Click on the start time text box.         |              | System should display a drop down list for students to choose the starting time of appointment.            | Drop down list is displayed.                             | Pass |  |
| 7.  | Select a desired appointment start time.  | "3.00PM"     | The selected time should display in the drop down list box.                                                | The selected time is displayed in the box given.         | Pass |  |
| 8.  | Click on the end time text box.           |              | System should display a drop down list for students to choose the ending time of appointment.              | Drop down list is displayed                              | Pass |  |
| 9.  | Select a desired appointment end time.    | "3.30PM"     | The selected time should display in the drop down list box.                                                | The selected time is displayed in the box given.         | Pass |  |
| 10. | Click on the number of students text box. |              | System should display a drop down list for students to choose values 1                                     | The drop down list with values 1 to 4 is shown.          | Pass |  |

|     |                                                            |                                                                     |                                                                                        |                                                          |      |  |
|-----|------------------------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------|------|--|
|     |                                                            |                                                                     | to 4.                                                                                  |                                                          |      |  |
| 11. | Select the number of students involved in the appointment. | "1"                                                                 |                                                                                        |                                                          |      |  |
| 12. | Click on the title text box.                               |                                                                     |                                                                                        |                                                          |      |  |
| 13. | Enter the title for requesting the appointment.            | "FYP C5 Check"                                                      |                                                                                        |                                                          |      |  |
| 14. | Click on the reason text box.                              |                                                                     |                                                                                        |                                                          |      |  |
| 15. | Enter the reason for requesting the appointment.           | "Greetings, Sir. I would like to get feedback on my FYP chapter 5." |                                                                                        |                                                          |      |  |
| 16. | Click on the submit button.                                |                                                                     | The system should display a message that the appointment booking request is submitted. | Message "The request is sent successfully" is displayed. | Pass |  |

**Post - conditions :** The appointment request form is saved into the database and the selected staff is able to view and update the appointment.

## Test Case 2 :

|                                                                                                                                                                           |                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Test Case ID :</b> Invalid_BookAppointment_01                                                                                                                          | <b>Test Designed By :</b> Chew Jane       |
| <b>Test Priority (Low / Medium / High) :</b> Medium                                                                                                                       | <b>Test Designed Date :</b> 23 May 2020   |
| <b>Module Name :</b> Appointment Booking Module                                                                                                                           | <b>Test Executed By :</b> Chew Jane       |
| <b>Test Title :</b> Make an invalid appointment request                                                                                                                   | <b>Test Execution Date :</b> 27 June 2020 |
| <b>Description :</b> Test the platform able to let students make an appointment request with an invalid booking date that is out of the staff available consultation day. |                                           |

**Pre-conditions :** User has a valid student account and had login to their student account successfully.

| Step | Test Steps                                                   | Test Data         | Expected Results                                                                   | Actual Result                                                                     | Status (Pass/Fail) | Notes |
|------|--------------------------------------------------------------|-------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------|-------|
| 1.   | Click on the “Book Appointment” button in the side bar menu. |                   |                                                                                    |                                                                                   |                    |       |
| 2.   | Select a tutor name.                                         | “Mr. Ocean Cheng” | System should display the selected staff available consultation hours.             | The available consultation hours of selected staff is displayed.                  | Pass               |       |
| 3.   | Click on the “Make Appointment” button.                      |                   | System should display the appointment booking form with the selected staff name.   | The appointment booking form is displayed with the selected staff name correctly. | Pass               |       |
| 4.   | Click on the booking date button.                            |                   | System should display a calendar form for students to choose the appointment date. | The calendar is displayed.                                                        | Pass               |       |
| 5.   | Select an invalid appointment date.                          | “11/07/2020”      | System should display an error message that                                        | Error message “Day selected is not under                                          | Pass               |       |

|  |  |  |                                                                            |                                    |  |  |
|--|--|--|----------------------------------------------------------------------------|------------------------------------|--|--|
|  |  |  | indicates the day selected is out of the staff available consultation day. | staff available consultation day.” |  |  |
|--|--|--|----------------------------------------------------------------------------|------------------------------------|--|--|

**Post - conditions :** The system is able to detect the selected appointment booking date is out of the range of staff available on consultation day and error message is displayed.

## Test Case 3 :

|                                                                                    |                                           |
|------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Test Case ID :</b> Valid_CancelAppointment_01                                   | <b>Test Designed By :</b> Chew Jane       |
| <b>Test Priority (Low / Medium / High) :</b> Low                                   | <b>Test Designed Date :</b> 24 May 2020   |
| <b>Module Name :</b> Appointment Booking Module                                    | <b>Test Executed By :</b> Chew Jane       |
| <b>Test Title :</b> Cancel an appointment                                          | <b>Test Execution Date :</b> 28 June 2020 |
| <b>Description :</b> Test to let students cancel an appointment that is requested. |                                           |

**Pre-conditions :** User has a valid student account and had login to their student account successfully.

| Step | Test Steps                                                      | Test Data                                       | Expected Results                                                             | Actual Result                                                    | Status (Pass/Fail) | Notes |
|------|-----------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------|--------------------|-------|
| 1.   | Click an appointment that is desired to cancel on the homepage. |                                                 | The appointment details will be displayed.                                   | The appointment detail is shown.                                 | Pass               |       |
| 2.   | Click the “Cancel Appointment” button.                          |                                                 | System should display a confirmation message.                                | A confirmation message is displayed.                             | Pass               |       |
| 3.   | Click “Yes” to cancel the appointment.                          |                                                 | System should prompt for appointment cancellation reasons.                   | Textbox to fill cancellation reasons is displayed.               | Pass               |       |
| 4.   | Enter the reason for appointment cancellation.                  | “Sorry, I had a replacement class on that day.” |                                                                              |                                                                  |                    |       |
| 5.   | Click the “Ok” button.                                          |                                                 | System should save the record and display a successful cancellation message. | Message “The appointment is cancelled successfully” is displayed | Pass               |       |

**Post - conditions :** The system updates the appointment status and cancellation reason into the database and the student is then redirected back to homepage.

## Test Case 4 :

|                                                                         |                                           |
|-------------------------------------------------------------------------|-------------------------------------------|
| <b>Test Case ID :</b> Valid_AcceptAppointment_01                        | <b>Test Designed By :</b> Chew Jane       |
| <b>Test Priority (Low / Medium / High) :</b> High                       | <b>Test Designed Date :</b> 24 May 2020   |
| <b>Module Name :</b> Appointment Booking Module                         | <b>Test Executed By :</b> Chew Jane       |
| <b>Test Title :</b> Accept an appointment                               | <b>Test Execution Date :</b> 28 June 2020 |
| <b>Description :</b> Test to let staff accept an appointment requested. |                                           |

**Pre-conditions :** Staff had login to their account successfully and the staff had received an appointment booking request.

| Step | Test Steps                                                      | Test Data | Expected Results                                                                         | Actual Result                                                                           | Status (Pass/Fail) | Notes |
|------|-----------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------|-------|
| 1.   | Click an appointment that is desired to accept on the homepage. |           | The appointment details will be displayed.                                               | The appointment detail is shown.                                                        | Pass               |       |
| 2.   | Click the “PENDING” status shown.                               |           | System should display a message to let staff choose to accept or reject the appointment. | A message is displayed for staff to choose to accept or reject the appointment request. | Pass               |       |
| 3.   | Click “Yes” to accept the appointment.                          |           | System should display a message that the appointment is accepted.                        | Message “Appointment Accepted” is displayed.                                            | Pass               |       |

**Post - conditions :** The system updates the appointment status into the database and the staff is then redirected back to homepage.

## Test Case 5 :

|                                                                                                                   |                                           |
|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Test Case ID :</b> Invalid_AcceptAppointment_01                                                                | <b>Test Designed By :</b> Chew Jane       |
| <b>Test Priority (Low / Medium / High) :</b> Medium                                                               | <b>Test Designed Date :</b> 24 May 2020   |
| <b>Module Name :</b> Appointment Booking Module                                                                   | <b>Test Executed By :</b> Chew Jane       |
| <b>Test Title :</b> Accept an appointment which the appointment status is invalid to accept                       | <b>Test Execution Date :</b> 28 June 2020 |
| <b>Description :</b> Test to let staff accept an appointment requested which the appointment status is cancelled. |                                           |

**Pre-conditions :** Staff had login to their account successfully and the staff had received an appointment booking request which was cancelled by the student.

| Step | Test Steps                                                       | Test Data | Expected Results                           | Actual Result                    | Status (Pass/Fail) | Notes                                                                                                 |
|------|------------------------------------------------------------------|-----------|--------------------------------------------|----------------------------------|--------------------|-------------------------------------------------------------------------------------------------------|
| 1.   | Click an appointment that status is "CANCELLED" on the homepage. |           | The appointment details will be displayed. | The appointment detail is shown. | Pass               |                                                                                                       |
| 2.   | Click the "CANCELLED" status shown.                              |           | No action should be taken.                 | No action or changes are taken.  | Pass               | Because the appointment is already cancelled. Hence, staff is disable from accepting or rejecting it. |

## Test Case 6 :

|                                                                         |                                           |
|-------------------------------------------------------------------------|-------------------------------------------|
| <b>Test Case ID :</b> Valid_RejectAppointment_01                        | <b>Test Designed By :</b> Chew Jane       |
| <b>Test Priority (Low / Medium / High) :</b> High                       | <b>Test Designed Date :</b> 25 May 2020   |
| <b>Module Name :</b> Appointment Booking Module                         | <b>Test Executed By :</b> Chew Jane       |
| <b>Test Title :</b> Reject an appointment                               | <b>Test Execution Date :</b> 29 June 2020 |
| <b>Description :</b> Test to let staff reject an appointment requested. |                                           |

**Pre-conditions :** Staff had login to their account successfully and the staff had received an appointment booking request.

| Step | Test Steps                                                      | Test Data                                                                          | Expected Results                                                                         | Actual Result                                                                           | Status (Pass/Fail) | Notes |
|------|-----------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------|-------|
| 1.   | Click an appointment that is desired to accept on the homepage. |                                                                                    | The appointment details will be displayed.                                               | The appointment detail is shown.                                                        | Pass               |       |
| 2.   | Click the "PENDING" status shown.                               |                                                                                    | System should display a message to let staff choose to accept or reject the appointment. | A message is displayed for staff to choose to accept or reject the appointment request. | Pass               |       |
| 3.   | Click "No" to reject the appointment.                           |                                                                                    | System should display a textbox for staff to enter the rejected reason.                  | Textbox is displayed for staff to enter rejected reason.                                | Pass               |       |
| 4.   | Enter appointment rejected reason.                              | "I had a meeting at that time. Please book the appointment another time again. TQ" |                                                                                          |                                                                                         |                    |       |
| 5.   | Click the "OK" button.                                          |                                                                                    | The system should display a message that the appointment is rejected.                    | Message "Appointment Rejected" is displayed.                                            | Pass               |       |

**Post - conditions :** The system updates the appointment status into the database and the staff is then redirected back to homepage.

Test Case 7 :

|                                                                    |                                           |
|--------------------------------------------------------------------|-------------------------------------------|
| <b>Test Case ID :</b><br>Valid_UpdateAppointmentVenue_01           | <b>Test Designed By :</b> Chew Jane       |
| <b>Test Priority (Low / Medium / High) :</b> Medium                | <b>Test Designed Date :</b> 25 May 2020   |
| <b>Module Name :</b> Appointment Booking Module                    | <b>Test Executed By :</b> Chew Jane       |
| <b>Test Title :</b> Update an appointment venue                    | <b>Test Execution Date :</b> 29 June 2020 |
| <b>Description :</b> Test to let staff update an appointment venue |                                           |

**Pre-conditions :** Staff had login to their account successfully and the staff had received an appointment booking request.

| Step | Test Steps                                                            | Test Data       | Expected Results                                                                   | Actual Result                                                                      | Status (Pass/Fail) | Notes |
|------|-----------------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------|-------|
| 1.   | Click an appointment that is desired to update the appointment venue. |                 | The appointment details will be displayed.                                         | The appointment detail is shown.                                                   | Pass               |       |
| 2.   | Click the “ - ” shown in Venue.                                       |                 | System should display a drop down list to let staff choose a booking venue.        | Drop down list is shown for staff to choose a booking venue.                       | Pass               |       |
| 3.   | Select a preferred booking venue from the drop down list.             | “Block B, B112” | System should display the selected venue in the field given.                       | The selected booking venue is displayed in the field given.                        | Pass               |       |
| 4.   | Click the “Update” button beside the venue selected.                  |                 | The system should display a message that the appointment booking venue is updated. | Message “Appointment booking venue is updated” is shown together with the selected | Pass               |       |

---

|  |  |  |  |                |  |  |
|--|--|--|--|----------------|--|--|
|  |  |  |  | booking venue. |  |  |
|--|--|--|--|----------------|--|--|

**Post - conditions :** The system updates the appointment booking venue into the database and the staff is then redirected back to homepage.

### 5.6.3 Test Cases for Time Schedule Module

Time schedule module contains test cases that are designed to ensure the platform developed is able to let students view staff available consultation time and the staff can update their available consultation time too.

#### Test Case 1 :

|                                                                        |                                           |
|------------------------------------------------------------------------|-------------------------------------------|
| <b>Test Case ID :</b> ViewStaffConsultationTime_01                     | <b>Test Designed By :</b> Chew Jane       |
| <b>Test Priority (Low / Medium / High) :</b> High                      | <b>Test Designed Date :</b> 26 May 2020   |
| <b>Module Name :</b> Time Schedule Module                              | <b>Test Executed By :</b> Chew Jane       |
| <b>Test Title :</b> View a staff available consultation time           | <b>Test Execution Date :</b> 30 June 2020 |
| <b>Description :</b> Test to view a staff available consultation time. |                                           |

**Pre-conditions :** User has a valid student account and had login to their student account successfully.

| Step | Test Steps                                                   | Test Data               | Expected Results                                                                                               | Actual Result                                                                                                  | Status (Pass/Fail) | Notes |
|------|--------------------------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------|-------|
| 1.   | Click on the “Book Appointment” button in the side bar menu. |                         | System should navigate to the page that displays the student currently taking courses and the staff in charge. | System should navigate to the page that displays the student currently taking courses and the staff in charge. | Pass               |       |
| 2.   | Select a tutor name.                                         | “Mr. Hetty Fitzpatrick” | System should display the staff’s available consultation hours.                                                | The available consultation hours of the staff is displayed.                                                    | Pass               |       |

**Post - conditions :** The system displays the selected staff available consultation hours.

## Test Case 2 :

|                                                                                                      |                                           |
|------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Test Case ID :</b><br>Valid_UpdateStaffConsultationTime_01                                        | <b>Test Designed By :</b> Chew Jane       |
| <b>Test Priority (Low / Medium / High) :</b> Medium                                                  | <b>Test Designed Date :</b> 26 May 2020   |
| <b>Module Name :</b> Appointment Booking Module                                                      | <b>Test Executed By :</b> Chew Jane       |
| <b>Test Title :</b> Update staff available consultation time                                         | <b>Test Execution Date :</b> 30 June 2020 |
| <b>Description :</b> Test to let staff update available consultation time for 1 day and 1 time slot. |                                           |

**Pre-conditions :** User has a valid staff account and had login to their staff account successfully.

| Step | Test Steps                                                       | Test Data  | Expected Results                                                                                                                  | Actual Result                                                                  | Status (Pass/Fail) | Notes |
|------|------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------|-------|
| 1.   | Click on the staff profile picture in the header.                |            | System should navigate to the page with the page title "Staff Profile".                                                           | The "Staff Profile" page is displayed.                                         | Pass               |       |
| 2.   | Click on the button "Update Consultation Time".                  |            | System should display the page with the page title "Update Profile" and current available consultation hours should be displayed. | The "Update Profile" page with content update consultation hours is displayed. | Pass               |       |
| 3.   | Select an available consultation day.                            | "Thursday" | The system should display the selected day in the field given.                                                                    | The day selected is displayed in the field given.                              | Pass               |       |
| 4.   | Enter start time of available consultation hours in time slot 1. | "01:00 PM" |                                                                                                                                   |                                                                                |                    |       |
| 5.   | Enter end time of available consultation hours in time slot 1.   | "03:00PM"  |                                                                                                                                   |                                                                                |                    |       |

|    |                                               |  |                                                            |                                                                                                         |      |  |
|----|-----------------------------------------------|--|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------|--|
| 6. | Click the “Update Consultation Hours” button. |  | The system should display a message “Update Successfully”. | The message “Update Successfully” is displayed together with the selected available consultation hours. | Pass |  |
|----|-----------------------------------------------|--|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------|--|

**Post - conditions :** The staff available consultation hours is updated and saved in the database.

## Test Case 3 :

|                                                                                                       |                                           |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Test Case ID :</b><br>Valid_UpdateStaffConsultationTime_02                                         | <b>Test Designed By :</b> Chew Jane       |
| <b>Test Priority (Low / Medium / High) :</b> Medium                                                   | <b>Test Designed Date :</b> 26 May 2020   |
| <b>Module Name :</b> Appointment Booking Module                                                       | <b>Test Executed By :</b> Chew Jane       |
| <b>Test Title :</b> Update staff available consultation time with more than 1 time slot               | <b>Test Execution Date :</b> 30 June 2020 |
| <b>Description :</b> Test to let staff update available consultation time for 2 day and 2 time slots. |                                           |

**Pre-conditions :** User has a valid staff account and had login to their staff account successfully.

| Step | Test Steps                                                       | Test Data  | Expected Results                                                                                                                  | Actual Result                                                                  | Status (Pass/Fail) | Notes |
|------|------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------|-------|
| 1.   | Click on the staff profile picture in the header.                |            | System should navigate to the page with the page title "Staff Profile".                                                           | The "Staff Profile" page is displayed.                                         | Pass               |       |
| 2.   | Click on the button "Update Consultation Time".                  |            | System should display the page with the page title "Update Profile" and current available consultation hours should be displayed. | The "Update Profile" page with content update consultation hours is displayed. | Pass               |       |
| 3.   | Select an available consultation day.                            | "Thursday" | The system should display the selected day in the field given.                                                                    | The day selected is displayed in the field given.                              | Pass               |       |
| 4.   | Enter start time of available consultation hours in time slot 1. | "09:00 AM" |                                                                                                                                   |                                                                                |                    |       |
| 5.   | Enter end time of available consultation hours in time slot 1.   | "12:00PM"  |                                                                                                                                   |                                                                                |                    |       |

|     |                                                                  |            |                                                                                                      |                                                                    |      |  |
|-----|------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|------|--|
| 6.  | Click the “+ Add Time Slot” button.                              |            | The system should display the second time slot field for staff to input under the same day selected. | Time slot 2 is displayed under the same consultation day selected. | Pass |  |
| 7.  | Enter start time of available consultation hours in time slot 1. | “01:00 PM” |                                                                                                      |                                                                    |      |  |
| 8.  | Enter end time of available consultation hours in time slot 1.   | “03:00PM”  |                                                                                                      |                                                                    |      |  |
| 9.  | Select the second available consultation day.                    | “Friday”   | The system should display the selected day in the field given.                                       | The day selected is displayed in the field given.                  | Pass |  |
| 10. | Enter start time of available consultation hours in time slot 1. | “09:30 AM” |                                                                                                      |                                                                    |      |  |
| 11. | Enter end time of available consultation hours in time slot 1.   | “11:30PM”  |                                                                                                      |                                                                    |      |  |
| 12. | Click the “+ Add Time Slot” button.                              |            | The system should display the second time slot field for staff to input under the same day selected. | Time slot 2 is displayed under the same consultation day selected. | Pass |  |
| 13. | Enter start time of available consultation hours in time slot 1. | “02:00 PM” |                                                                                                      |                                                                    |      |  |
| 14. | Enter end time of available consultation hours in time slot 1.   | “03:30PM”  |                                                                                                      |                                                                    |      |  |

|     |                                               |  |                                                            |                                                                                                         |      |  |
|-----|-----------------------------------------------|--|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------|--|
| 15. | Click the “Update Consultation Hours” button. |  | The system should display a message “Update Successfully”. | The message “Update Successfully” is displayed together with the selected available consultation hours. | Pass |  |
|-----|-----------------------------------------------|--|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------|--|

**Post - conditions :** The staff available consultation hours is updated and saved in the database.

### 5.6.4 Test Cases for Feedback Module

Feedback module developed in the online booking appointment platform is to let staff write feedback or a short conclusion of the appointment which can be used as a future appointment booking reference. Hence, the test case for the feedback module is to ensure the staff is allowed to write feedback for an appointment request.

#### Test Case 1 :

|                                                                           |                                           |
|---------------------------------------------------------------------------|-------------------------------------------|
| <b>Test Case ID :</b> Valid_WriteFeedback_01                              | <b>Test Designed By :</b> Chew Jane       |
| <b>Test Priority (Low / Medium / High) :</b> Medium                       | <b>Test Designed Date :</b> 27 May 2020   |
| <b>Module Name :</b> Feedback Module                                      | <b>Test Executed By :</b> Chew Jane       |
| <b>Test Title :</b> Write and save the feedback for an appointment        | <b>Test Execution Date :</b> 01 July 2020 |
| <b>Description :</b> Test to let staff write feedback for an appointment. |                                           |

**Pre-conditions :** Staff has logged into their account successfully and the staff had an appointment request that has been conducted.

| Step | Test Steps                                            | Test Data | Expected Results                                                                                     | Actual Result                                                      | Status (Pass/Fail) | Notes                                                     |
|------|-------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|--------------------|-----------------------------------------------------------|
| 1.   | Click on the “History” tab on the homepage.           |           | System should display all the appointment history records.                                           | The appointment history records for the staff are displayed.       | Pass               |                                                           |
| 2.   | Select a desired appointment to write a feedback.     |           | System should display a message to let staff choose to add feedback or view the appointment details. | A message is displayed with the title “Add Feedback Confirmation”. | Pass               | Only appointment status is “ACCEPTED” can write feedback. |
| 3.   | Click “Yes” on the message displayed to add feedback. |           | System should display a textbox for staff to enter the feedback.                                     | Textbox is displayed for staff to enter feedback.                  | Pass               |                                                           |

|    |                                     |                      |                                                                   |                                          |      |  |
|----|-------------------------------------|----------------------|-------------------------------------------------------------------|------------------------------------------|------|--|
| 4. | Enter feedback for the appointment. | “Test add feedback.” |                                                                   |                                          |      |  |
| 5. | Click the “OK” button.              |                      | The system should display a message that the feedback is written. | Message “Feedback Written” is displayed. | Pass |  |

**Post - conditions :** The system updates the feedback of the appointment into the database and the staff is then redirected back to homepage.

## Test Case 2 :

|                                                                                                 |                                           |
|-------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Test Case ID :</b> Invalid_WriteFeedback_01                                                  | <b>Test Designed By :</b> Chew Jane       |
| <b>Test Priority (Low / Medium / High) :</b> Medium                                             | <b>Test Designed Date :</b> 27 May 2020   |
| <b>Module Name :</b> Feedback Module                                                            | <b>Test Executed By :</b> Chew Jane       |
| <b>Test Title :</b> Write a feedback which the appointment status is invalid to accept feedback | <b>Test Execution Date :</b> 01 July 2020 |
| <b>Description :</b> Test to let staff write feedback for an appointment.                       |                                           |

**Pre-conditions :** Staff has logged into their account successfully and the staff had an appointment request that has been conducted.

| Step | Test Steps                                    | Test Data | Expected Results                                                      | Actual Result                                                                  | Status (Pass/Fail) | Notes                                                     |
|------|-----------------------------------------------|-----------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------|
| 1.   | Click on the "History" tab on the homepage.   |           | System should display all the appointment history records.            | The appointment history records for the staff are displayed.                   | Pass               |                                                           |
| 2.   | Select an appointment with status "REJECTED". |           | System will navigate to view the appointment details page as default. | The appointment details page is displayed and no any feedback action is taken. | Pass               | Only appointment status is "ACCEPTED" can write feedback. |

### 5.6.5 Test Cases for Chat Box Module

Test cases created for the chat box module is the platform developed should enable a student message to a staff and the staff can reply to the message received.

#### Test Case 1 :

|                                                               |                                           |
|---------------------------------------------------------------|-------------------------------------------|
| <b>Test Case ID :</b> MessageStaff_01                         | <b>Test Designed By :</b> Chew Jane       |
| <b>Test Priority (Low / Medium / High) :</b> Medium           | <b>Test Designed Date :</b> 28 May 2020   |
| <b>Module Name :</b> Chat Box Module                          | <b>Test Executed By :</b> Chew Jane       |
| <b>Test Title :</b> Write a message to a staff                | <b>Test Execution Date :</b> 02 July 2020 |
| <b>Description :</b> Test to let students to message a staff. |                                           |

**Pre-conditions :** Students had logged into their account successfully.

| Step | Test Steps                                                   | Test Data                                                 | Expected Results                                                       | Actual Result                                                    | Status (Pass/Fail) | Notes |
|------|--------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------|--------------------|-------|
| 1.   | Click on the “Book Appointment” button in the side bar menu. |                                                           |                                                                        |                                                                  |                    |       |
| 2.   | Select a tutor name.                                         | “Mr. Ocean Cheng”                                         | System should display the selected staff available consultation hours. | The available consultation hours of selected staff is displayed. | Pass               |       |
| 3.   | Click on the “Message him / her” button.                     |                                                           | System should display the chat room between the student and the staff. | The chat room is displayed.                                      | Pass               |       |
| 4.   | Write a message in the textbox given.                        | “Hi, Sir, I want to ask a question about the assignment.” |                                                                        |                                                                  |                    |       |

|    |                      |  |                                                        |                                                 |      |  |
|----|----------------------|--|--------------------------------------------------------|-------------------------------------------------|------|--|
| 5. | Click the send icon. |  | The message sent should be displayed in the chat room. | The message sent is displayed in the chat room. | Pass |  |
|----|----------------------|--|--------------------------------------------------------|-------------------------------------------------|------|--|

**Post - conditions :** The system saves the message sent into the database and the chat room is auto refreshed to show the new message.

## Test Case 2 :

|                                                                      |                                           |
|----------------------------------------------------------------------|-------------------------------------------|
| <b>Test Case ID :</b> ReplyStudent_01                                | <b>Test Designed By :</b> Chew Jane       |
| <b>Test Priority (Low / Medium / High) :</b> Medium                  | <b>Test Designed Date :</b> 28 May 2020   |
| <b>Module Name :</b> Chat Box Module                                 | <b>Test Executed By :</b> Chew Jane       |
| <b>Test Title :</b> Reply a message to a student                     | <b>Test Execution Date :</b> 02 July 2020 |
| <b>Description :</b> Test to let staff reply a message to a student. |                                           |

**Pre-conditions :** Staff had logged into their account successfully.

| Step | Test Steps                                      | Test Data | Expected Results                                                                              | Actual Result                                                                     | Status (Pass/Fail) | Notes |
|------|-------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------|-------|
| 1.   | Click on the Chat icon on the header.           |           | System should display all the students' profile photos that have been chatted with the staff. | The students' profile photo, name and last chat sentence are shown.               | Pass               |       |
| 2.   | Select a desired student to reply to a message. |           | System should display a chat room with all chat records between the student and the staff.    | A chat room with all chat records between the student and the staff is displayed. | Pass               |       |
| 3.   | Write a message in the textbox given.           | "Yes?"    |                                                                                               |                                                                                   |                    |       |
| 4.   | Click the send icon.                            |           | The message sent should be displayed in the chat room.                                        | The message sent is displayed in the chat room.                                   | Pass               |       |

**Post - conditions :** The system saves the message sent into the database and the chat room is auto refreshed to show the new message.

## 5.7 Chapter Summary and Evaluation

*Chapter 5 had discussed brief software requirements to implement the platform which the details of system deployment will be written in the next chapter. In this chapter, the software testing aspect is included which a detailed test plan is created and testing process is defined. Moreover, test data and test cases are created for each module to check the data requirements of each text box field and to ensure each unit in application is working properly as expected.*

*The next chapter, chapter 6 will discuss the system deployment which step by step guideline will be written for users can follow the guidance to install and deploy the online academic staff booking appointment platform.*

## Chapter 6

# **System Deployment**

## 6 System Deployment

Software deployment is defined as the processes needed for preparing a software application to run and operate in an ideal environment (Williams, 2020). In this chapter, we had included the methods and steps to deploy the online academic staff booking appointment platform for both student and staff versions.

### 6.1 Deployment Environment

In order to let the online academic staff booking appointment platform be used by all students and staff, the system is required to deploy in the desktop or laptop at TARUC institution as the server environment. Furthermore, the server environment is required to install software Microsoft Visual Studio of version 2017 or a higher software version. This is to reduce the version conflict among the Microsoft Visual Studio software and to ensure the system is able to run on the server environment smoothly.

Moreover, XAMPP local server is needed as the MySQL database management system for the online academic staff booking appointment platform. Apart from that, the web-based platform is required to run on a web browser which is preferable to use Google Chrome. Last but not least, the web-based platform required internet connection during its operation hour to operate the XAMPP local server in a real-time manner.

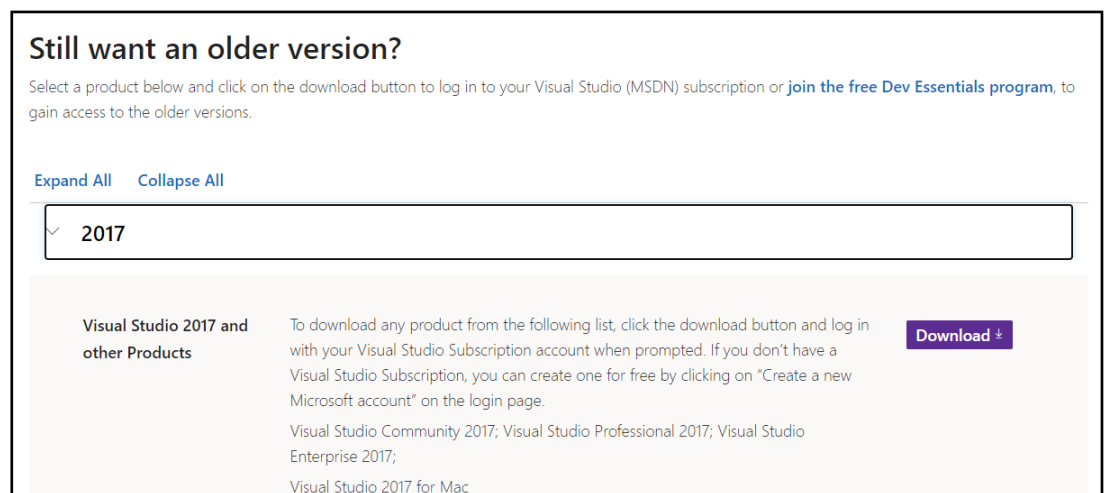
Below shows the summary of software and hardware required to deploy the system:

- Requirements:
  - Desktop or Laptop
  - Microsoft Visual Studio 2017 or higher version
  - Web Browser (Preferable Google Chrome)
  - XAMPP local server
  - Internet Connection

## 6.2 Deployment Process or Steps

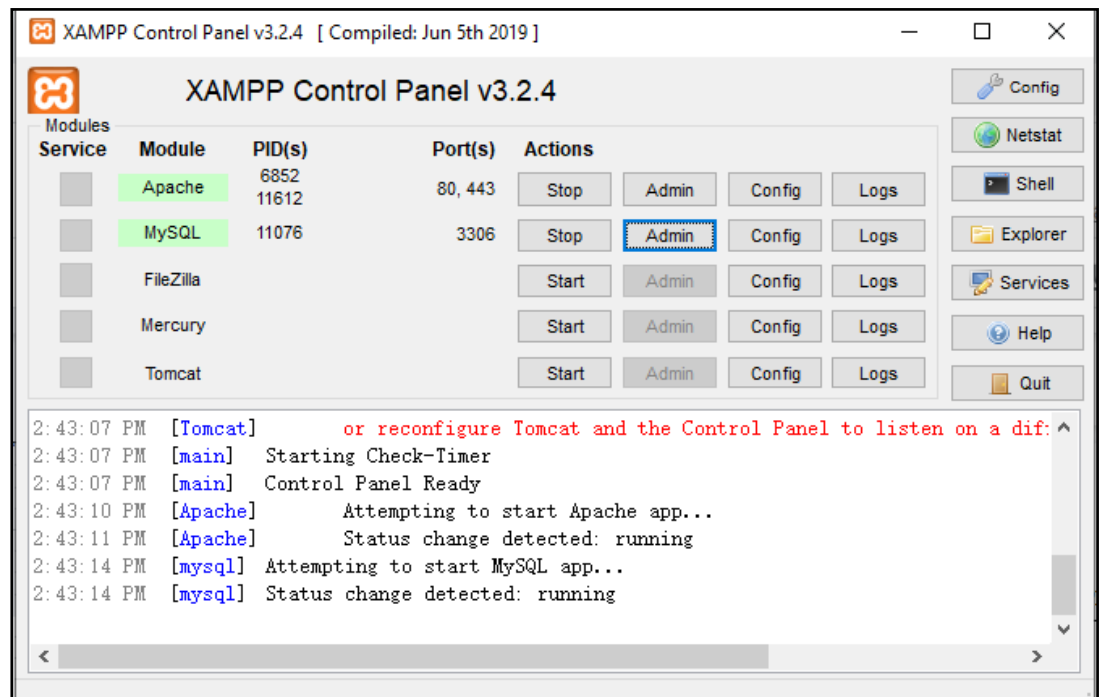
The deployment steps are written to provide the guidance to the user on how to install and deploy the online academic staff booking appointment platform.

1. Install Microsoft Visual Studio 2017 software. The official website to install Microsoft Visual Studio is: <https://visualstudio.microsoft.com/vs/older-downloads/>



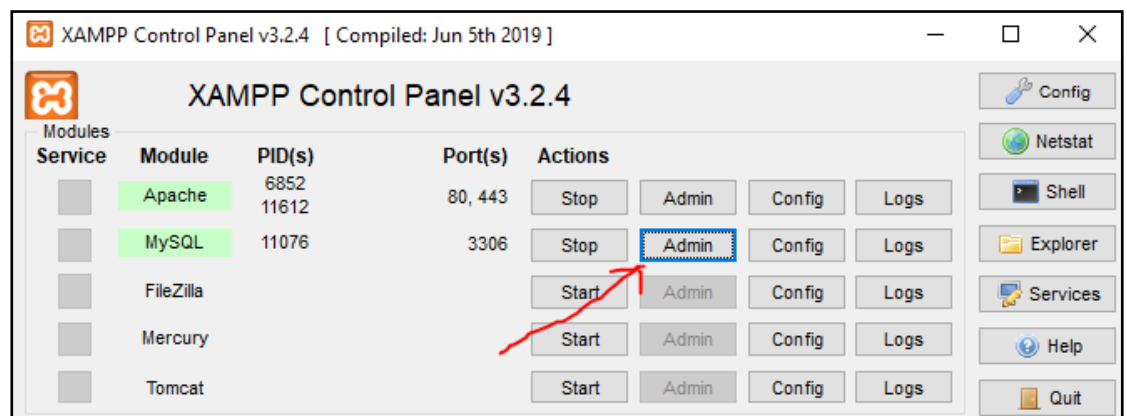
*Preferable to Install Microsoft Visual Studio 2017 version*

2. Install XAMPP local server. The official website to install XAMPP local server is : <https://www.apachefriends.org/download.html>
3. After installing the XAMPP completely. Open the XAMPP control panel. Then, start the Apache service and MySQL service as shown in image below.



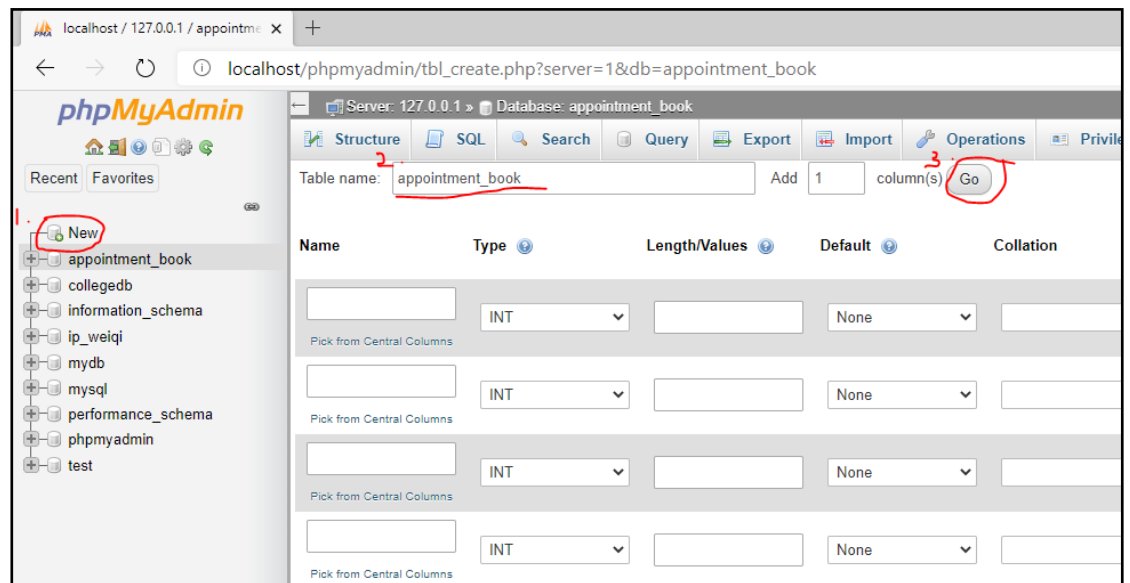
*Start Apache Service and MySQL service in XAMPP Control Panel*

- Click the Admin button of MySQL to open the phpMyAdmin web page.



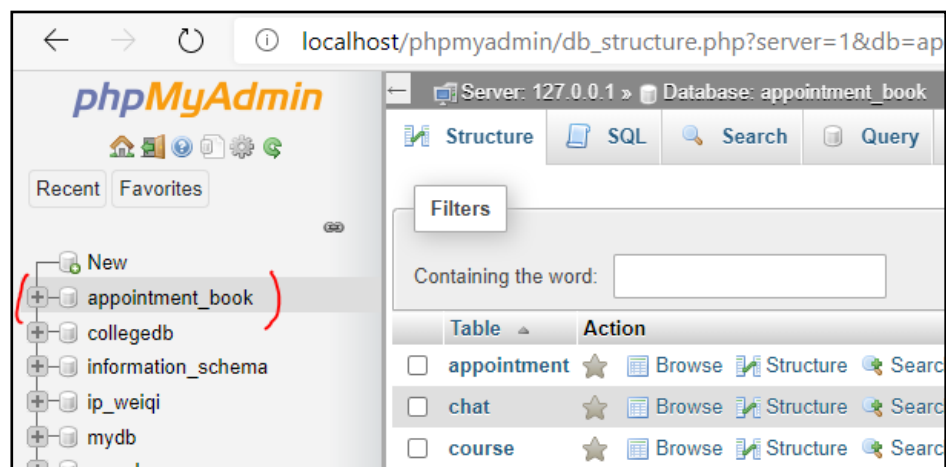
*Example of open MySQL phpMyAdmin web page*

- Click the “New” button to create a database and name it as appointment\_book.



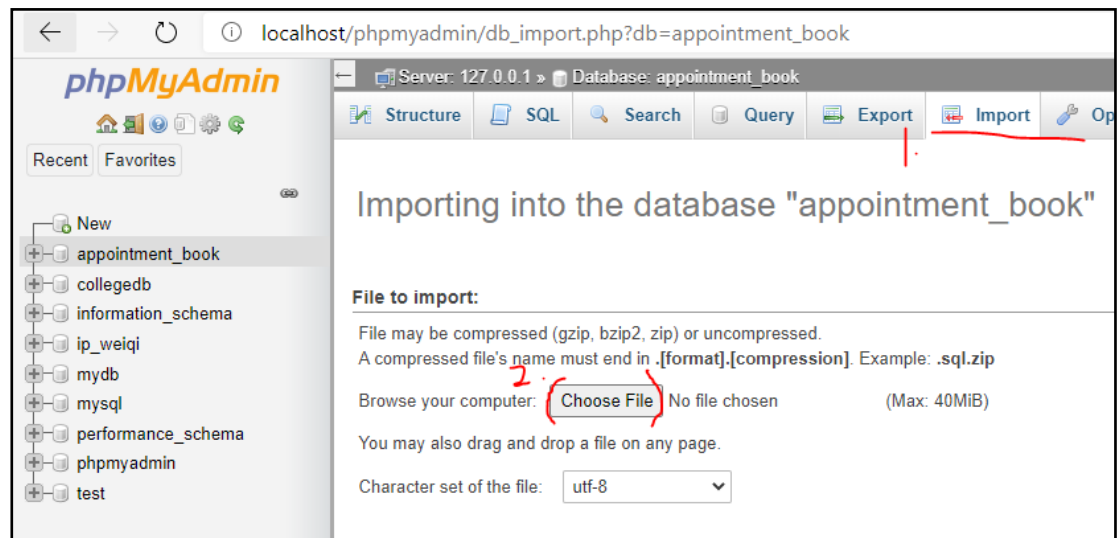
*Create a database appointment\_book in phpMyAdmin*

6. Do not close the phpMyAdmin web page. Open a new web page and open the Google Drive. Then, download the appointment\_book MySQL text file, AppointmentBooking folder and AppointmentBooking\_Staffv folder.
7. After the folders are downloaded, go back to the phpMyAdmin web page and click on the database created just now.



*Click on database created just now*

8. Click the "Import" button and choose the appointment\_book MySQL text file downloaded before.



*Import the appointment\_book MySql text file downloaded*

After that, browse to the bottom of the phpMyAdmin web page and click the “Go” button.

9. Unzip the folders of AppointmentBooking and AppointmentBooking\_Staffv that were downloaded before.

\*AppointmentBooking is the online academic staff booking appointment platform for student.

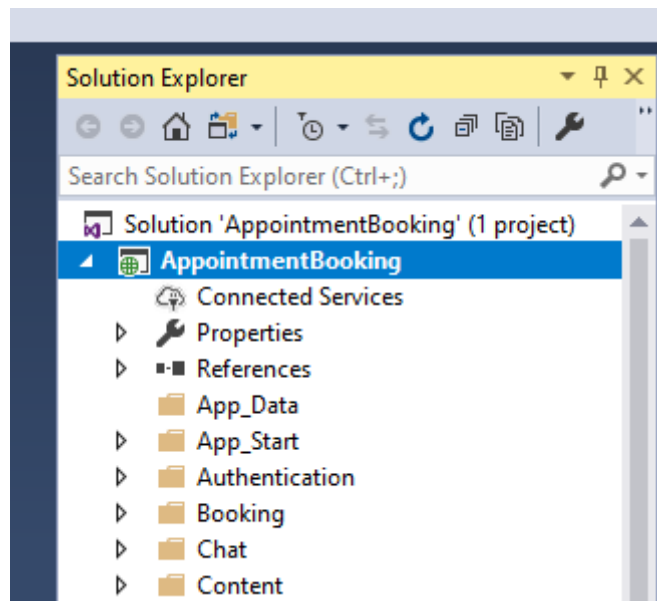
\*AppointmentBooking\_Staffv is the online academic staff booking appointment platform for staff.

10. Open the AppointmentBooking.sln file.

| Name                   | Date modified      | Type                  | Size |
|------------------------|--------------------|-----------------------|------|
| AppointmentBooking     | 13/7/2020 12:33 PM | File folder           |      |
| packages               | 6/6/2020 1:35 PM   | File folder           |      |
| AppointmentBooking.sln | 21/3/2020 9:14 PM  | Visual Studio Solu... | 2 KB |

*Open the AppointmentBooking.sln file*

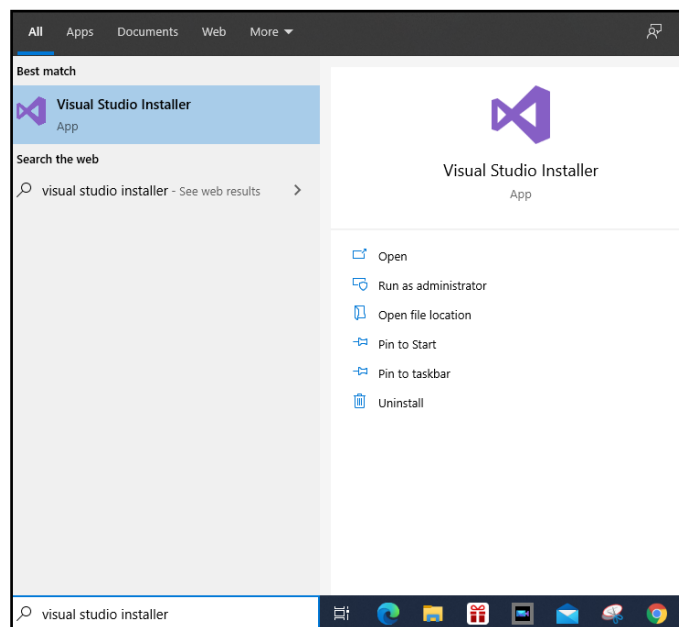
11. After the application is done loading, the programme structure should look like this at the solution explorer window on the right side.



*Example of AppointmentBooking application programme structure*

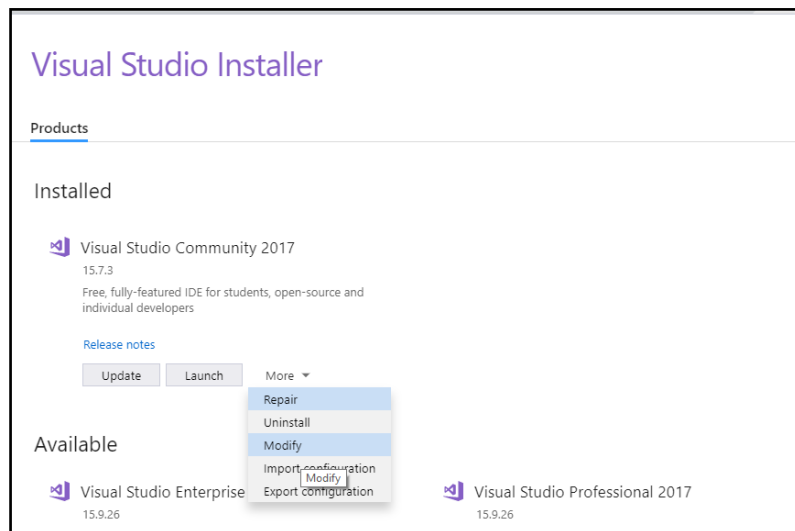
If you encounter a problem of programme structure is not shown as above but shown as unavailable. Try the solution written below:

- On your computer, search visual studio installer and open it.



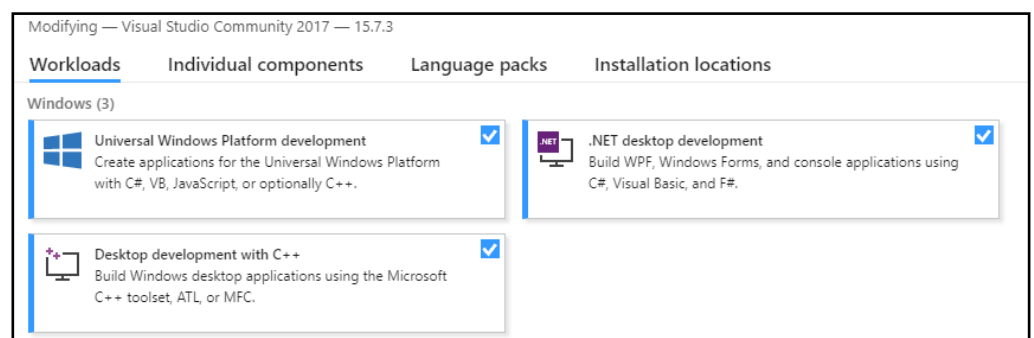
*Search and open visual studio installer*

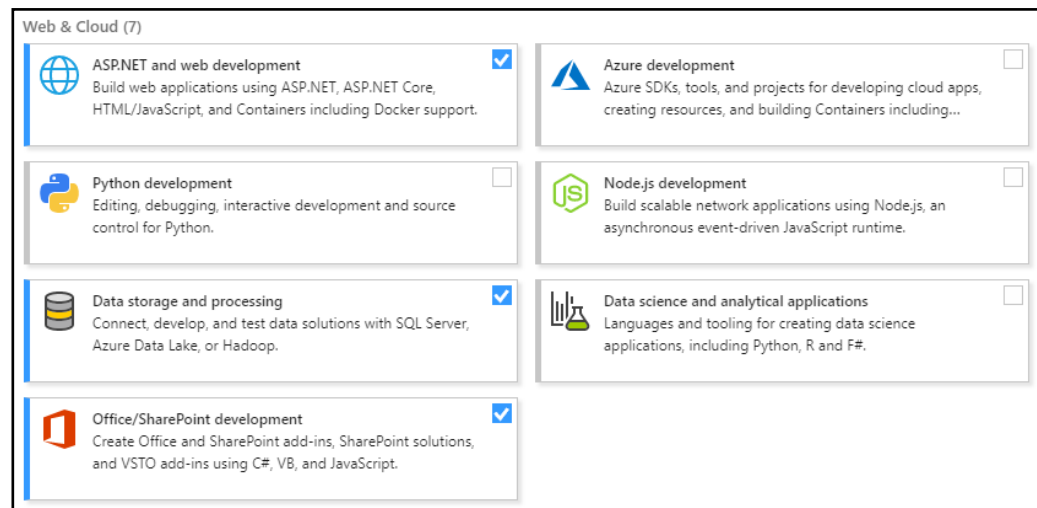
- Click “OK” if a message pops up to allow it to make changes on your computer.
- Click on the drop down arrow shown and select “modify” option.



*Select modify to make changes on visual studio*

- Refer the images below and tick the component if it is untick.  
\*No need to make changes on the specific component if some of the component is ticked in your visual studio installer but it does not tick on images below.

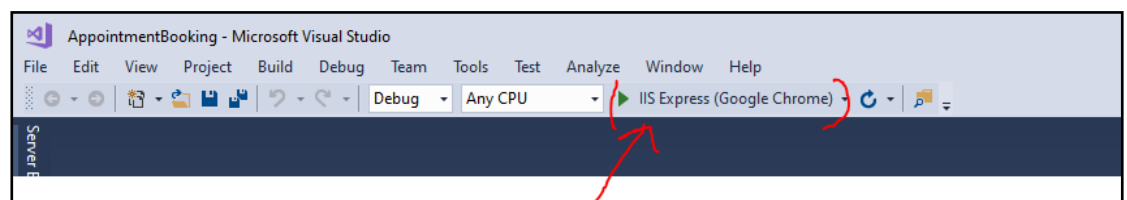




*Components that are needed to deploy the application*

- After that, click the “Modify” button at right bottom and wait for the software to update.
- Once the visual studio software is modified, open the AppointmentBooking.sln file as shown in step 10 and you should see the programme structure of the application as referred to the image in step 11.

12. Click on the “IIS Express (Google Chrome)” button to run the application on the web browser as shown in image below.



*Click the button as shown in image to run the application*

13. The login web page should show as below and you can login as a student by referring to the database student ID in the student table and password in the user table.

Online Academic Staff Booking

localhost:60228/Authentication/loginPage

**TARUC** TUNKU ABDUL RAHMAN UNIVERSITY COLLEGE

**TARUC** Appointment Booking Platform

**Login**

**Student ID**

**Password**

**Login**

[← I forgot my password](#)

*Login web page of AppointmentBooking file (Student Version)*

\*User\_ID in the student table is needed to refer to the user\_ID in the user table to know the password for the respective student\_ID.

Showing rows 0 - 4 (5 total, Query took 0.0023 seconds.)

SELECT \* FROM `student`

Number of rows: 25 Filter rows: Search this table Sort by key: None

|                          | student_ID | tutorial_grp | session_joined | programme_code | user_ID | advisor_ID |
|--------------------------|------------|--------------|----------------|----------------|---------|------------|
| <input type="checkbox"/> | 2000000    | 1            | 201905         | B_SSD          | 1       | 1001       |
| <input type="checkbox"/> | 2000001    | 1            | 201905         | B_SSD          | 2       | 1001       |
| <input type="checkbox"/> | 2000002    | 3            | 202005         | D_MAD          | 3       | 1008       |
| <input type="checkbox"/> | 2000003    | 3            | 202005         | D_MAD          | 4       | 1008       |
| <input type="checkbox"/> | 2000004    | 2            | 201905         | F_C            | 5       | 1010       |

With selected: Edit Copy Delete Export

*Student table in phpMyAdmin web page*

Showing rows 0 - 24 (66 total, Query took 0.0335 seconds.)

SELECT \* FROM `user`

1 > >> | ☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

+ Options

|                          |                  | user_ID | user_name        | user_email                       | user_role | user_password |
|--------------------------|------------------|---------|------------------|----------------------------------|-----------|---------------|
| <input type="checkbox"/> | Edit Copy Delete | 1       | Chew Jane        | chewj-wm18@student.tarc.edu.my   | student   | JC123456      |
| <input type="checkbox"/> | Edit Copy Delete | 2       | Cheng Qing Xiang | chengqx-wt14@student.tarc.edu.my | student   | QX123456      |
| <input type="checkbox"/> | Edit Copy Delete | 3       | Xue Fang He      | xuefh-wm17@student.tarc.edu.my   | student   | FH123456      |
| <input type="checkbox"/> | Edit Copy Delete | 4       | Chang Yue Wu     | changyw-wm17@student.tarc.edu.my | student   | YW123456      |
| <input type="checkbox"/> | Edit Copy Delete | 5       | Chin Fu          | chinf-wm18@student.tarc.edu.my   | student   | CF123456      |
| <input type="checkbox"/> | Edit Copy Delete | 6       | Ms. Jane Chew    | chewjane999@gmail.com            | staff     | JC123456      |
| <input type="checkbox"/> | Edit Copy Delete | 7       | Mr. Ocean Cheng  | xiangzdragan@hotmail.com         | staff     | OC123456      |

*User table in phpMy*

14. Repeat step 10 until 12 for AppointmentBooking\_Staffv for online academic staff booking appointment staff version.
15. Then, the student and staff version of online academic staff booking appointment is considered deployed successfully.

## 6.3 Issues and Solutions

The issue faced is the online academic staff booking appointment platform might be hard to deploy due to there are some extra libraries being used. Hence, a solution is given on step 11 as shown above in order to help solve the problem of unavailability of application.

## 6.4 Chapter Summary and Evaluation

*In this chapter, we had included the software and hardware requirements that are needed to deploy the online academic staff booking appointment platform. Furthermore, deployment steps are written as a guidance for users to install and deploy the application. We had also included the issues and solutions might faced during system deployment in the section of deployment process or steps.*

*Chapter 7 is related to the discussion and conclusion of the online academic staff booking appointment platform. In this chapter, a summary of the application and future improvements will be included.*

## Chapter 7

# **Discussions and Conclusion**

## 7 Discussions and Conclusion

Chapter 7 is focused on the evaluation of project and system developed. Furthermore, this chapter also included the contribution of the system developed towards the involved users.

### 7.1 Summary

In short, this project had proposed a web-based online academic staff booking appointment platform which is aimed to replace the manually appointment booking procedures that are time consuming and inefficient. The targeted users of the platform are staff and undergraduate students in higher education institutions. This platform enables the consultation time booking process to become computerized which is more time saving and its advantage is able to minimize the cases of overbookings occurred. Thus, this proposed platform is able to solve the problem faced using the manual booking process and it can provide a better user experience to the academic staff and the students in an academic institution.

The tools that had been used to develop the platform included Microsoft Visual Studio 2017 and XAMPP. We had used ASP.NET and C# programming language to develop the platform because C# is a high-level programming language which enables the developer to develop the web application system easier and time saving. This project is built based on the Incremental software development methodology which core modules are developed at the first version and the supplementary modules are developed in the next version until the desired system is released. Furthermore, the five stages in the testing process are carried out in the system to ensure the system meet the requirements and behaves correctly.

## 7.2 Achievements

In overall, the system developed has achieved all of the objectives and requirements stated in earlier chapters. Following shows the objectives of online academic staff booking appointment platform objectives that are achieved:

1. Meet all the requirements stated in all modules.
  - Authentication Module:
    - The system is allowed to verify a user by validating the user login details and redirect to the user authorized page according to the user roles and the system is allowed to let the user log out their session.
  - Appointment Booking Module:
    - The system is able to allow students to make appointment requests with the academic staff and also allows the staff to accept or reject a consultation time request.
  - Time Schedule Module:
    - The system is able to let staff manage their available consultation time and their preferred consultation venue.
    - The system also allowed the students to view a staff available consultation time and also let the users manage their appointments which included view appointment details and cancel an appointment.
  - Feedback Module:
    - The system is able to let staff write a feedback or conclusion after an appointment is completed and the student is allowed to read the feedback given by the staff.

- Chatbox Module:
  - The system is able to let users communicate with each other by leaving a message and reply to a message.
- 2. Simplify the process of booking a consultation time appointment.
  - The system developed has achieved the most important objective which is to simplify the appointment booking process so that the students no need to approach the staff through WhatsApp or Email which normally used up a large amount of time and even might be ignored by the staff.
- 3. Help the staff and students can manage their appointment booking easier.
  - The system developed also had achieved the objective which the appointments can be filtered by date in the system and this had helped the users can manage their appointments booking.
  - Moreover, appointment booking status has assigned different colors in the system which enables the users to easily identify an appointment is currently in what status. This function has helped a staff a lot because the staff can easily identify which appointment is in pending status and the staff can make changes on it through accept or reject the appointment.

### 7.3 Contribution

The significance of this framework is that an online academic staff booking appointment platform is proposed and developed for the academic institutions. The platform developed is a new facility and it can bring convenience and benefit to Tunku Abdul Rahman University College (TARUC) staff and students. This platform enables the consultation time booking process to become computerized which is more time saving and its advantage is able to minimize the cases of overbookings occurred.

From the perspective of the TARUC student, students are benefited from the system in which time taken needed to book an appointment is reduced due to the computerized system. Moreover, they are able to choose their desired appointment time based on the available consultation time range of the respective staff.

Apart from that, TARUC staff are benefited from the system that they can manage the appointment bookings more efficiently. By using the system developed, they can manage and handle the appointments easier compared to the manual appointment booking procedures. This is because the manual appointment booking procedures burden the staff that students are using different approaches to request for an appointment such as WhatsApp, Email, SMS or during the class. These different ways of appointment requests also make the staff hard to refer back. Hence, the system developed is able to solve the problem and it also enables the staff to update their consultation time from time to time.

## 7.4 Limitations and Future Improvements

The limitation of the web application of the online academic staff booking appointment platform is that it does not link and store the appointments to the google calendar. In the future, the system can link and store the appointment details in users google calendar by connecting to the google calendar's API using C# programming language. This helps to notify the users about the appointment before the appointment starts.

## 7.5 Issues and Solutions

The issue faced in this framework is the literature review part where we need to compare the proposed online academic staff booking appointment platform with another related system. The solution is we consult with our supervisor to get ideas and guidelines on what needs to be compared and what are the article keywords that we can search for. After the consultation, we are able to find two journal articles related to online booking appointments platform in academic institutions and enables us can make functions and implementations method comparison.

Moreover, the issue encountered is the time constraint to develop the system. The online academic staff booking appointment needs to create 2 versions which are student version and staff version. Each version contains 5 modules with different functionalities. Hence, it is a big challenge for us to develop the system within the deadline given. The solution is to plan our daily schedule carefully and set a weekly target on what requirements we needed to complete that week using the tool Pivotal Tracker. Fortunately, we successfully completely developed the system before the deadline was given.

## 7.6 Conclusion

In a nutshell, Chapter 7 had included the summary of this framework and the evaluation towards this project. The online academic staff booking appointment platform is developed according to the requirements listed in the framework and some extra features are added based on the feedback given by the supervisor. Through the development of the system, I had gained a lot of knowledge and skills as well as the preparation of framework documentation. Hence, I would like to show my gratitude to my supervisor and my teammate that had helped me a lot when preparation for framework and system development.

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